Sampling contacts from sequence

A figure-first writeup on existing predictors, oracle contacts, MarinFold and useful sampling diversity.

Fixed model: exp277 step 266344, 1.47B parameters, 248.584B raw training tokens. Natural proteins lead: 97 validation + 217 publication test. Designs remain separate.
AF2/AF3/Boltz-2 comparators use shared archived MSAs and no templates, with confidence selection.

Predictor generation → cached analysis tables → fast static and interactive rendering.
Every numerical figure element maps back to a source CSV row.

Regrouping by effective MSA depth

Neff80 is measured on all 333 exact frozen predictor alignments; no predictor is rerun. Each sequence contributes the inverse of its >=80%-identity neighbor count. Identity uses columns non-gap in both sequences; duplicate records retain their weights.

Same cutoffs: <10 / 10–<100 / 100–<1000 / >=1000. Contact tier sizes: 15 / 32 / 79 / 188 (314 natural proteins). Structural tier sizes: 14 / 31 / 76 / 184 (305 natural proteins). The lowest tier formerly contained five proteins by raw count.

At Neff<10: MarinFold P@L/2 = 0.401; Helico + MarinFold L/2 GDT-TS = 0.285. L/2 oracle GDT-TS = 0.872. Overall values and all per-protein scores stay unchanged. The draft now uses Neff panels; original raw-depth plots remain in the appendix. Extended AF3 sampling is still only the original five-protein subset, labeled explicitly.

Source: NEFF_ANALYSIS.md; msa_neff.csv; neff_figure_rows.csv; neff_summary.csv.

What the contact and confidence analyses show

AF2 / AF3 / Boltz-2 GDT-TS: 0.854 / 0.860 / 0.861 on 305 natural proteins. At MSA depth <10: 0.196 / 0.332 / 0.239, with only five proteins.

MarinFold contact R-precision: 0.561 overall; 0.345 at MSA depth <10 versus 0.617 at ≥1000. Only five natural proteins occupy the shallowest tier.

Precision @ L/2 contacts: MarinFold 0.713 overall, 0.449 at depth <10. Same nine predictors and 314 proteins as P@L/5; cached source scores, no new inference. Source: PL2_ANALYSIS.md; pl2_summary.csv → pl2_figure_rows.csv → original source rows.

Oracle pTM ranks 1, 5, 1, 1, 1 out of 101 maps for the five low-depth proteins. Highest pTM selects each map's Helico sample; no ipTM or clash penalty. 100 ESMFold2 maps/protein, a shared eligible-pair mask, three Helico samples/map. Original ESMFold2 median TM-score: 0.848, 0.860, 0.881, 0.649, 0.839. pTM–source TM-score Spearman: 0.127, 0.083, 0.254, 0.288, 0.613. Protein order: 8ii8_A, 8oxk_A, 8qoh_A, 8ux2_A, 8wrx_A. Only five biological examples.

Top-L contacts raise Helico GDT-TS from 0.150 to 0.504 on 305 matched natural proteins. The paired gain is 0.354 [0.323, 0.387]. The original top-L recipe was fixed before test inference.

Can more AlphaFold3 sampling recover accurate folds?

5,000 predictions: 1,000 independent full runs for each of five natural test proteins. One diffusion sample per seed, ten recycles, fixed archived MSA, no templates.

8oxk_A: best TM 0.516 in the original 25 → 0.818 in 100 fresh runs → 0.947 in 1,000. pTM selects the 0.947 structure (seed 10470, draw 471). 20 / 1,000 predictions reach TM >=0.8; the first is draw 85. At budget 100, pTM selected a different structure with TM 0.643.

8qoh_A / 8ux2_A: no TM >=0.8 in 1,000; best TM 0.502 / 0.635. 8ii8_A / 8wrx_A: already accurate; best TM 0.881 / 0.936.

Per-protein plots separate oracle best-of-N from pTM-selected TM. The first 100 runs are unchanged; original baseline scores reproduce. Five biological examples, fixed generation settings, no model tuning.

AF3 sampling in predictor and MSA-depth context

8oxk_A: ESMFold2 already reaches TM 0.940; AF3 at 1,000 reaches 0.947. Five low-depth proteins, mean TM: ESMFold2 0.854; AF3 baseline 0.603; AF3 at 1,000: 0.681 with official ranking, 0.719 with pTM, 0.780 oracle best. TM >=0.8: ESMFold2 4/5; original AF3 2/5; expanded AF3 3/5.

At depths 10–99 / 100–999 / >=1000, baseline AF3 means: 0.891 / 0.932 / 0.955. ESMFold2: 0.691 / 0.874 / 0.944. Same 305 proteins; bin n=5 / 20 / 60 / 220. Different proteins occupy each bin; extended sampling only exists at depth <10.

Official defaults: ten recycles, five diffusion samples per user-specified seed. Our baseline: 5 full seeds x 5 samples; expanded: 1,000 full seeds x 1 sample. Both use fixed archived MSAs and no templates, not the native search pipeline. 40x structures, 200x trunk evaluations; neither is a measured runtime ratio. Official ranking includes disorder/clash terms; pTM ranking is shown separately.

Useful diversity is the open question

314 proteins × 100 samples: consensus R-precision 0.561; oracle best individual map 0.528. Paired difference −0.032, 95% bootstrap interval [−0.038, −0.025].

All 100 maps are distinct for every protein; mean Jaccard 0.277, true-contact union recall 0.943. This is not an absence of contact-set diversity, and it does not directly measure distinct folds.

Next: better whole-map candidates, inference-time search and post-training.

Low benchmark MSA depth versus ESMC training sources

ESMC documents UniRef 2023_02, MGnify 2023_02 and JGI (July 2023), not Logan. Low retrieved alignment depth does not establish novelty to the language model.

New historical UniRef search, with MSA filtering disabled: Significant hits covering at least half the query: 1, 4, 1, 15, 1. Protein order: 8ii8_A, 8oxk_A, 8qoh_A, 8ux2_A, 8wrx_A. 14 of the 15 well-covered 8ux2_A hits are 97.9–100% identical to the query. Counts exclude the added query; they are not Neff or final training-cluster counts.

Online MGnify30 hits count heavily clustered representatives, not training-source depth. ESMC uses 70%-identity clustering; historical MGnify90 is the closer public search target. Full historical MGnify and JGI are not yet searched. Complete training depth is unknown. See TRAINING_SOURCE_SEARCH.md and data/training_source_depths.csv for details. Offline reduction: uv run python generation/search_training_sources.py prepare

L/2 oracle contacts across all MSA depths

The oracle dots in Figures 02 and 05 now use L/2 randomly sampled true contacts. Same 305 natural proteins: 5 / 20 / 60 / 220 across the four MSA tiers. Mean GDT-TS: 0.906 / 0.841 / 0.853 / 0.839; overall 0.843 versus full-map oracle 0.893. No MSA or negative contacts. Average two subsets per protein after three-sample confidence selection. 610 maps and 1,830 structures complete; exact low-depth results reused. One protein has 109 available positives versus 110 requested. Source: ORACLE_L2.md; oracle_l2_per_protein.csv → helico_oracle_l2_samples.csv.

Match MarinFold and oracle contact counts

Same archived 248B-model rankings; exact top-L/2 positives, no oracle seeds or MSA. 305 proteins, three Helico samples/map, original ranking-score selection.

Mean GDT-TS, top-L → top-L/2: <10: 0.261 → 0.297; 10–99: 0.325 → 0.341; 100–999: 0.369 → 0.370; >=1000: 0.563 → 0.524. Overall: 0.504 → 0.478; paired change −0.026, 95% interval [−0.041, −0.012]. Shallower-tier paired intervals include zero; only five proteins have depth <10. L/2 oracle remains 0.843 overall. Matching count leaves a large accuracy gap.

915 new structures audited; old predictor points remain unchanged. Figure 05 shows L/2 as its only MarinFold variant alongside every existing predictor. Source: MARINFOLD_L2.md; marinfold_l2_per_protein.csv → raw sample rows.

Reducing predicted contacts further to L/5

Same archived rankings, 305 proteins, Helico recipe and confidence selector. Each exact L/5 map is a subset of the frozen L/2 input; no oracle seeds or MSA.

Mean GDT-TS, L / L/2 / L/5: <10: 0.261 / 0.297 / 0.245; 10–99: 0.325 / 0.341 / 0.331; 100–999: 0.369 / 0.370 / 0.304; >=1000: 0.563 / 0.524 / 0.402. Overall: 0.504 / 0.478 / 0.376. The L/2 oracle remains 0.843. Paired L/5 minus L/2: −0.103, 95% interval [−0.123, −0.083]. L/5 minus L: −0.129, 95% interval [−0.151, −0.108]. Shallow-tier intervals include zero; only five proteins have depth <10.

All 915 new structures and direct timings retained; original scores unchanged. Exploratory follow-up requested after earlier results; no model tuning. Source: MARINFOLD_L5.md; marinfold_l5_per_protein.csv → raw sample rows.

Sparse oracle contacts at very low MSA depth

All five natural proteins with MSA depth <10; Helico receives no MSA. Random positive contacts only; every unselected pair remains unknown. Two random subsets/budget/protein, three diffusion samples/map, confidence selection. Mean GDT-TS: none 0.244; 5 contacts 0.296; 10 contacts 0.362; L/5 0.622; L/2 0.906. All true contacts 0.918; full positive/negative oracle 0.932. ESMFold2 baseline 0.680. Per-protein panels retain both subset results. L/2 comes close to the full map. Only five proteins; two subsets do not fully characterize map-sampling variability. Source: ORACLE_BUDGET.md, data/oracle_budget_*.csv; render.py uses cached tables.

Can MarinFold extend known contacts?

Five natural proteins with MSA depth <10; fixed 248B-token model, no MSA input. Same 0, 5, 10 or L/5 true seed contacts in both arms; two subsets when nonzero. Direct Helico versus retaining the seeds and filling to L/2 with 100-rollout consensus.

Mean GDT-TS, direct → completed: 0 seeds: 0.244 → 0.301; 5: 0.296 → 0.379; 10: 0.362 → 0.469; L/5: 0.622 → 0.747. At 10 seeds, 8oxk_A / 8qoh_A / 8wrx_A improve; 8ii8_A / 8ux2_A worsen. At L/5 all five improve, with the largest gains on 8oxk_A and 8qoh_A. Mean TM at 10 seeds: 0.573 → 0.712. Mean lDDT: 0.491 → 0.604. Added-contact precision alone: 44.6% / 56.6% / 63.1% / 81.3% across seed budgets.

All 3,500 rollouts and 105 new structures retained; direct predictions reused exactly. Confidence selection per map, subset averaging within protein, protein bootstrap. Exploratory oracle-seeded diagnostic; only five biological examples. Source: SEED_COMPLETION.md; seed_completion_per_protein.csv → selected source rows.

Precision at L/5 and the KNN comparison

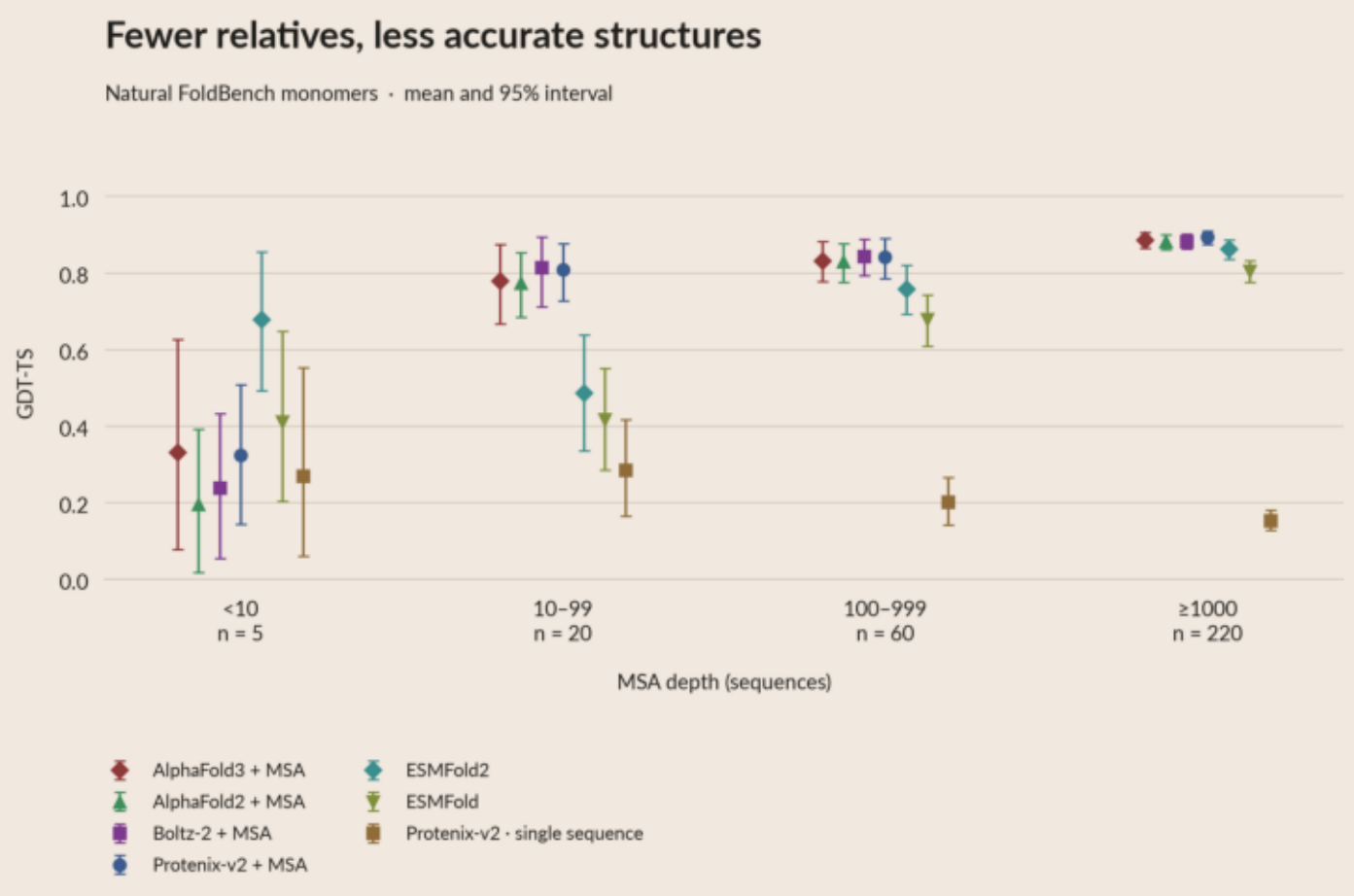
k=max(1,floor(L/5)); L is the input sequence length, not the resolved-residue count. Same 314 natural proteins, nine predictors, candidate pairs and archived predictions. All-range / long-range P@L/5: MarinFold 0.830 / 0.761; sequence-KNN 0.657 / 0.595. At depth <10 (five proteins): MarinFold 0.595; KNN 0.102; ESMFold2 0.727. KNN transfers from ten native decontaminated-corpus neighbors; no redesign index.

Same 100-rollout pools: mean individual 0.400; consensus 0.829; oracle best 0.609. Oracle reselected by P@L/5. Emission order ranks individual maps; short maps retain k. Invalid individual maps receive zero. Consensus follows the original vote rules. All 5,994 predictor and 1,884 sampling R-precision cells reproduce before rescoring.

Every static plot has native vector PDF/SVG and a 7,200-pixel-wide PNG. At 300 dpi the raster supports 24-inch-wide panels; vector PDFs scale further. POSTER.md links the public collection. Preprocessing remains separate from rendering.

01_predictors.png

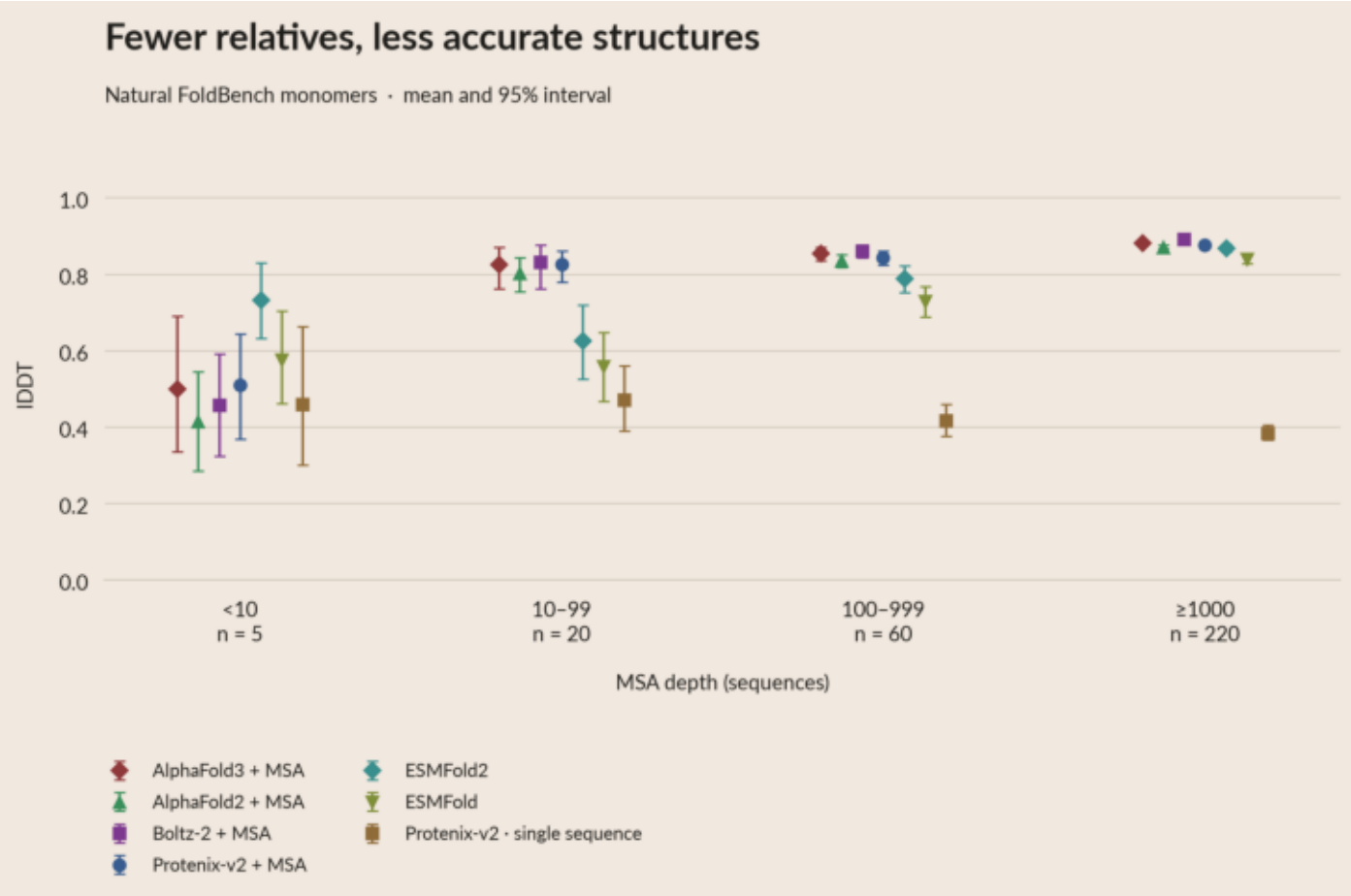
Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



$ render.py

01_predictors_lddt.png

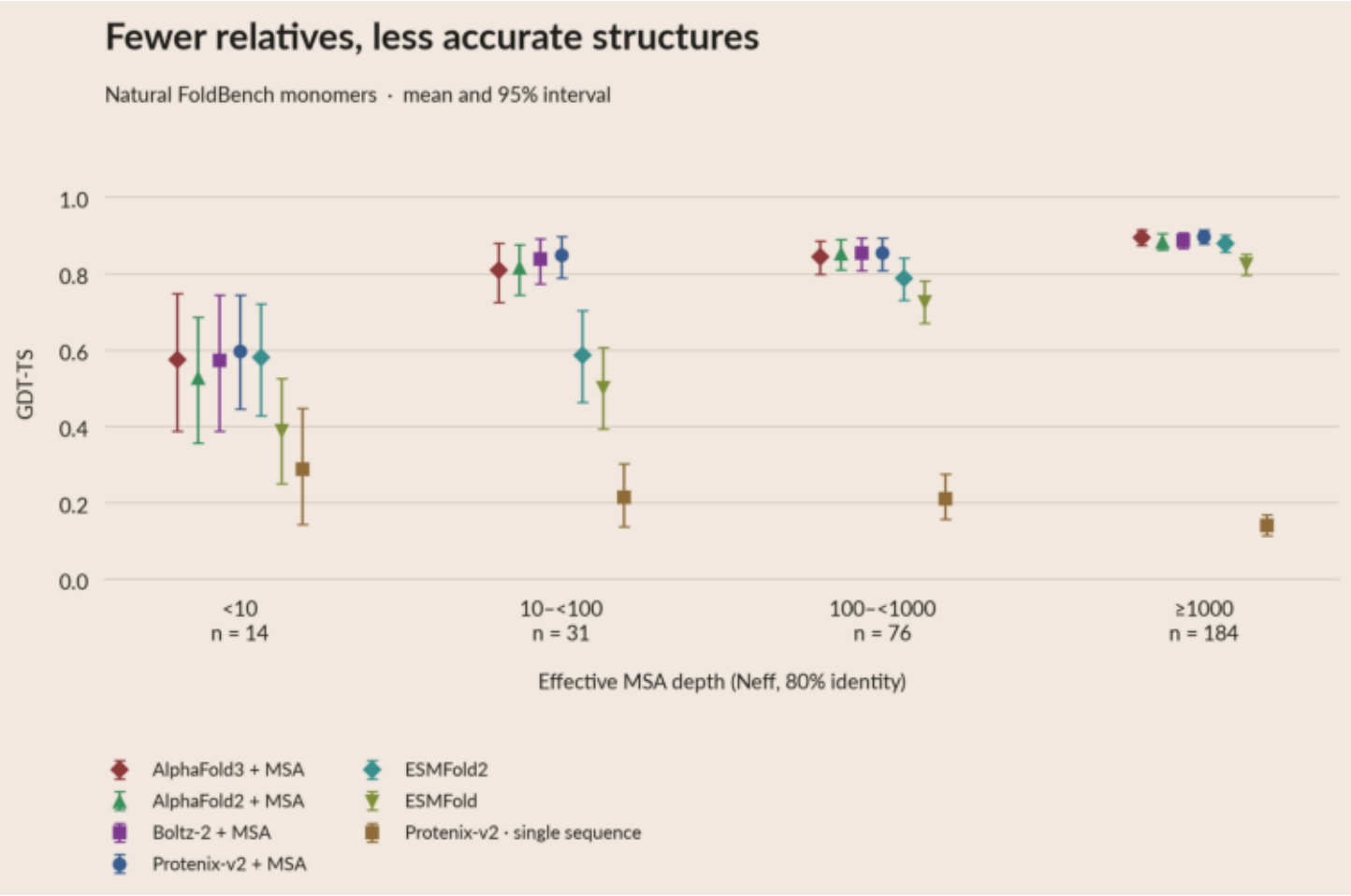
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01_predictors_neff.png

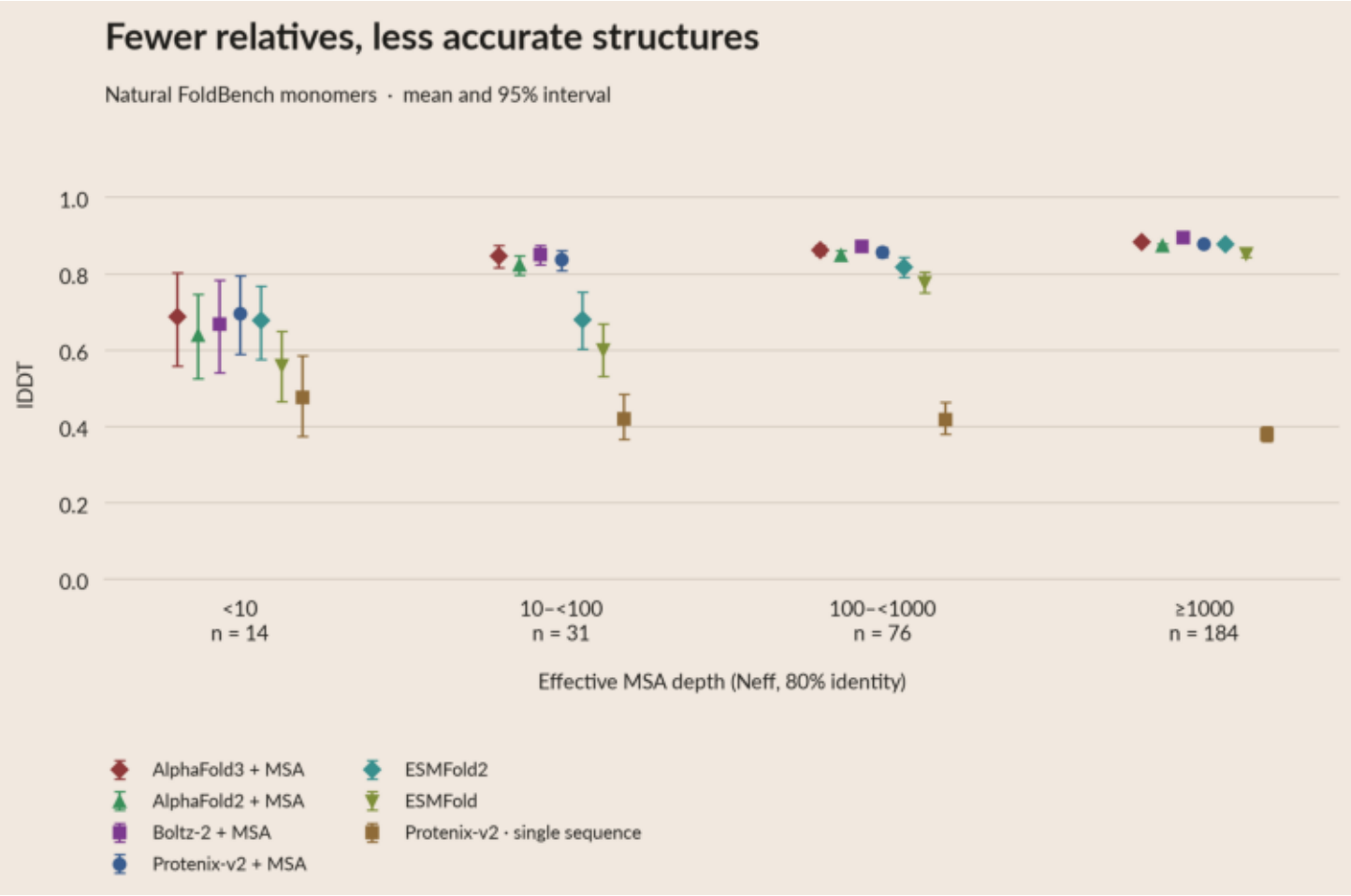
Same seven archived structure predictors; 305 matched natural proteins. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

01_predictors_neff_lddt.png

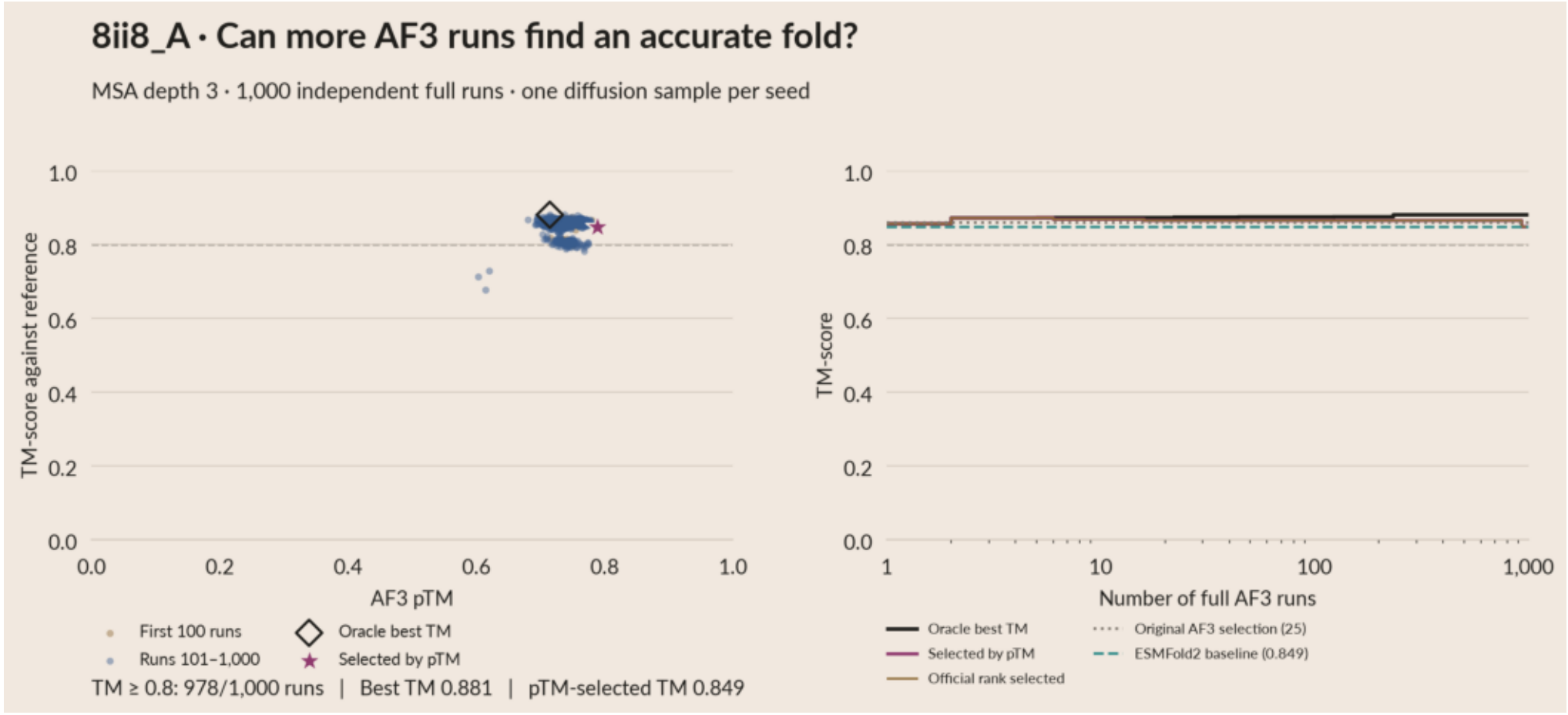
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$ render.py

01b_af3_sampling_8ii8_A.png

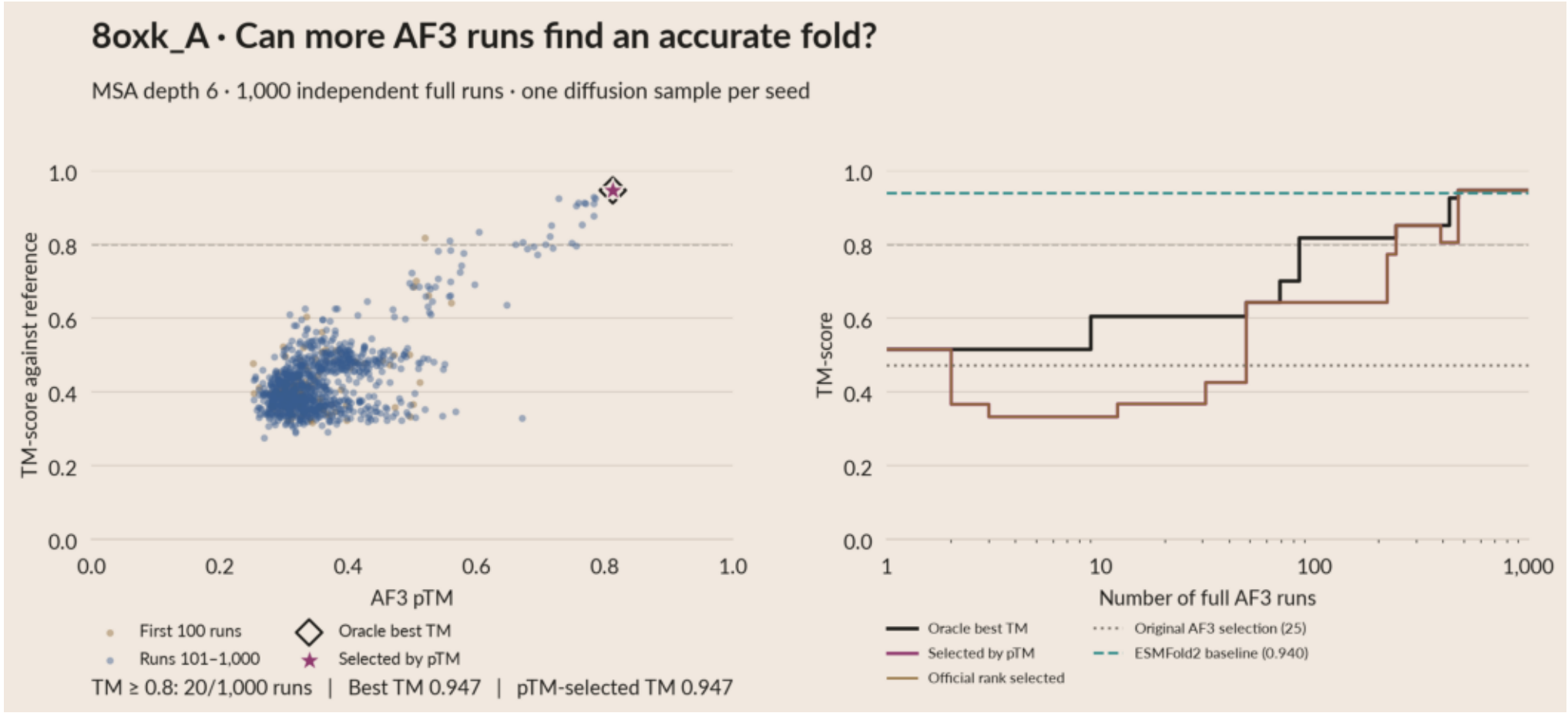
8ii8_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM >=0.8. Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking_score. Data: af3_sampling_samples.csv and af3_sampling_curves.csv; original baseline: af3_sampling_original.csv. ESMFold2 baseline: af3_context_rows.csv. Official ranking_score selection is shown separately from pTM. All are in data/; preprocessing: prepare_af3_sampling.py and prepare_af3_context.py.




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$ render_af3_sampling.py
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01b_af3_sampling_8oxk_A.png

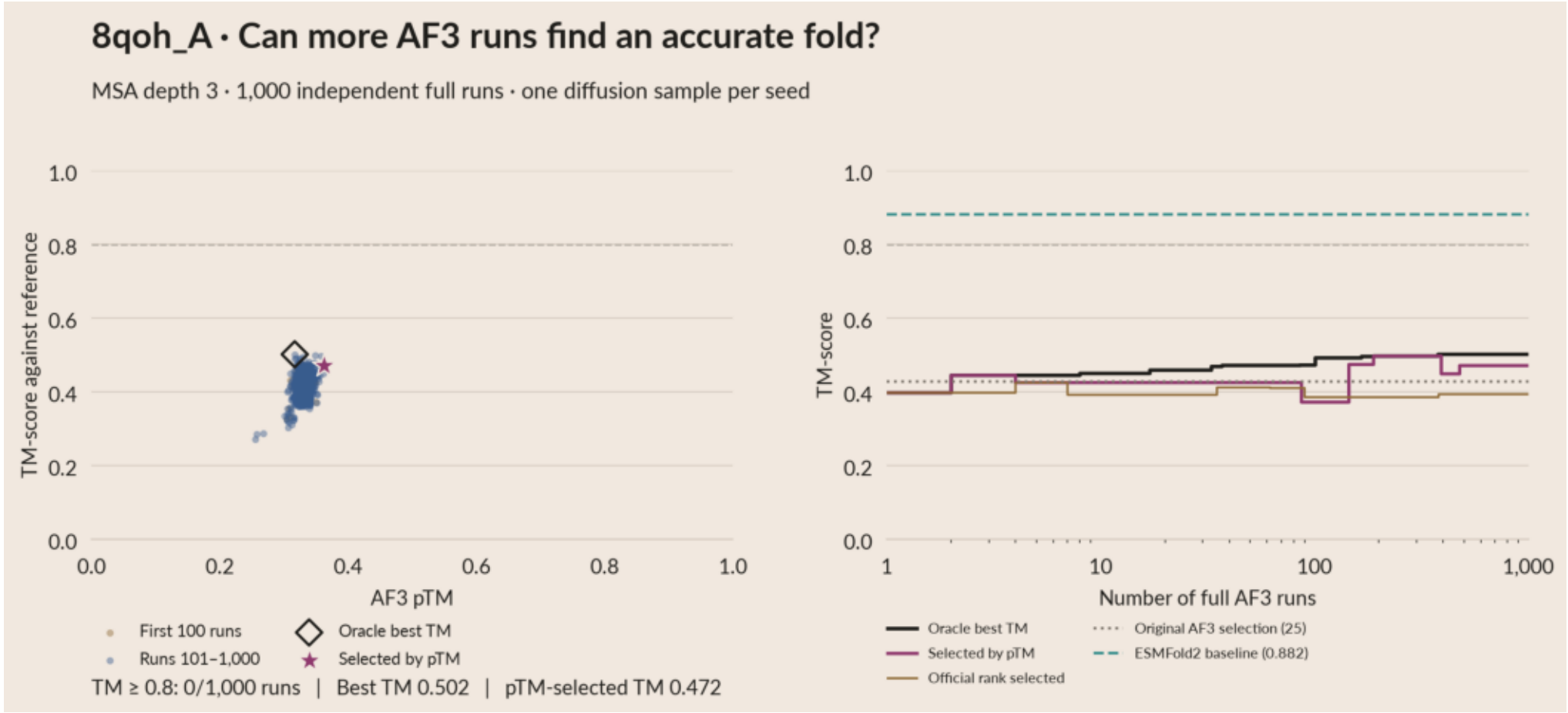
8oxk_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM >=0.8. Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking_score. Data: af3_sampling_samples.csv and af3_sampling_curves.csv; original baseline: af3_sampling_original.csv. ESMFold2 baseline: af3_context_rows.csv. Official ranking_score selection is shown separately from pTM. All are in data/; preprocessing: prepare_af3_sampling.py and prepare_af3_context.py.




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$ render_af3_sampling.py
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01b_af3_sampling_8qoh_A.png

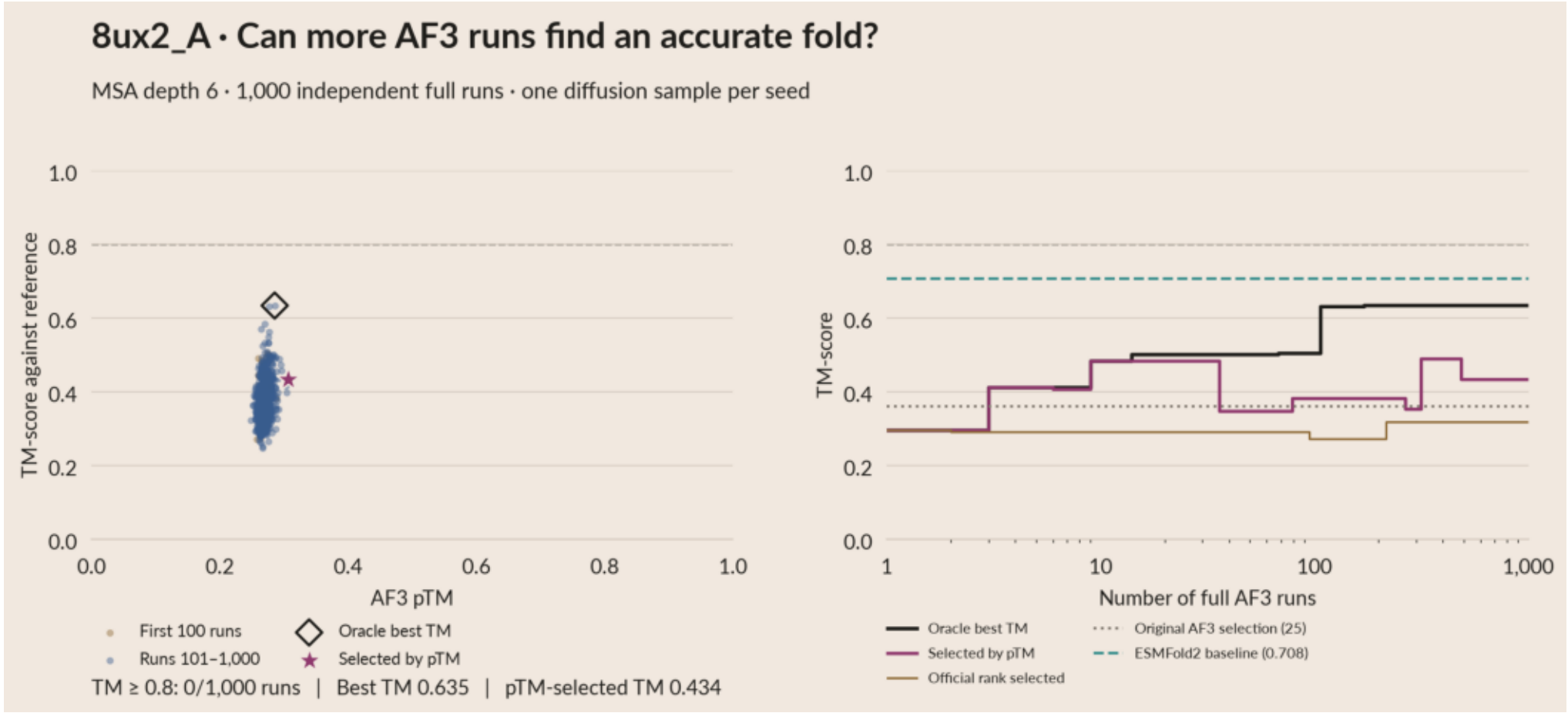
8qoh_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM >=0.8. Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking_score. Data: af3_sampling_samples.csv and af3_sampling_curves.csv; original baseline: af3_sampling_original.csv. ESMFold2 baseline: af3_context_rows.csv. Official ranking_score selection is shown separately from pTM. All are in data/; preprocessing: prepare_af3_sampling.py and prepare_af3_context.py.




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$ render_af3_sampling.py
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01b_af3_sampling_8ux2_A.png

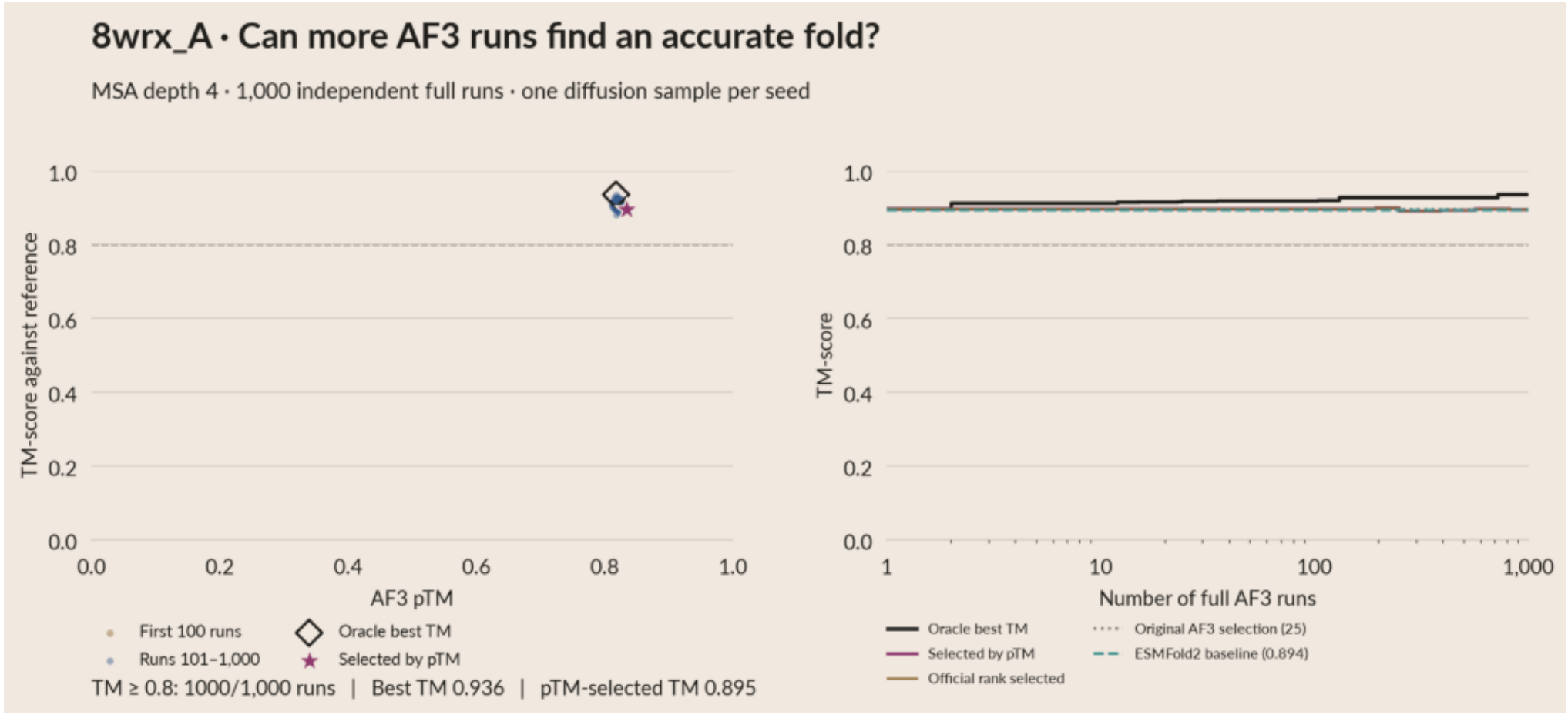
8ux2_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM >=0.8. Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking_score. Data: af3_sampling_samples.csv and af3_sampling_curves.csv; original baseline: af3_sampling_original.csv. ESMFold2 baseline: af3_context_rows.csv. Official ranking_score selection is shown separately from pTM. All are in data/; preprocessing: prepare_af3_sampling.py and prepare_af3_context.py.




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$ render_af3_sampling.py
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01b_af3_sampling_8wrx_A.png

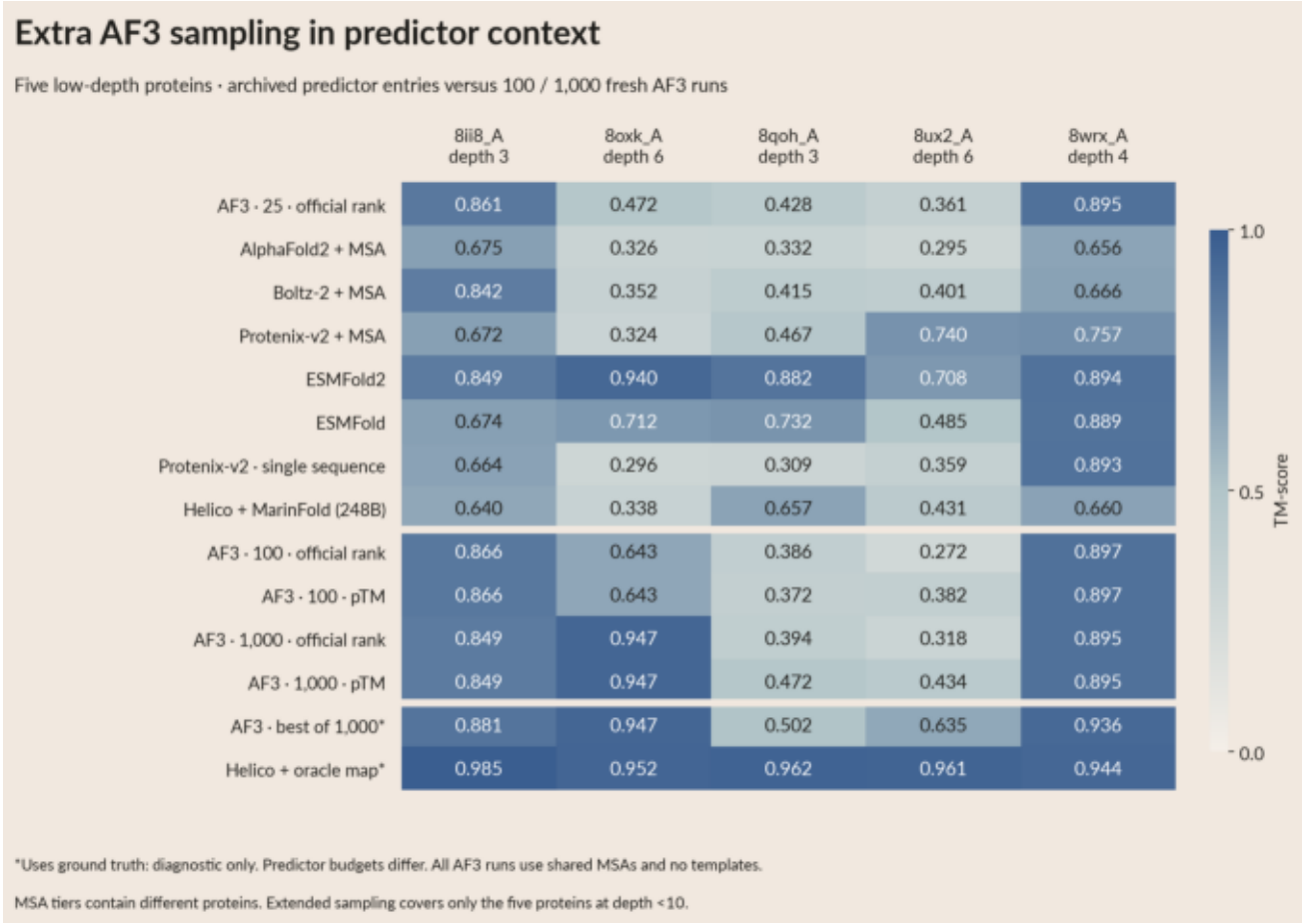
8wrx_A: each point is one independent AF3 trunk + diffusion run, seeds starting at 10000, ten recycles, fixed archived MSA and no templates. Full-precision pTM selects without using truth. Oracle best TM uses truth only for this diagnostic. Dashed line: prespecified TM >=0.8. Curves use the ascending-seed prefix; original dotted baseline selected among five seeds x five samples by official ranking_score. Data: af3_sampling_samples.csv and af3_sampling_curves.csv; original baseline: af3_sampling_original.csv. ESMFold2 baseline: af3_context_rows.csv. Official ranking_score selection is shown separately from pTM. All are in data/; preprocessing: prepare_af3_sampling.py and prepare_af3_context.py.




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$ render_af3_sampling.py
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01c_af3_context.png

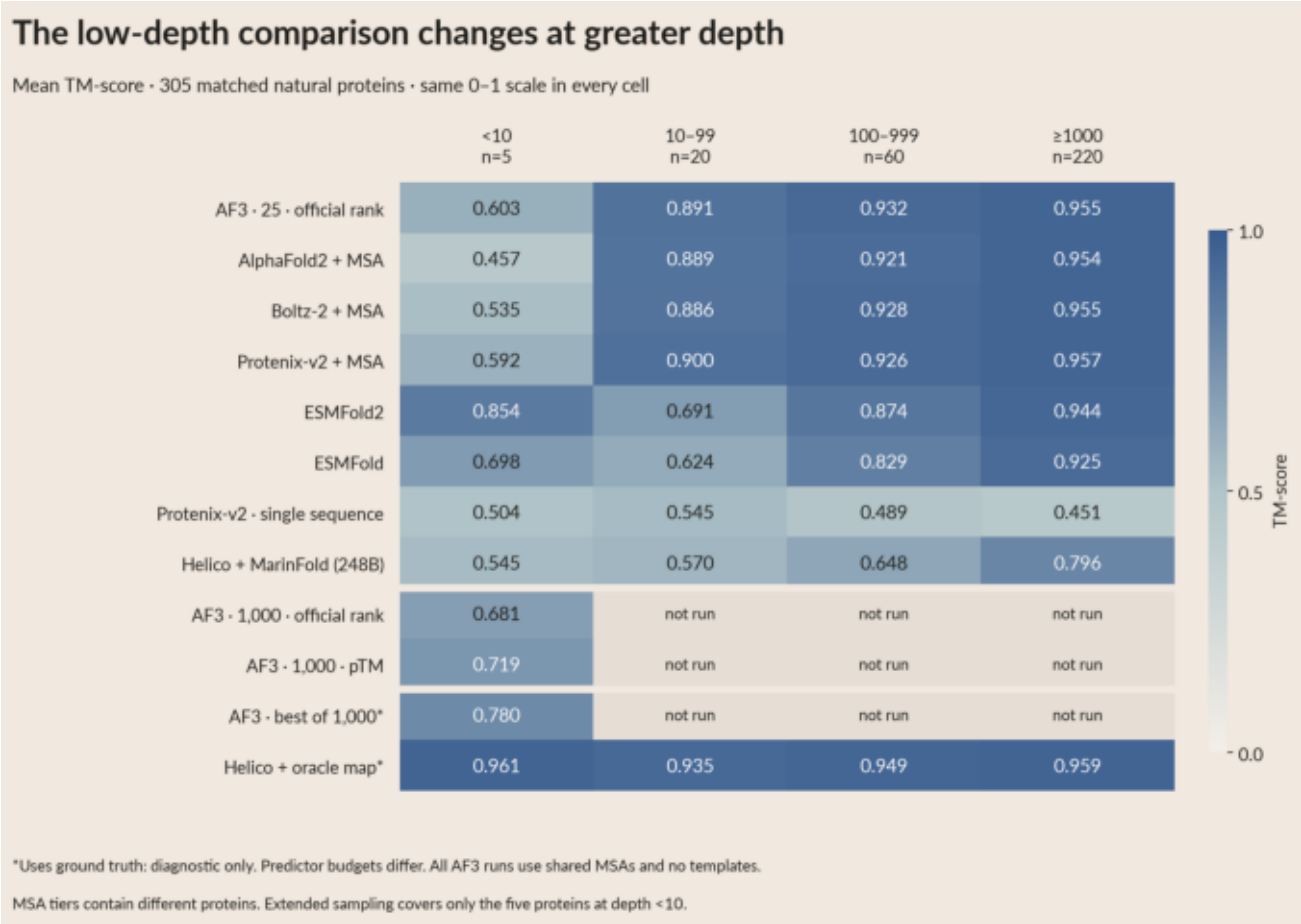
TM-score for the same five natural test proteins with MSA depth <10. Top block: archived predictor entries, including the 248B-token MarinFold top-L contacts through Helico. Middle: 100 or 1,000 fresh AF3 full runs, one diffusion sample per seed, selected by official ranking_score or pTM. Bottom: ground-truth diagnostics, oracle best AF3 TM and Helico conditioned on the oracle map. ESMFold2 is the archived baseline, not a selected member or average of the 100-contact-map decoy pool. AF3 baseline uses five seeds times five diffusion samples, ten recycles, shared archived MSAs and no templates. Expanded runs retain the same inputs and recycles. Compute budgets differ. Hover identifies the exact source CSV cell.




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$ render_af3_context.py
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01d_af3_depth_context.png

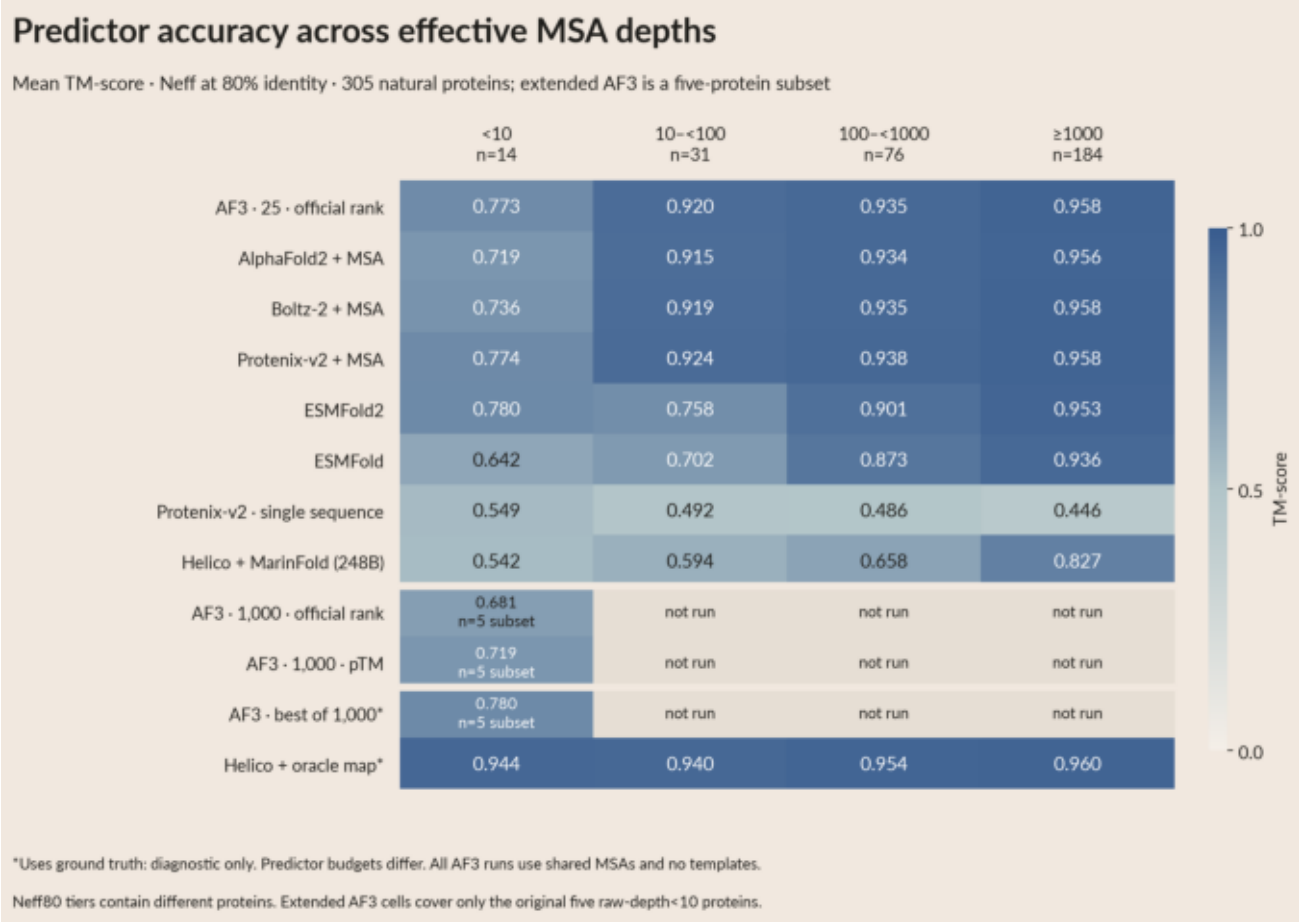
Mean TM-score on the same 305 natural proteins as the main structural figures: 5, 20, 60 and 220 proteins across the four MSA bins. All archived predictors cover exactly this population; each protein receives equal weight. Hover shows 95% protein-bootstrap intervals and success counts; the prepared CSV retains every contributing row. The 1,000-run AF3 study covers only the five <10 proteins; other cells are not run. Stars mark ground-truth diagnostics. Different bins contain different proteins, so this does not estimate the causal effect of adding MSA sequences. All AF3 results use shared benchmark MSAs and no templates, not the native search pipeline. Budgets and training data differ between predictors.




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01d_af3_depth_context_neff.png

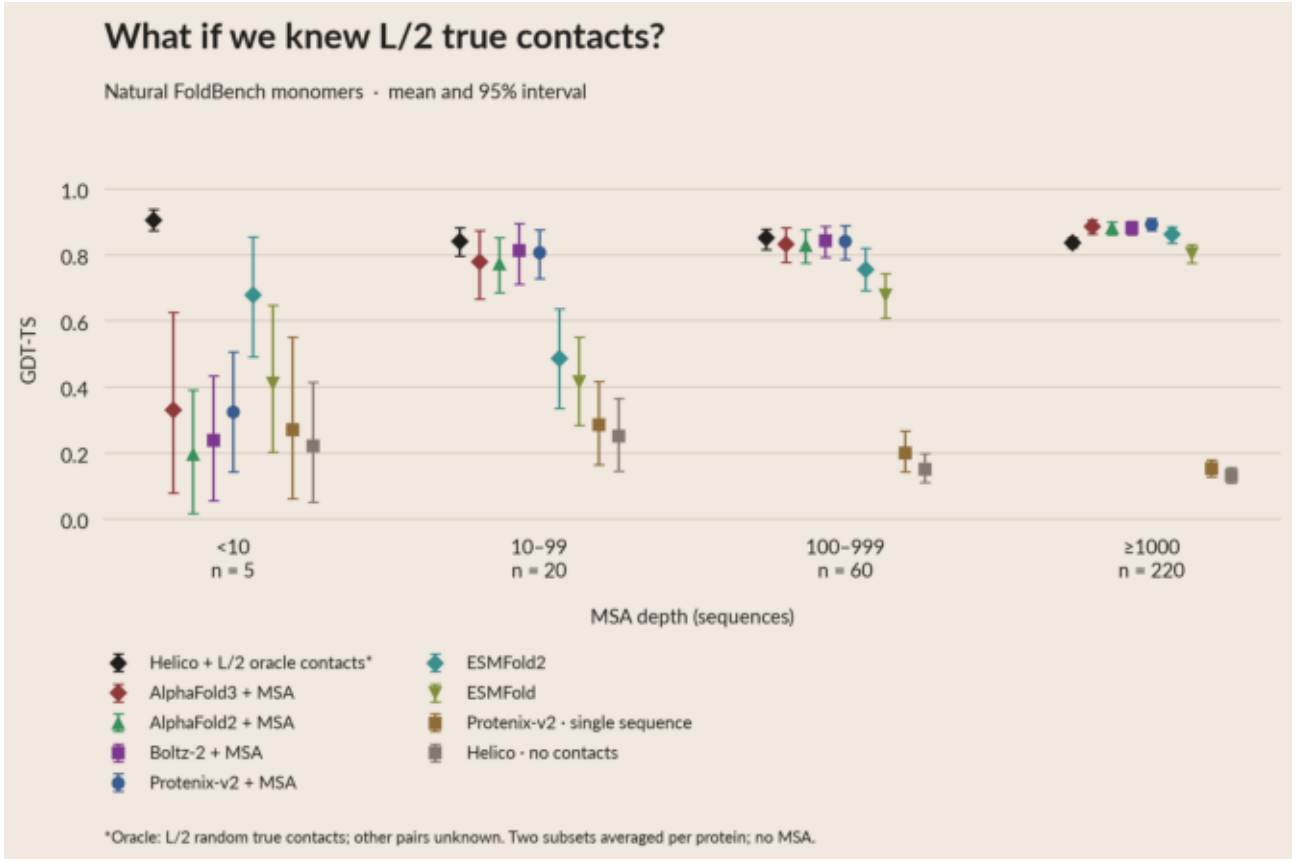
Mean TM-score regrouped by Neff80 on the same 305 natural proteins and frozen alignments. Archived predictor rows retain their original protocols, including top-L MarinFold contacts and the full-map oracle in this diagnostic. Extended AF3 sampling still covers only the original five raw-depth<10 proteins; its Neff<10 cells explicitly show n=5, a subset of the full tier. Other extended-sampling cells are not run. These are not matched tier-wide AF3 sampling comparisons. Every source score and protein is preserved in neff_af3_context_rows.csv and neff_af3_context_summary.csv. See NEFF_ANALYSIS.md and AF3_SAMPLING.md.




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$ render_af3_context.py
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02_oracle.png

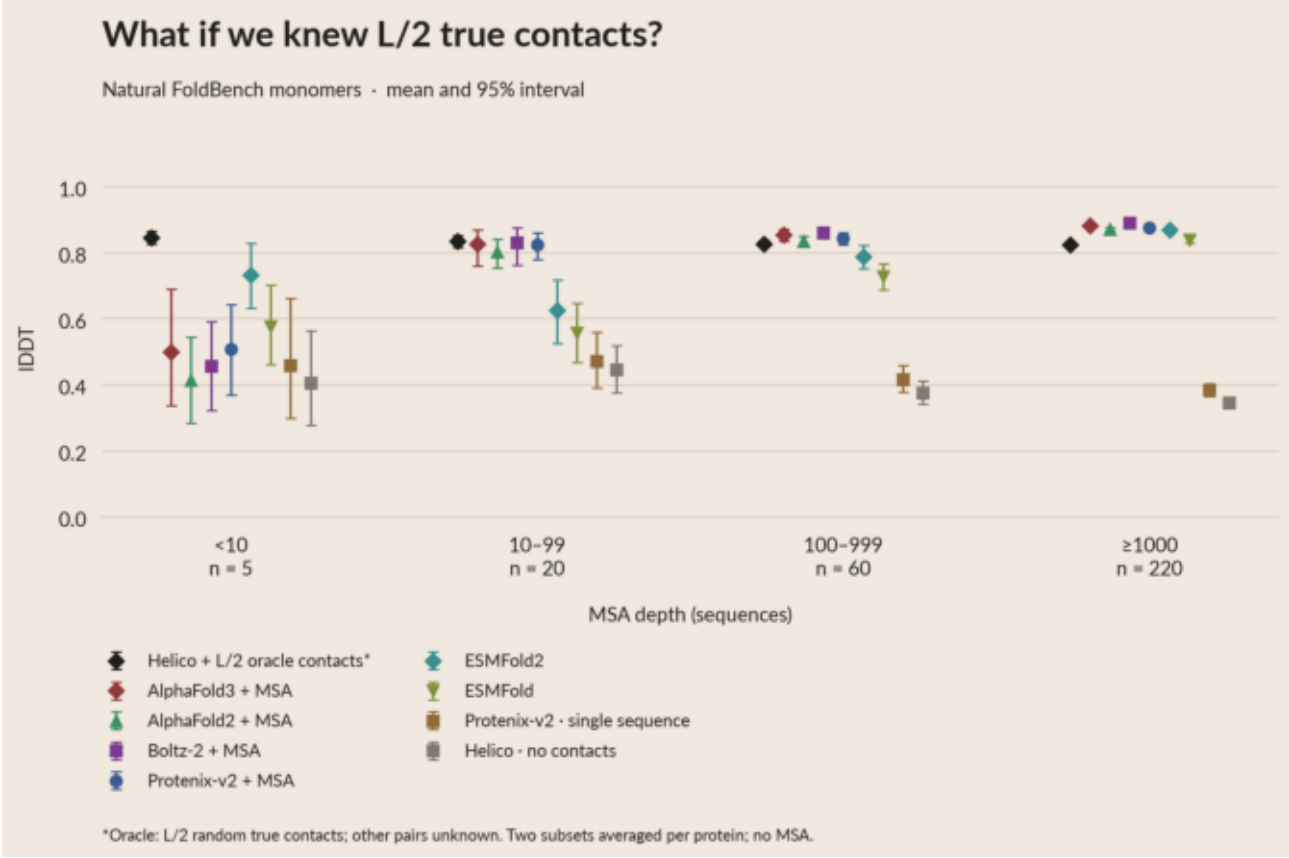
*Oracle = two random subsets of floor(L/2) true contacts per protein; every unselected pair is unknown and no non-contacts are supplied. L is the frozen input sequence length; counts cap at available positives (see ORACLE_L2.md). Three diffusion samples/subset, highest ranking_score selection, then average both subset accuracies within protein. Helico receives no MSA. Same 305 natural proteins as Figure 1, with all seven archived predictors and no-contact Helico; 95% protein-bootstrap intervals. This uses ground truth and is not a deployable predictor. The full-map oracle remains in separate confidence/budget diagnostics. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



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02_oracle_lddt.png

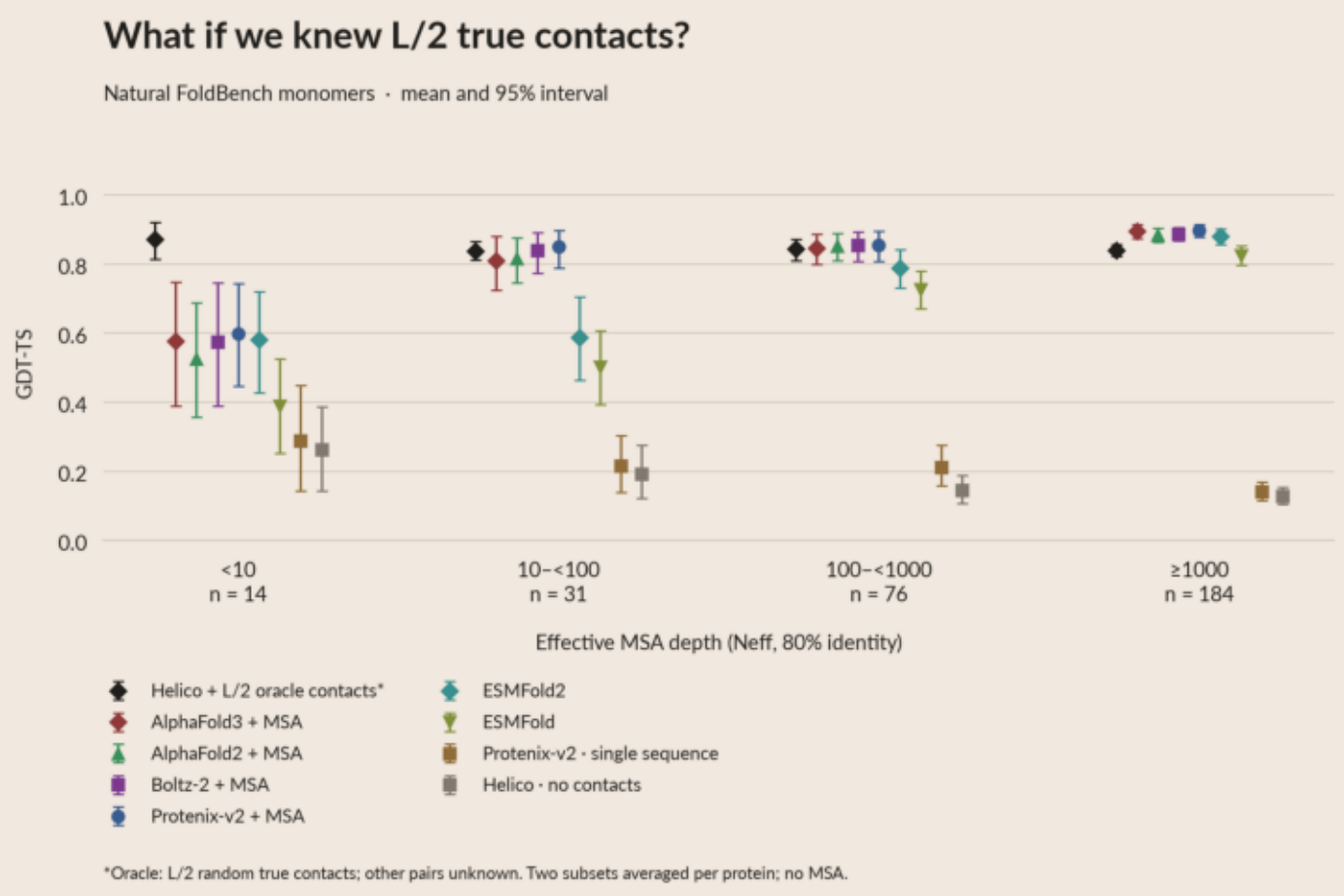
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$ render.py

02_oracle_neff.png

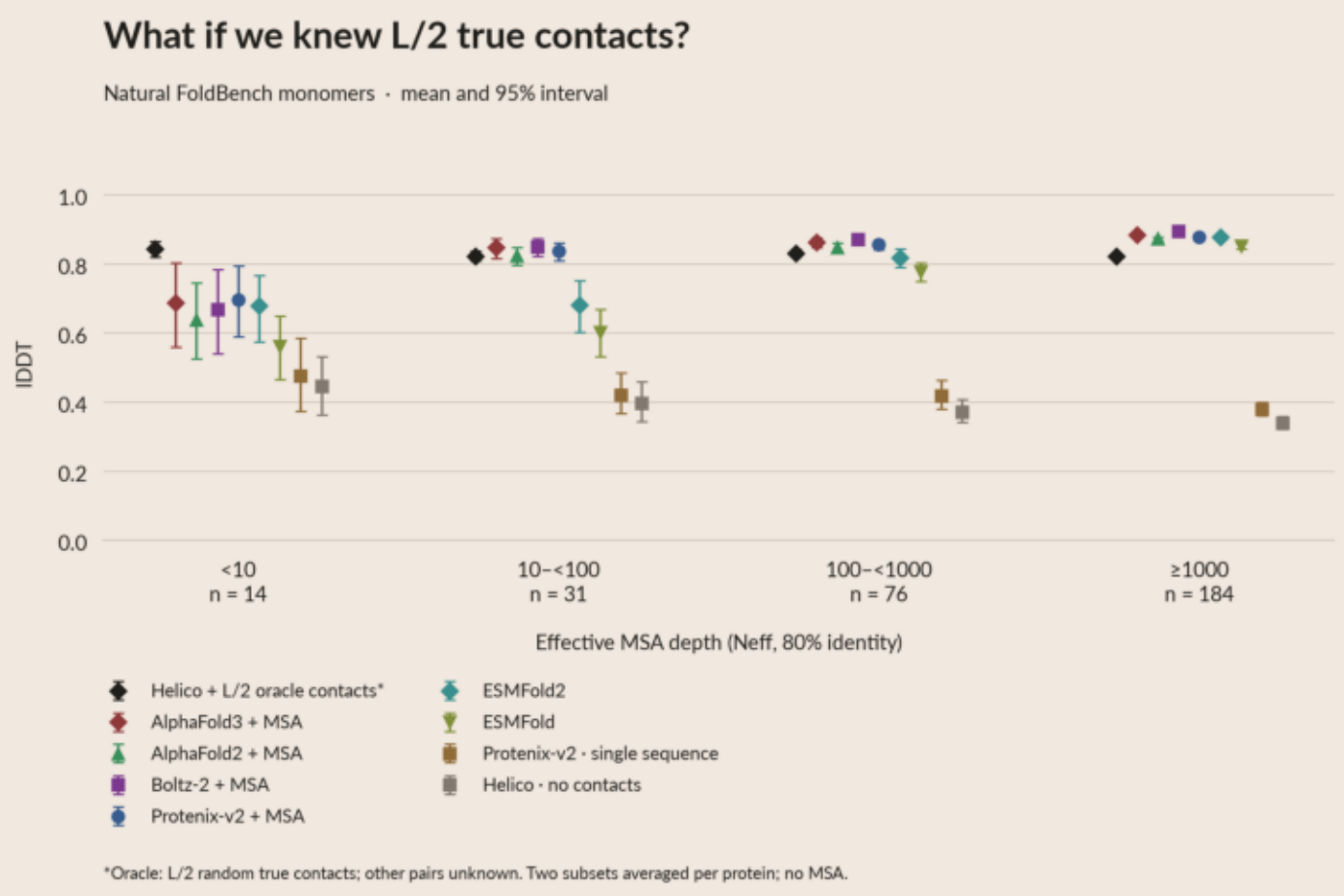
Same 305 natural proteins, all seven predictors and no-contact Helico. Oracle: two random L/2 true-contact subsets, three samples/subset, ranking_score selection then subset averaging; no MSA. See ORACLE_L2.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

02_oracle_neff_lddt.png

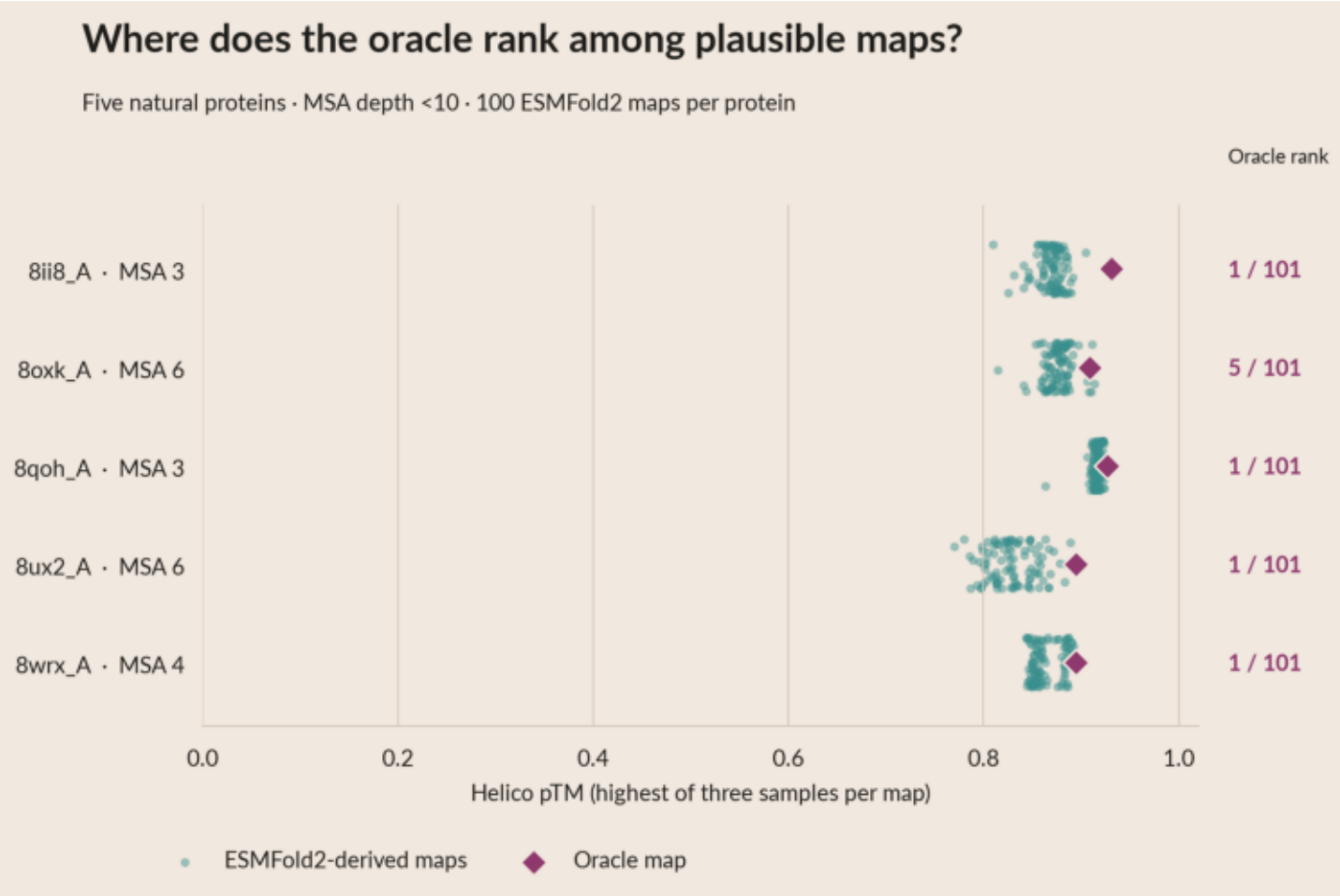
Same 305 natural proteins, all seven predictors and no-contact Helico. Oracle: two random L/2 true-contact subsets, three samples/subset, ranking_score selection then subset averaging; no MSA. See ORACLE_L2.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

02b_confidence.png

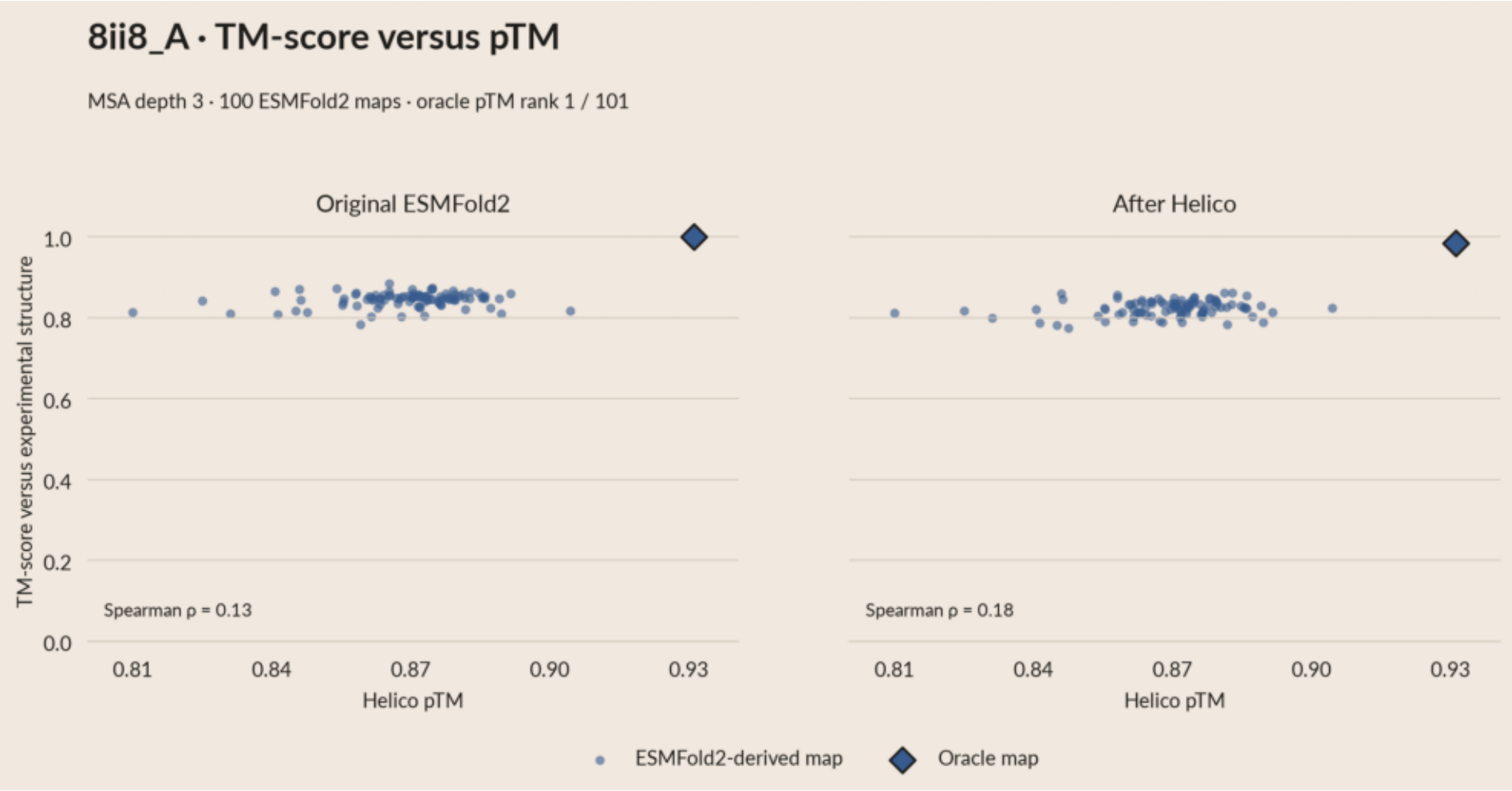
All five natural FoldBench proteins with MSA depth <10 (test split). Each dot is one of 100 single-sequence ESMFold2 predictions, converted to a full contact/non-contact map and folded by Helico; diamonds use the ground-truth oracle map. Every map has the same eligible-pair mask and three Helico diffusion samples. Highest pTM selects a sample per map; that pTM also ranks the 101 maps. No ipTM term, clash penalty or map filtering is used. Contact counts may vary. Rank 1 is highest pTM; ties are shown as rank intervals. Vertical jitter only separates points. These 500 decoys represent five proteins.



$ render.py

02c_accuracy_confidence_protein_8ii8_A.png

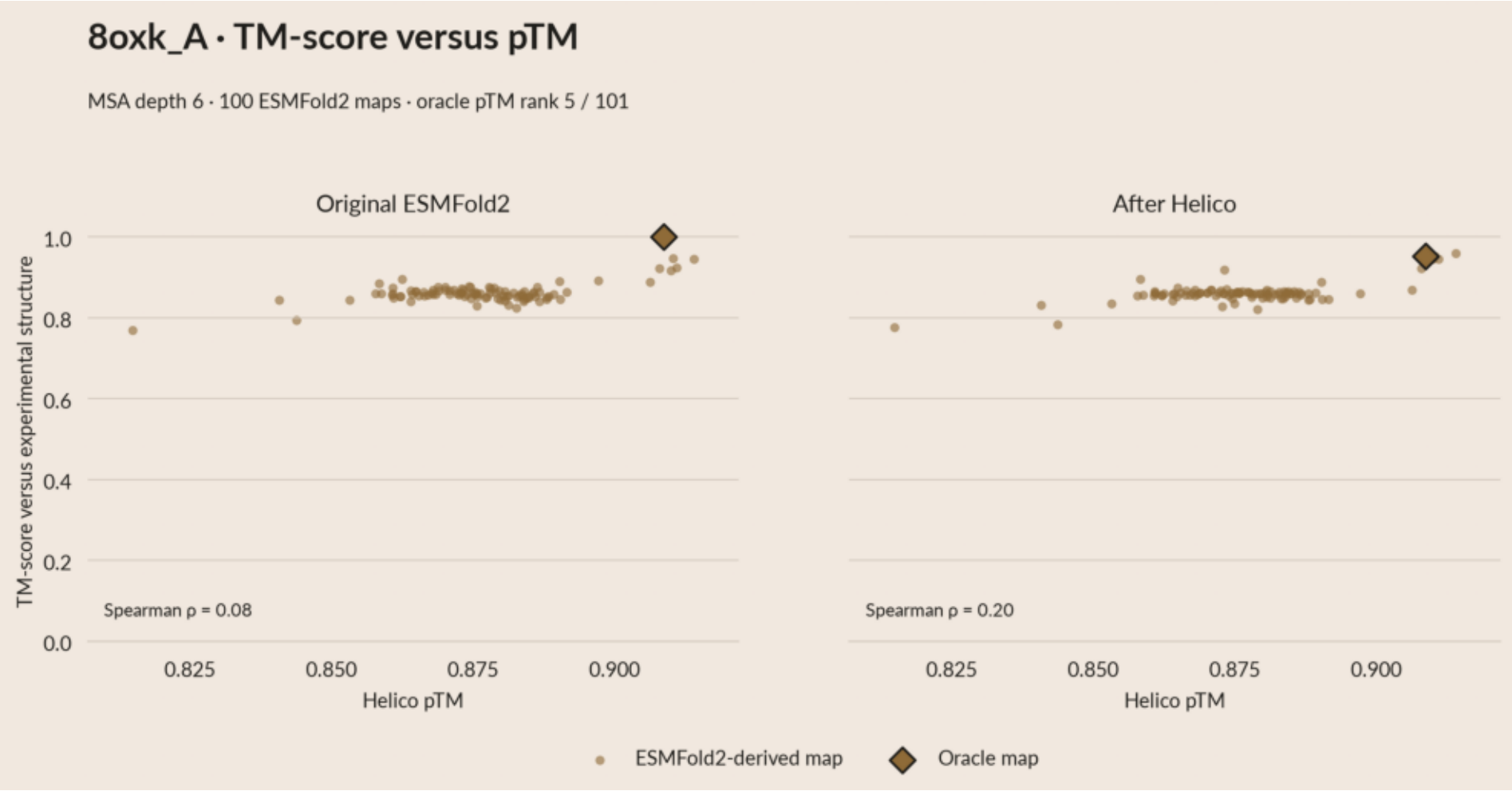
Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



$ render.py

02c_accuracy_confidence_protein_8oxk_A.png

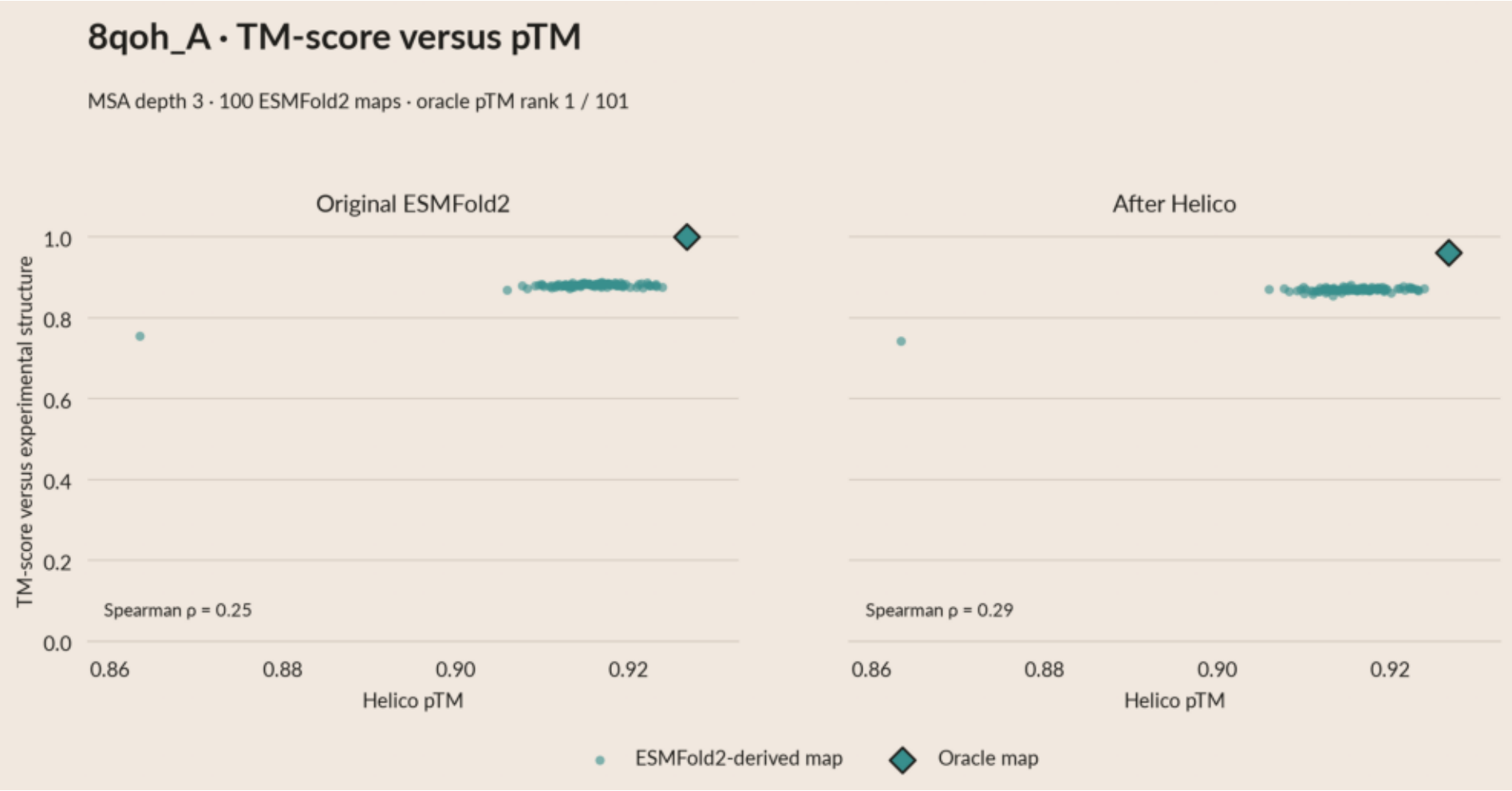
Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.




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$ render.py
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02c_accuracy_confidence_protein_8qoh_A.png

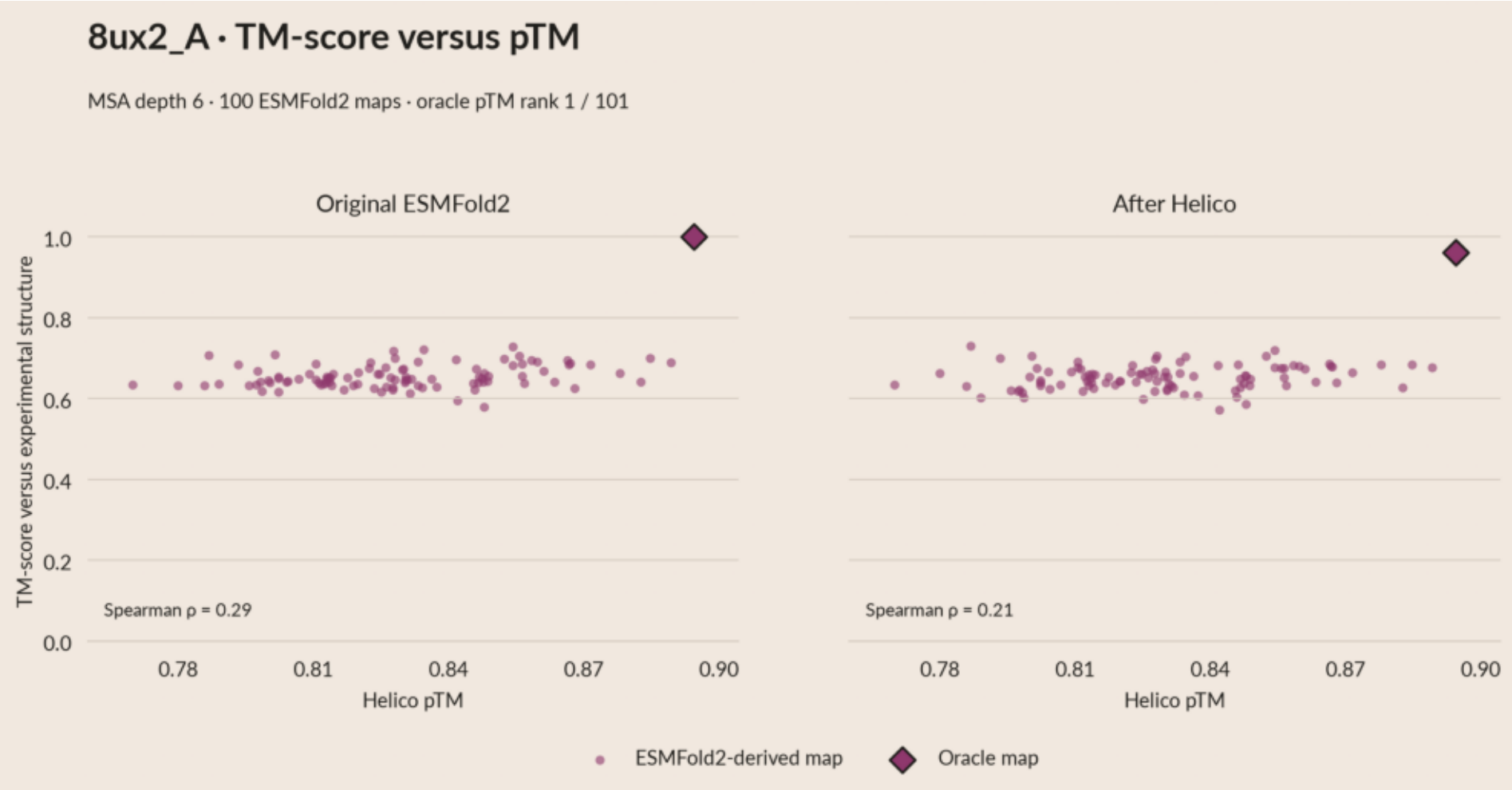
Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.



$ render.py

02c_accuracy_confidence_protein_8ux2_A.png

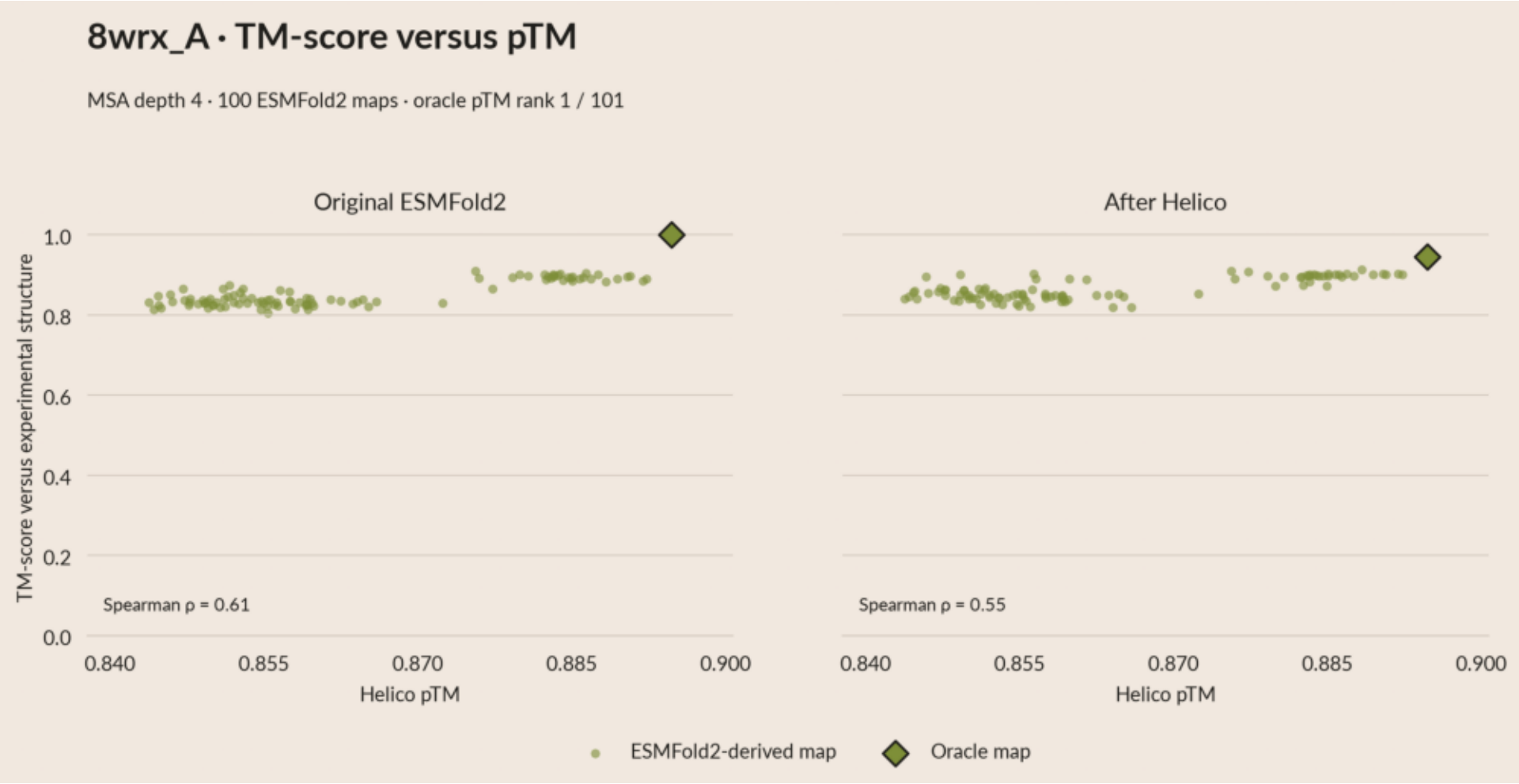
Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.




```
$ render.py
```

02c_accuracy_confidence_protein_8wrx_A.png

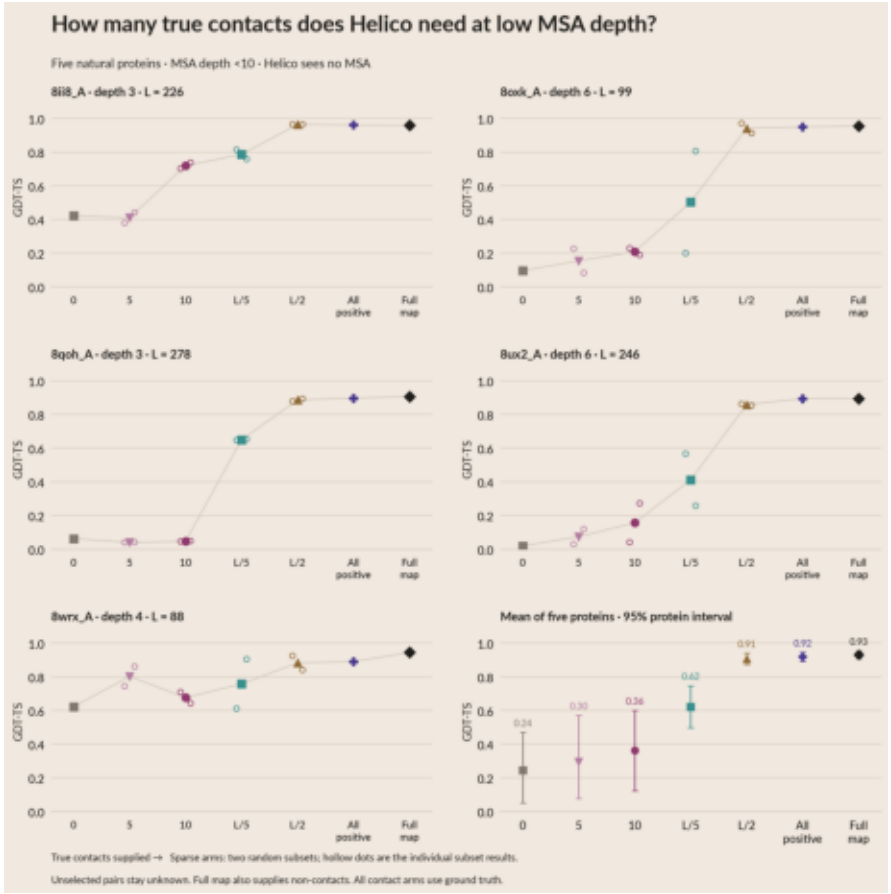
Highest pTM selects one of three Helico samples for every map; pTM also ranks the 101 maps. Left: original ESMFold2 TM-score (oracle reference = 1). Right: selected Helico reconstruction TM-score. Diamonds identify oracle contacts; correlations use only the 100 ESMFold2 maps. TM-score uses matched protein CA atoms; pTM retains all Helico input tokens. No ipTM or clash penalty.




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$ render.py
```

02e_oracle_budget.png

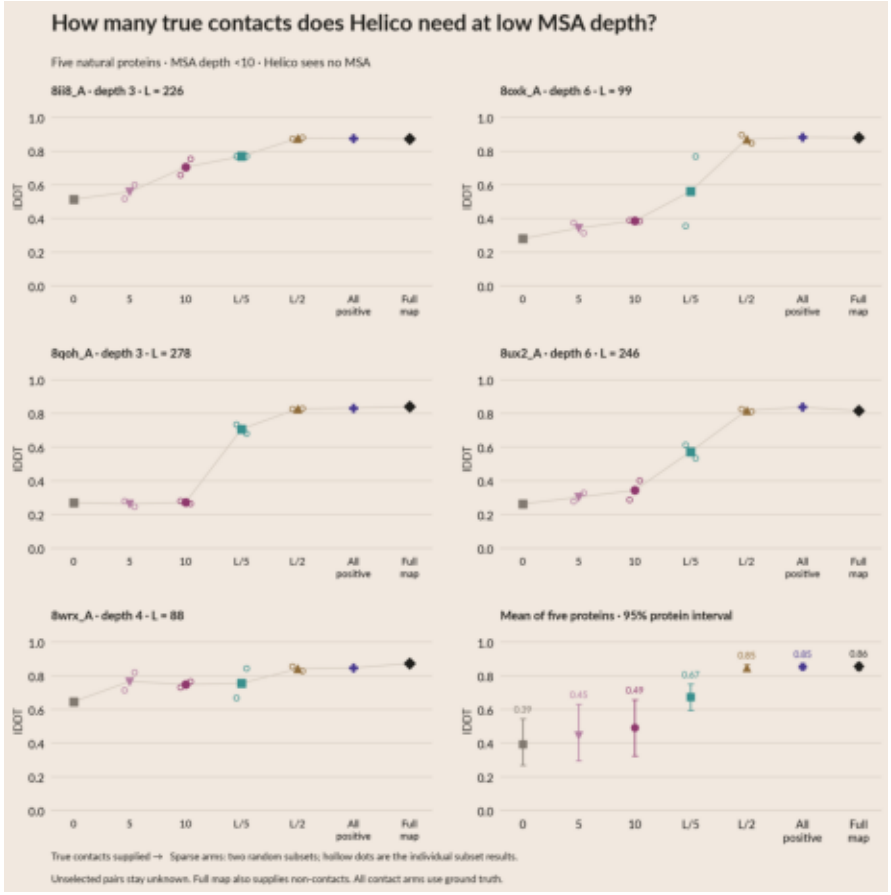
All five natural Figure 02 proteins with MSA depth <10. Random subsets of 5, 10, floor(L/5) and floor(L/2) true contacts; L is frozen input sequence length. Unselected pairs stay unknown. Two independent subsets per budget, nested across budgets within each draw. Three diffusion samples/map, six recycles, seed42, no MSA. Highest ranking_score selects one sample per subset, then average the two subset accuracies within protein. Hollow dots show both subsets; filled markers show their mean. The sixth panel averages the five proteins with 95% protein-bootstrap intervals. Fresh no-contact, all-positive and full positive/negative controls. All oracle contacts use ground truth. Menus expose each protein and pTM selection. See ORACLE_BUDGET.md and oracle_budget_per_protein.csv for counts and source rows.



$ render.py

02e_oracle_budget_lddt.png

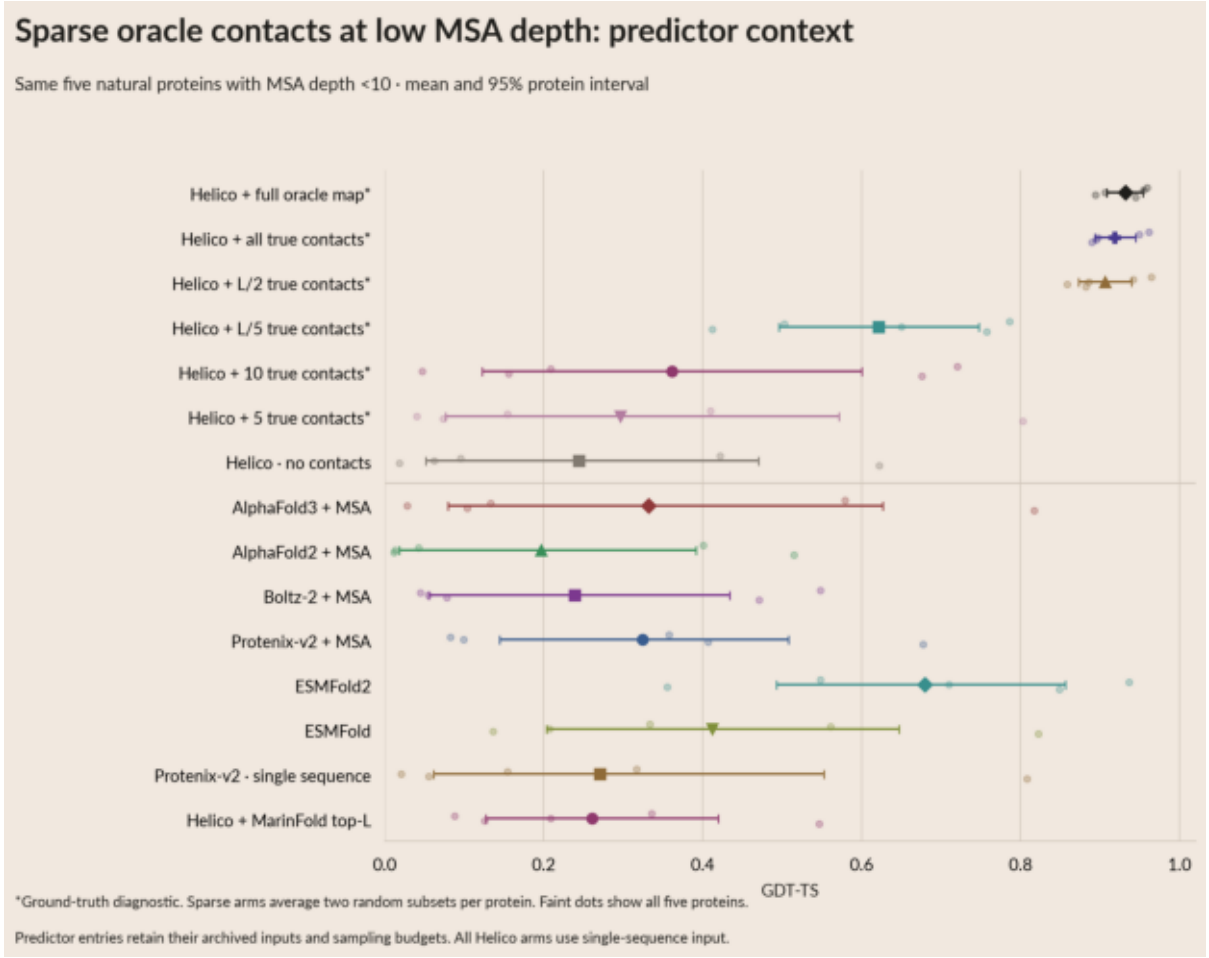
All five natural Figure 02 proteins with MSA depth <10. Random subsets of 5, 10, floor(L/5) and floor(L/2) true contacts; L is frozen input sequence length. Unselected pairs stay unknown. Two independent subsets per budget, nested across budgets within each draw. Three diffusion samples/map, six recycles, seed42, no MSA. Highest ranking_score selects one sample per subset, then average the two subset accuracies within protein. Hollow dots show both subsets; filled markers show their mean. The sixth panel averages the five proteins with 95% protein-bootstrap intervals. Fresh no-contact, all-positive and full positive/negative controls. All oracle contacts use ground truth. Menus expose each protein and pTM selection. See ORACLE_BUDGET.md and oracle_budget_per_protein.csv for counts and source rows.



$ render.py

02f_oracle_budget_context.png

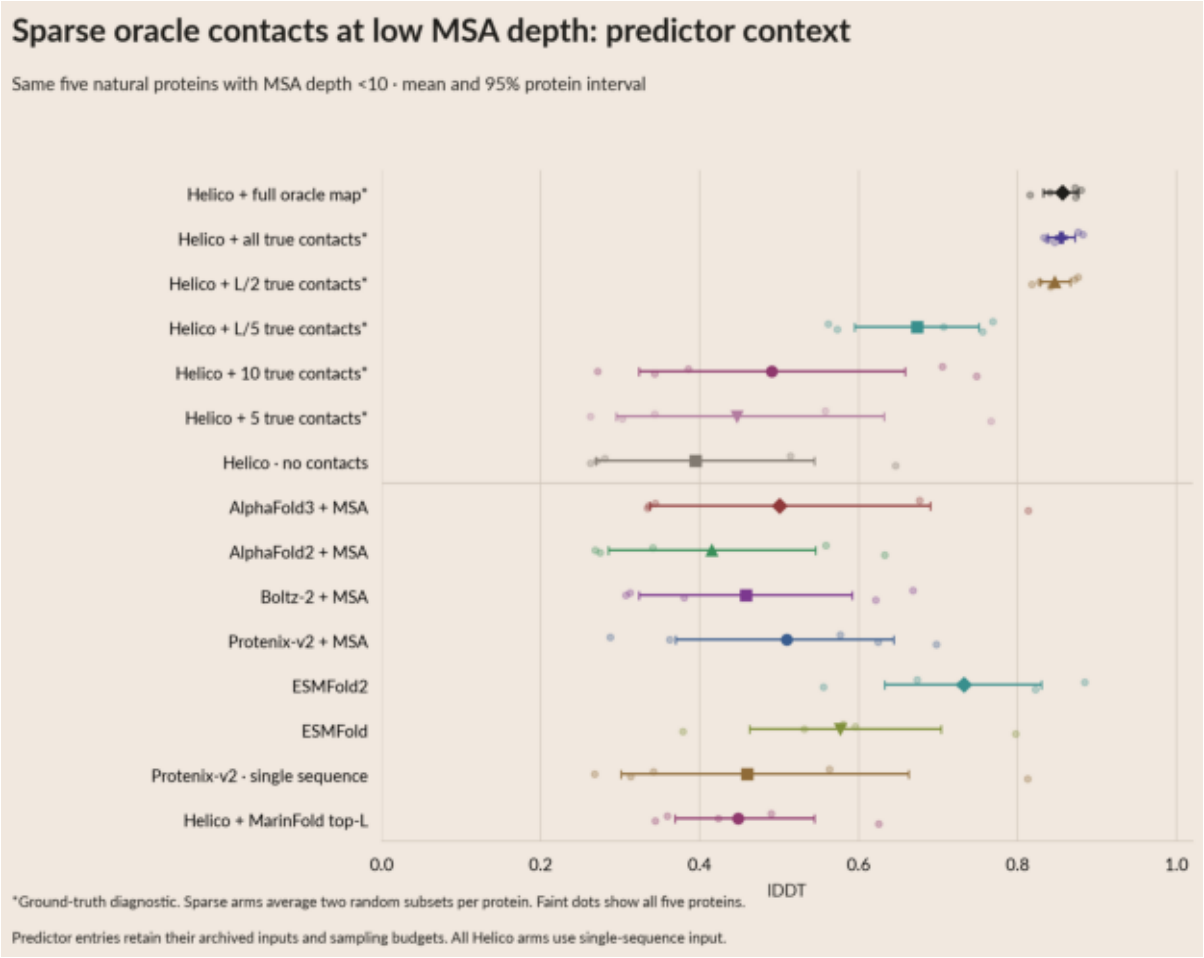
The same five low-depth natural proteins in every arm: sparse oracle conditioning versus archived AF3, AF2, Boltz-2, Protenix-v2 with/without MSA, ESMFold/ESMFold2 and Helico with 248B-token MarinFold contacts. Predictor inputs and sampling budgets retain their original protocols. Faint dots show individual proteins; means and 95% intervals bootstrap the five proteins. Oracle subsets use ground truth and are averaged within protein. The selector menu changes only the new Helico oracle-budget sweep; archived predictors and MarinFold folding retain their original selection. See ORACLE_BUDGET.md.




```
$ render.py
```

02f_oracle_budget_context_lddt.png

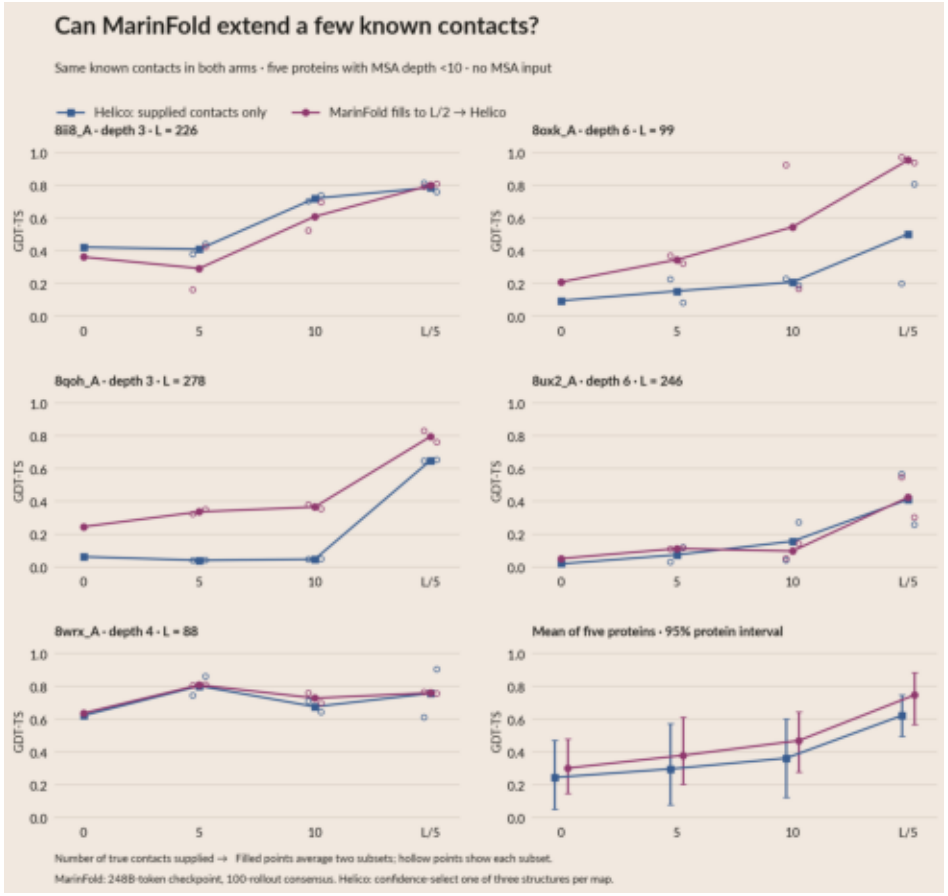
The same five low-depth natural proteins in every arm: sparse oracle conditioning versus archived AF3, AF2, Boltz-2, Protenix-v2 with/without MSA, ESMFold/ESMFold2 and Helico with 248B-token MarinFold contacts. Predictor inputs and sampling budgets retain their original protocols. Faint dots show individual proteins; means and 95% intervals bootstrap the five proteins. Oracle subsets use ground truth and are averaged within protein. The selector menu changes only the new Helico oracle-budget sweep; archived predictors and MarinFold folding retain their original selection. See ORACLE_BUDGET.md.



$ render.py

02g_seed_completion.png

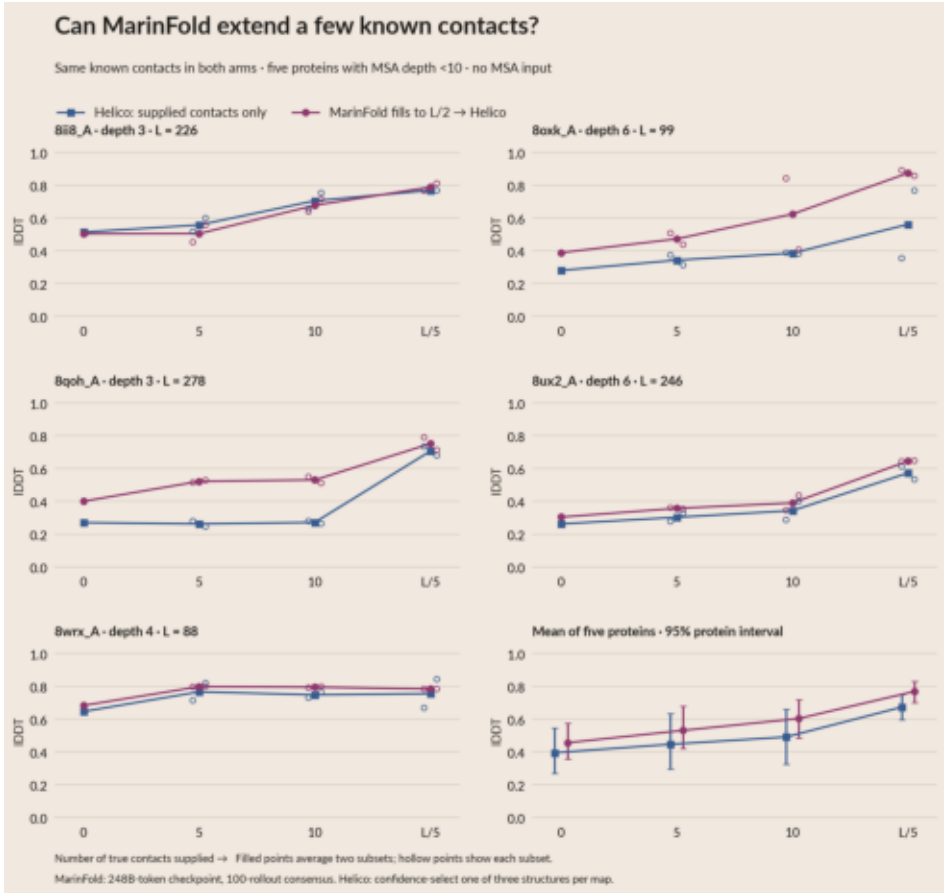
Same five natural eval-test proteins with MSA depth <10. Both arms receive identical random subsets of 0, 5, 10 or floor(L/5) ground-truth contacts. Direct: Helico sees only those pairs. Completion: the fixed 248B-token MarinFold model receives those pairs in its prompt, then 100 resampled rollouts vote on additions; keep every seed and fill to floor(L/2) total contacts. Unselected pairs stay unknown. Helico uses no MSA, six recycles and three diffusion samples/map. Primary selection uses ranking_score, matching the sparse-oracle comparison; menus also expose pTM. Average two subsets within protein (one map at zero seeds). Hollow points show subsets; cohort intervals bootstrap five proteins. Menus expose all proteins, GDT-TS, TM-score and lDDT. This is an exploratory oracle-seeded diagnostic, not sequence-only prediction. See SEED_COMPLETION.md.



$ render.py

02g_seed_completion_lddt.png

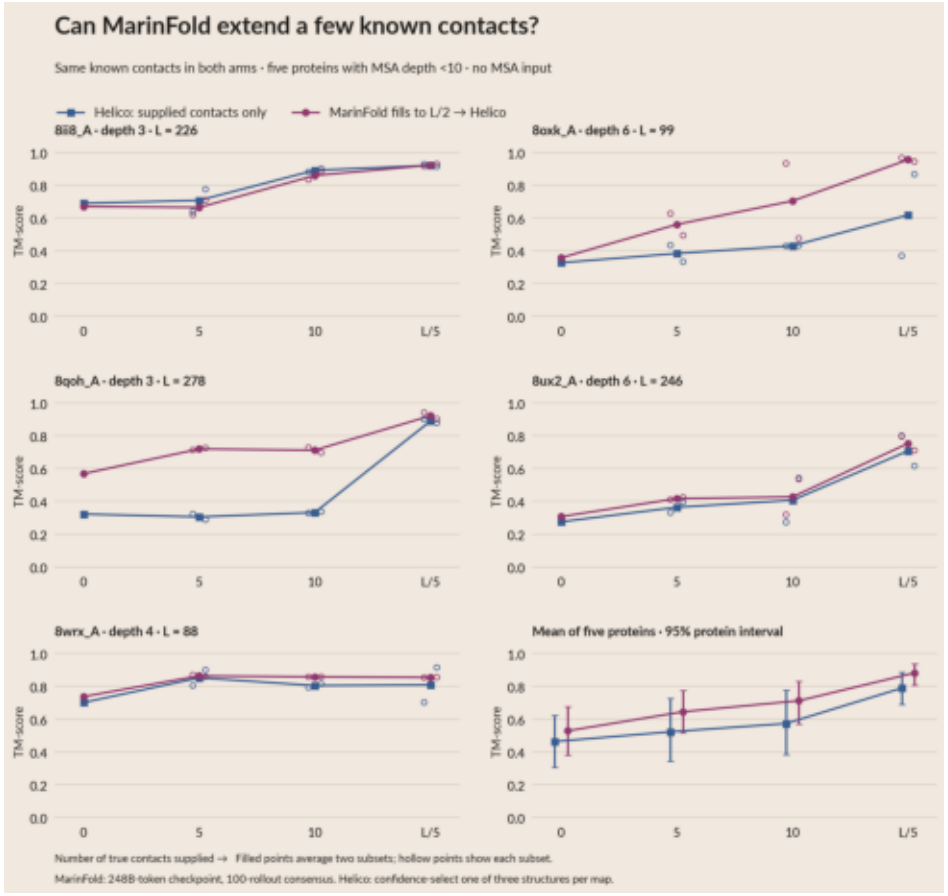
Same five natural eval-test proteins with MSA depth <10. Both arms receive identical random subsets of 0, 5, 10 or floor(L/5) ground-truth contacts. Direct: Helico sees only those pairs. Completion: the fixed 248B-token MarinFold model receives those pairs in its prompt, then 100 resampled rollouts vote on additions; keep every seed and fill to floor(L/2) total contacts. Unselected pairs stay unknown. Helico uses no MSA, six recycles and three diffusion samples/map. Primary selection uses ranking_score, matching the sparse-oracle comparison; menus also expose pTM. Average two subsets within protein (one map at zero seeds). Hollow points show subsets; cohort intervals bootstrap five proteins. Menus expose all proteins, GDT-TS, TM-score and lDDT. This is an exploratory oracle-seeded diagnostic, not sequence-only prediction. See SEED_COMPLETION.md.



$ render.py

02g_seed_completion_tm_score.png

Same five natural eval-test proteins with MSA depth <10. Both arms receive identical random subsets of 0, 5, 10 or floor(L/5) ground-truth contacts. Direct: Helico sees only those pairs. Completion: the fixed 248B-token MarinFold model receives those pairs in its prompt, then 100 resampled rollouts vote on additions; keep every seed and fill to floor(L/2) total contacts. Unselected pairs stay unknown. Helico uses no MSA, six recycles and three diffusion samples/map. Primary selection uses ranking_score, matching the sparse-oracle comparison; menus also expose pTM. Average two subsets within protein (one map at zero seeds). Hollow points show subsets; cohort intervals bootstrap five proteins. Menus expose all proteins, GDT-TS, TM-score and lDDT. This is an exploratory oracle-seeded diagnostic, not sequence-only prediction. See SEED_COMPLETION.md.



$ render.py

02h_seed_contact_precision.png

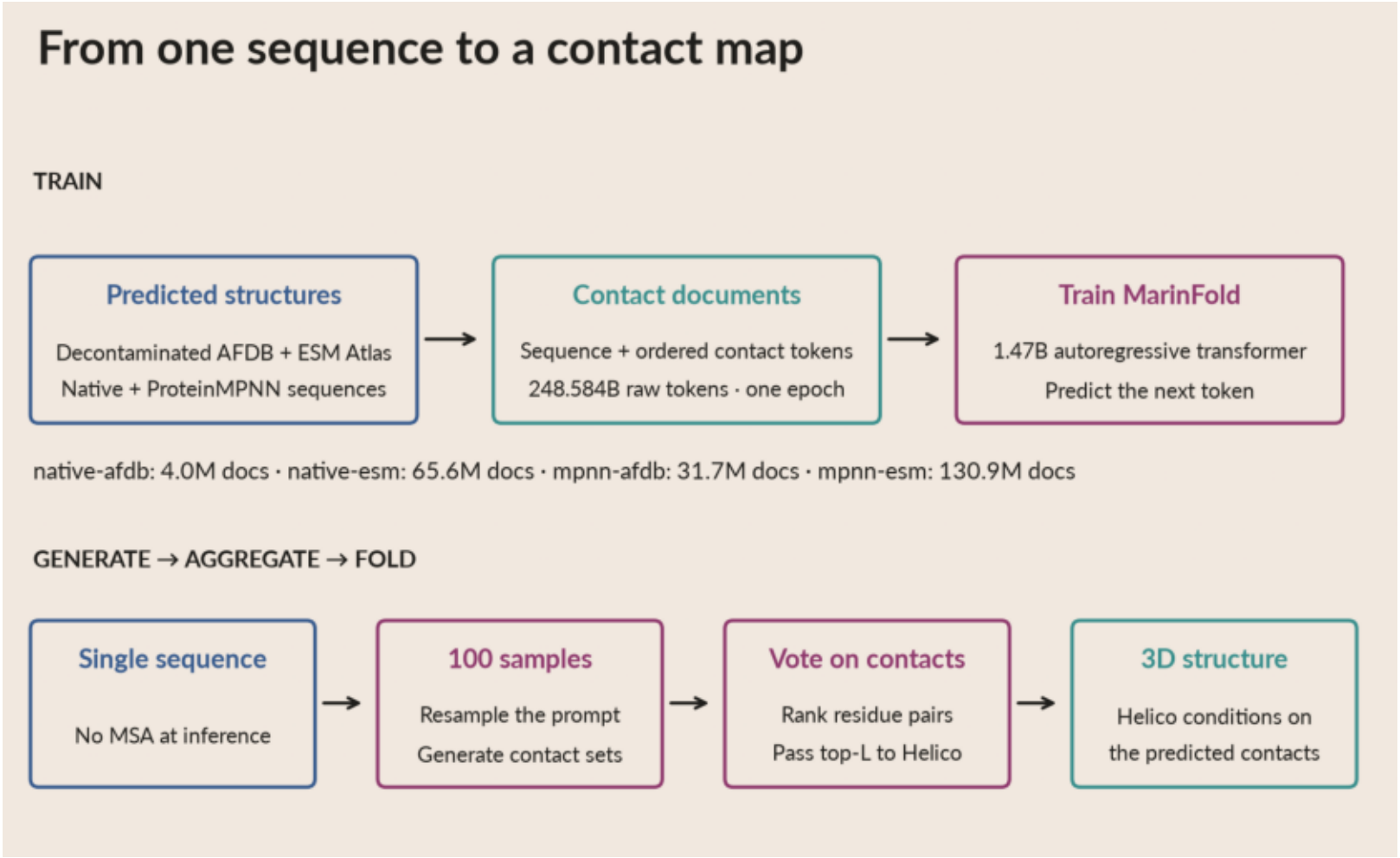
Precision of MarinFold additions alone versus the complete L/2 contact set including guaranteed-correct seeds. Pairs are selected solely by votes from normally terminated rollouts, with valid residue mappings and separation >=6; ties break by sequence indices. Ground truth is consulted for scoring only after selecting additions. Two random subsets are averaged per protein, then proteins receive equal weight. n=0 has one map. Each point traces to seed_completion_contact_precision.csv; exact selected pairs, votes and correctness labels are in seed_completion_pairs.csv.



$ render.py

03_method.png

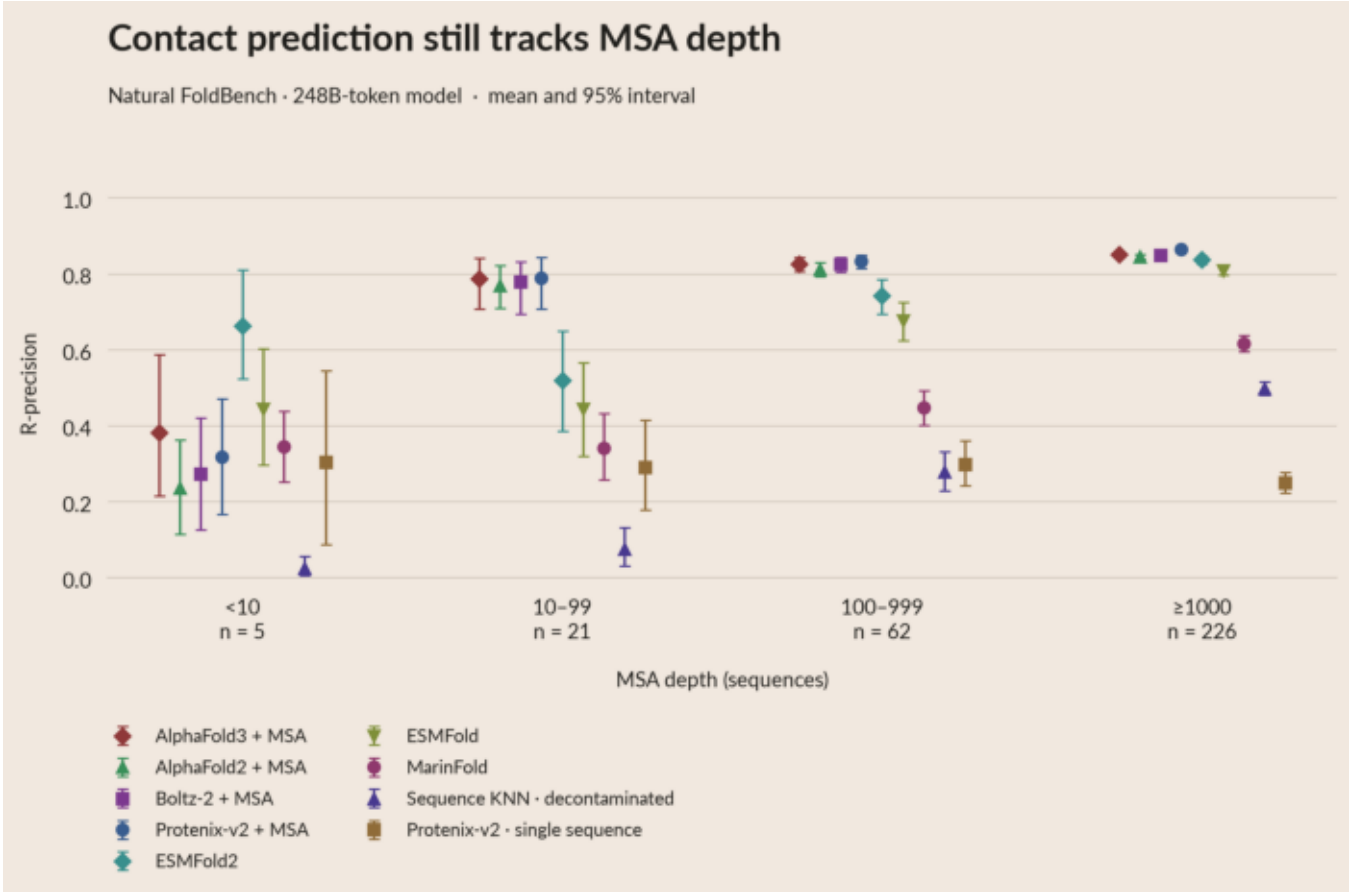
Schematic; arrow widths do not encode amounts. Exp277 step 266,344: a scratch-trained 1.47B-parameter Qwen3, one epoch over 232,090,905 contact documents, totaling 248,583,762,834 raw tokens. Native and ProteinMPNN-redesigned sequences use AFDB / ESM Atlas source structures from the decontaminated corpus. Raw tokens differ from padded training slots.



$ render.py

04_contacts.png

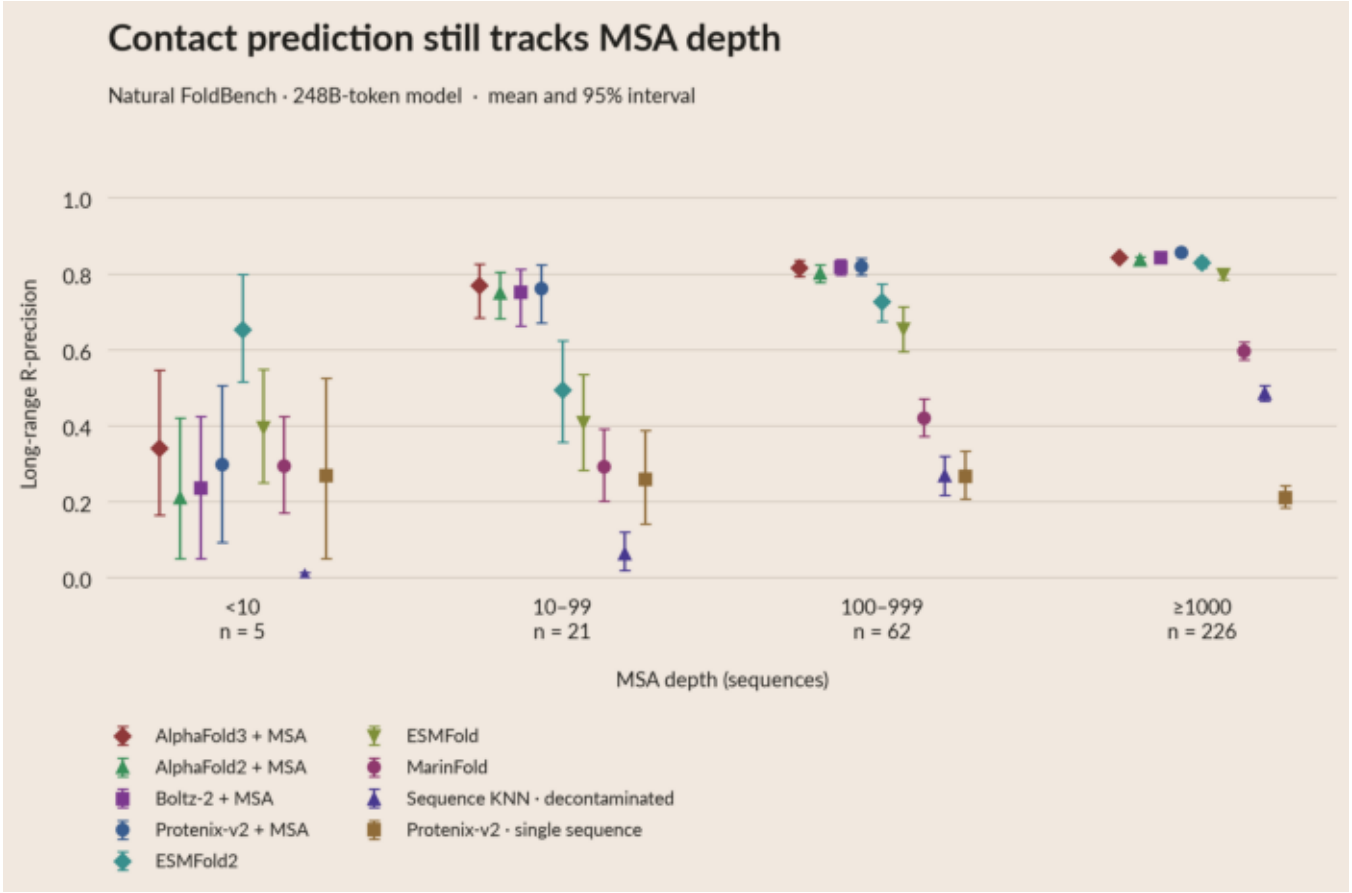
Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



$ render.py

04_contacts_long.png

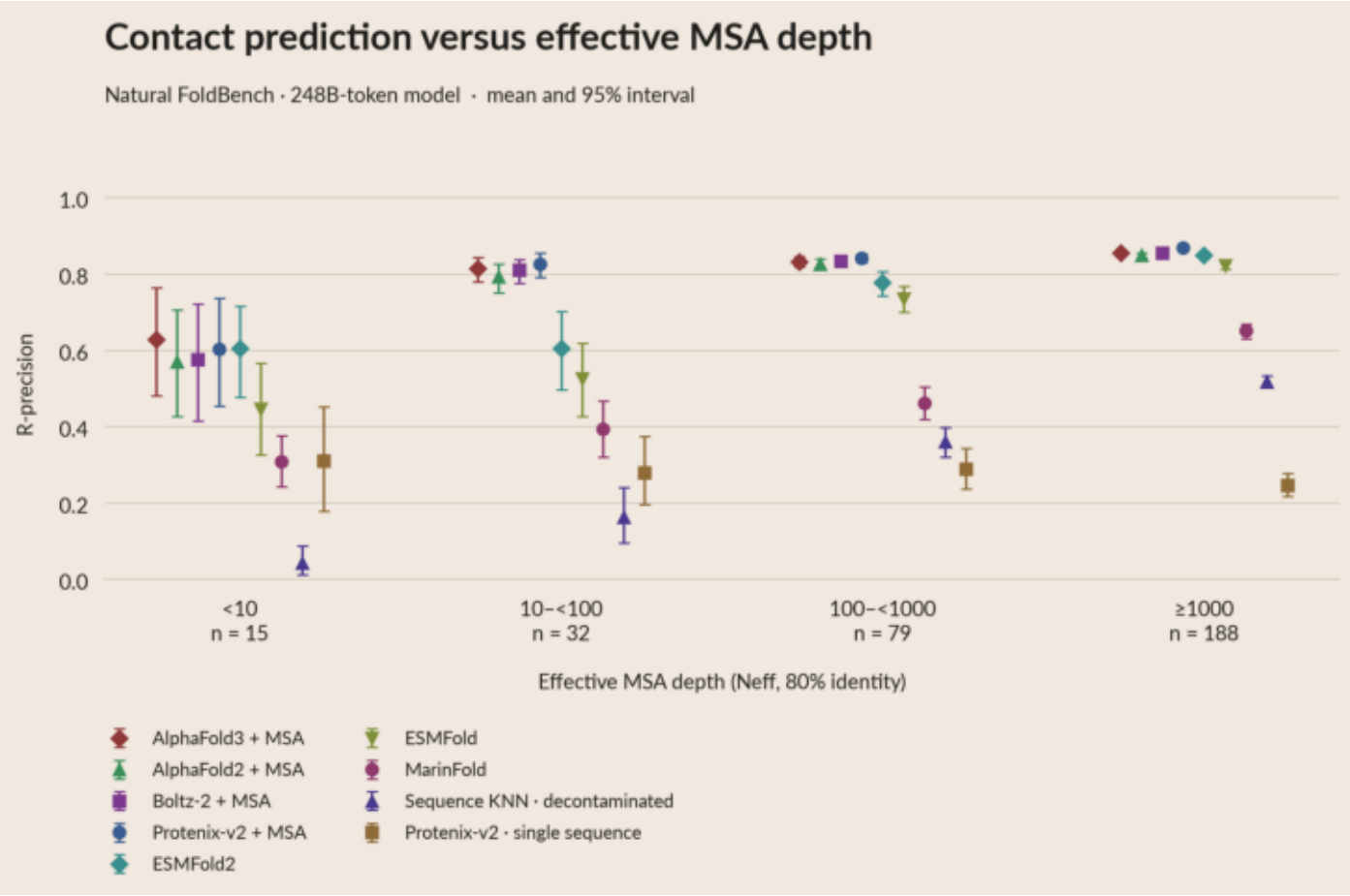
Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.




```
$ render.py
```

04_contacts_neff.png

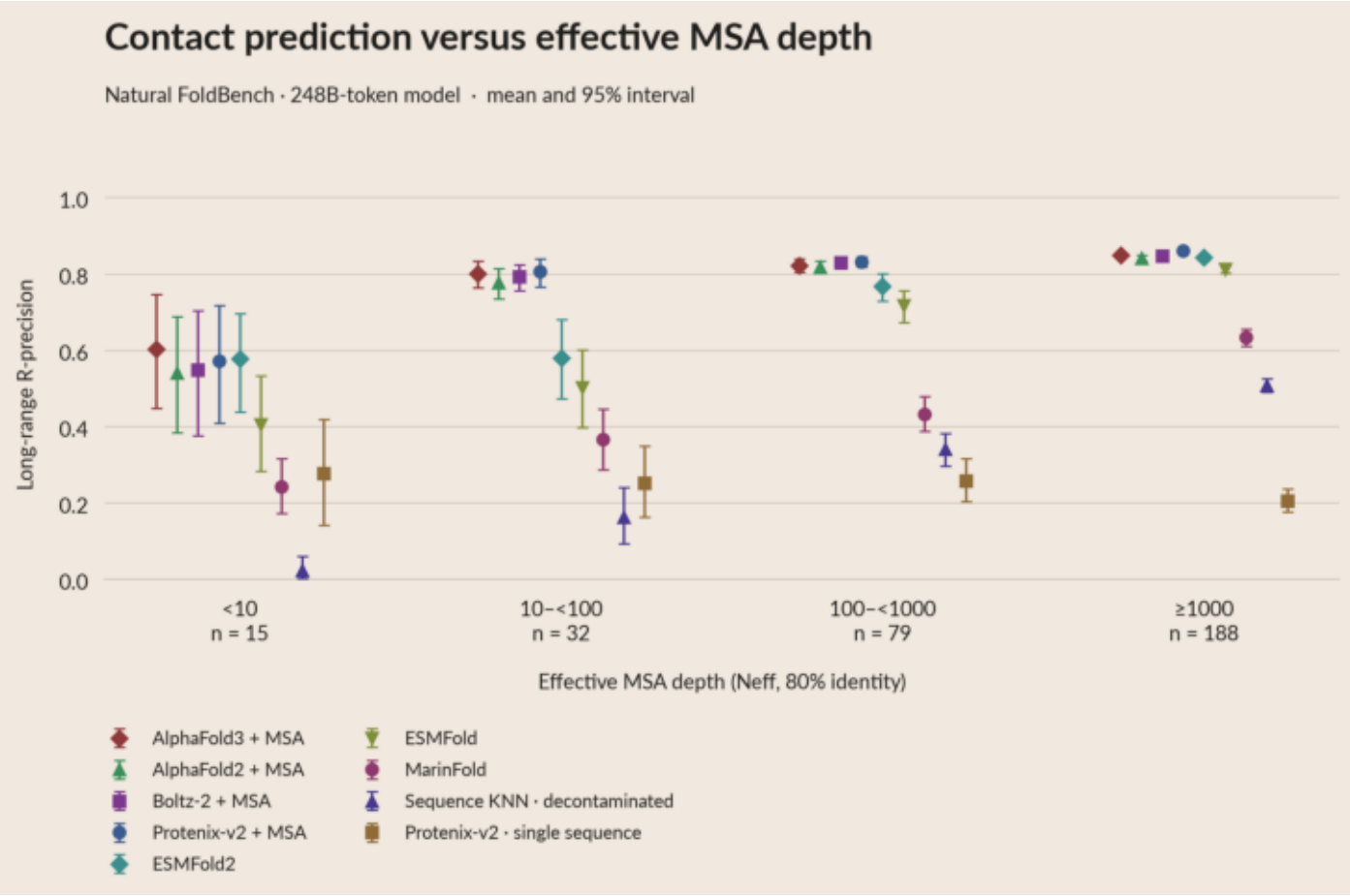
Same nine contact predictors and 314 natural proteins; fixed 248B-token model; R-precision on resolved pairs with separation >=6 or >=24 for long-range. Same contact definition and rankings as the original panel. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

04_contacts_neff_long.png

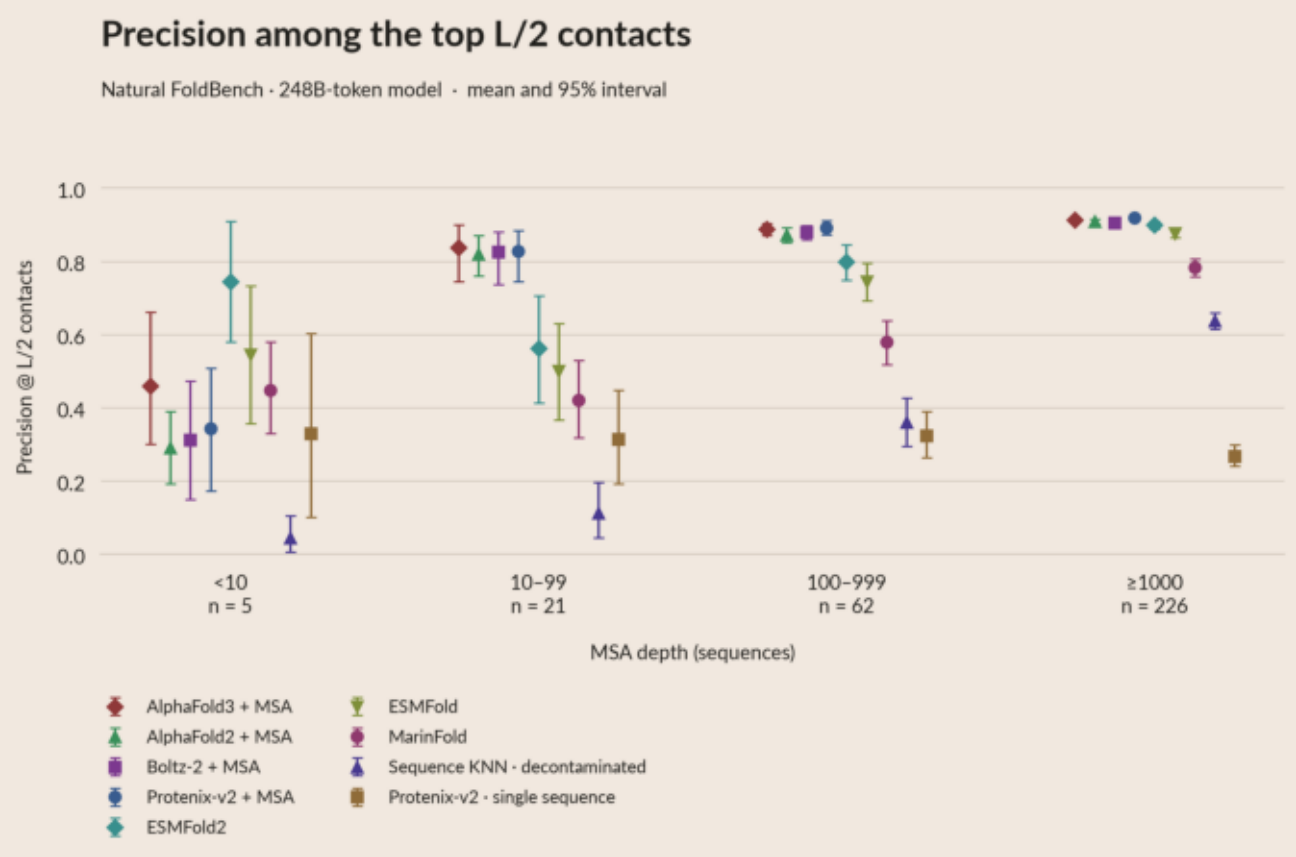
Same nine contact predictors and 314 natural proteins; fixed 248B-token model; R-precision on resolved pairs with separation >=6 or >=24 for long-range. Same contact definition and rankings as the original panel. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

04_contacts_pl2.png

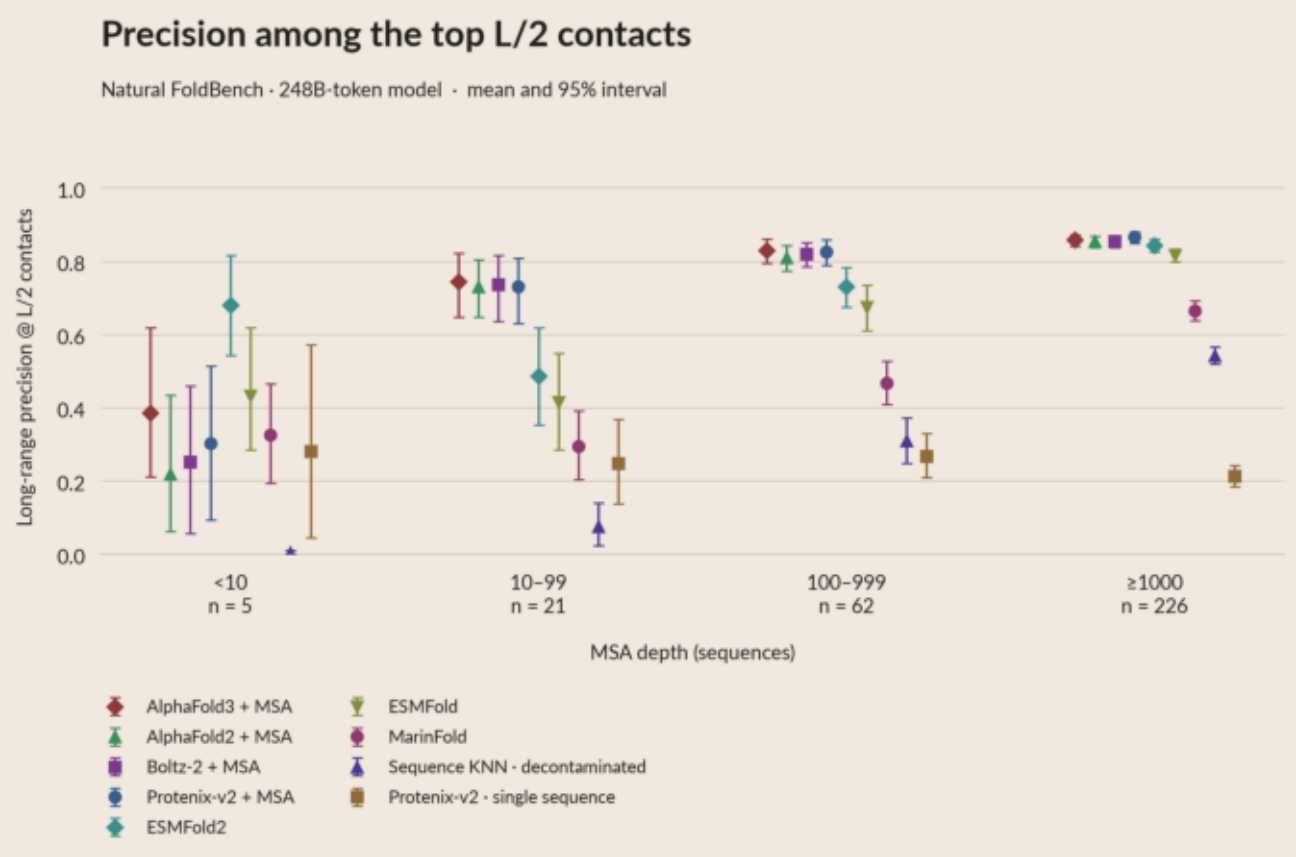
Precision among the top floor(L/2) contacts on the same 314 natural FoldBench proteins as P@L/5 (97 validation + 217 test); L is the frozen input sequence length, including unresolved input residues. Same nine predictors, archived rankings and fixed 248B-token MarinFold model. True contacts have pyconfind degree >=0.001; eligible pairs have resolved residues and sequence separation >=6 (all-range) or >=24 (long-range). Structure predictors rank contact degree, MarinFold ranks consensus votes, and sequence-KNN uses transferred scores; stable candidate-order ties. Every natural-protein score uses the full L/2 denominator. Mean and 95% protein-bootstrap intervals. The menu exposes both ranges, splits, viral status and designs separately; the designed protein 5sbj_A has only 10 long-range candidates versus 15 requested. All original R-precision scores reproduce before selecting archived L/2 scores. See PL2_ANALYSIS.md.




```
$ render.py
```

04_contacts_pl2_long.png

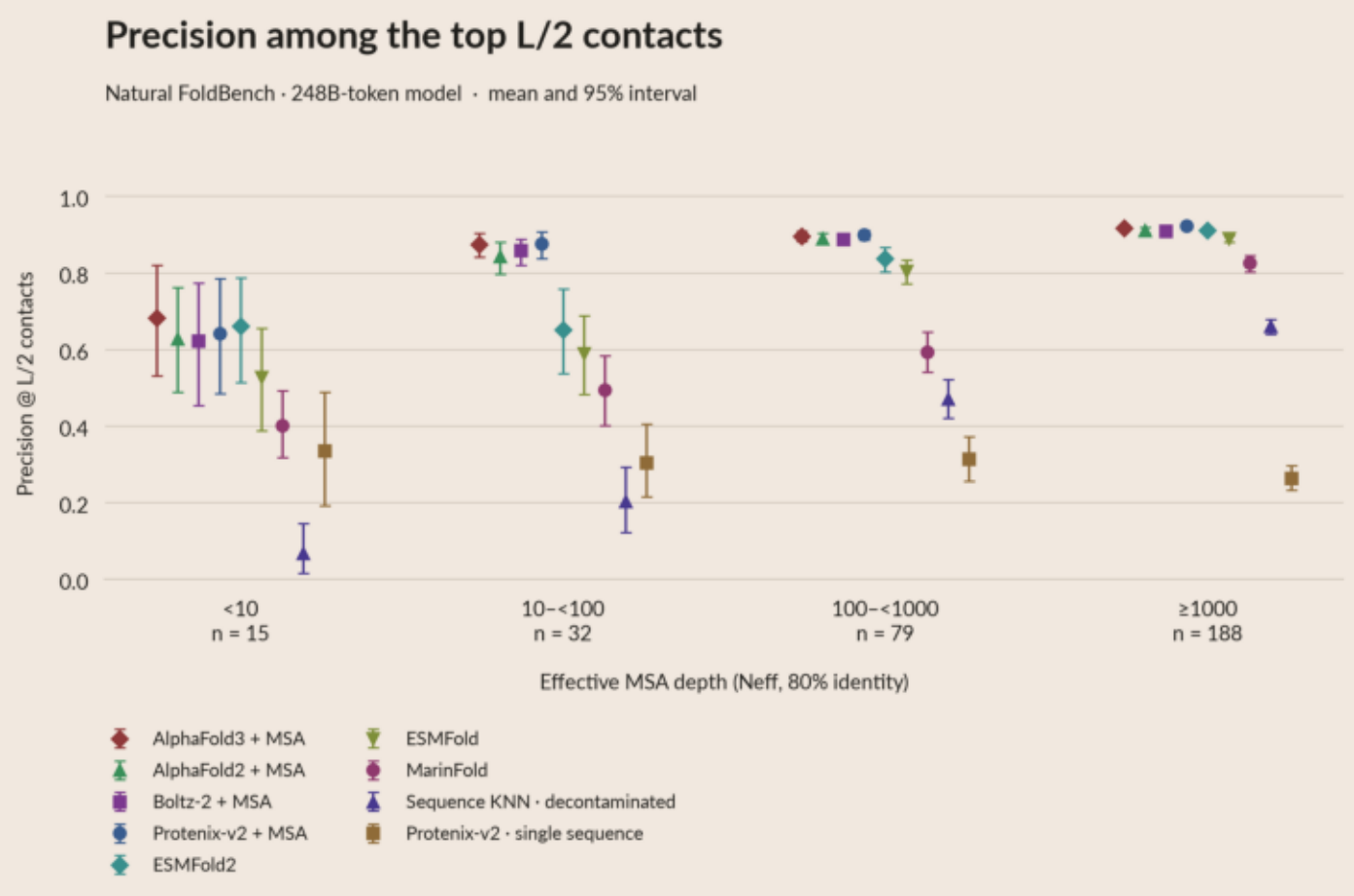
Precision among the top floor(L/2) contacts on the same 314 natural FoldBench proteins as P@L/5 (97 validation + 217 test); L is the frozen input sequence length, including unresolved input residues. Same nine predictors, archived rankings and fixed 248B-token MarinFold model. True contacts have pyconfind degree >=0.001; eligible pairs have resolved residues and sequence separation >=6 (all-range) or >=24 (long-range). Structure predictors rank contact degree, MarinFold ranks consensus votes, and sequence-KNN uses transferred scores; stable candidate-order ties. Every natural-protein score uses the full L/2 denominator. Mean and 95% protein-bootstrap intervals. The menu exposes both ranges, splits, viral status and designs separately; the designed protein 5sbj_A has only 10 long-range candidates versus 15 requested. All original R-precision scores reproduce before selecting archived L/2 scores. See PL2_ANALYSIS.md.




```
$ render.py
```

04_contacts_pl2_neff.png

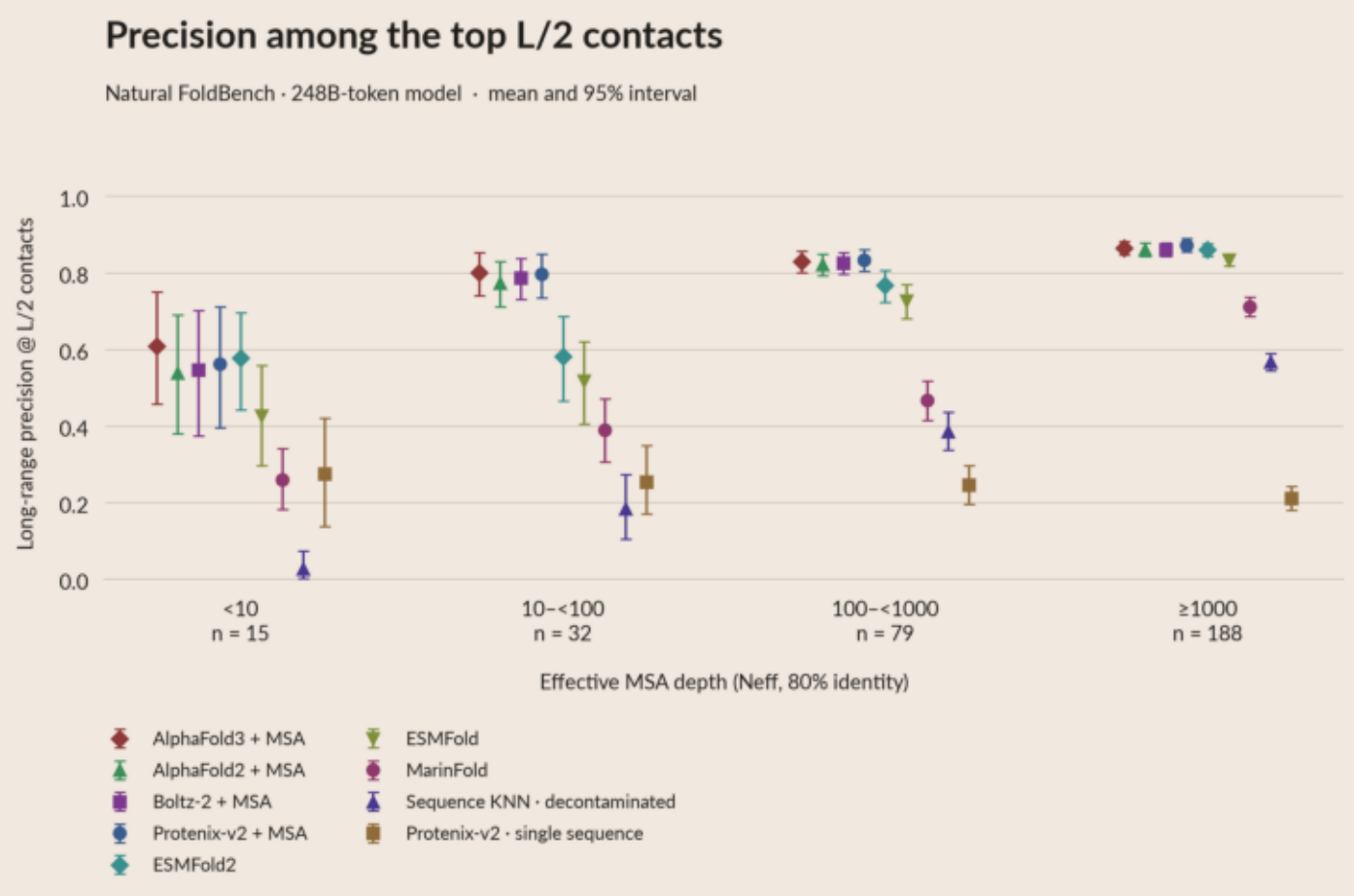
Same nine contact predictors and 314 natural proteins; Precision @ L/2 contacts, L=frozen input length. Every natural row uses the full floor(L/2) denominator. See PL2_ANALYSIS.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.




```
$ render.py
```

04_contacts_pl2_neff_long.png

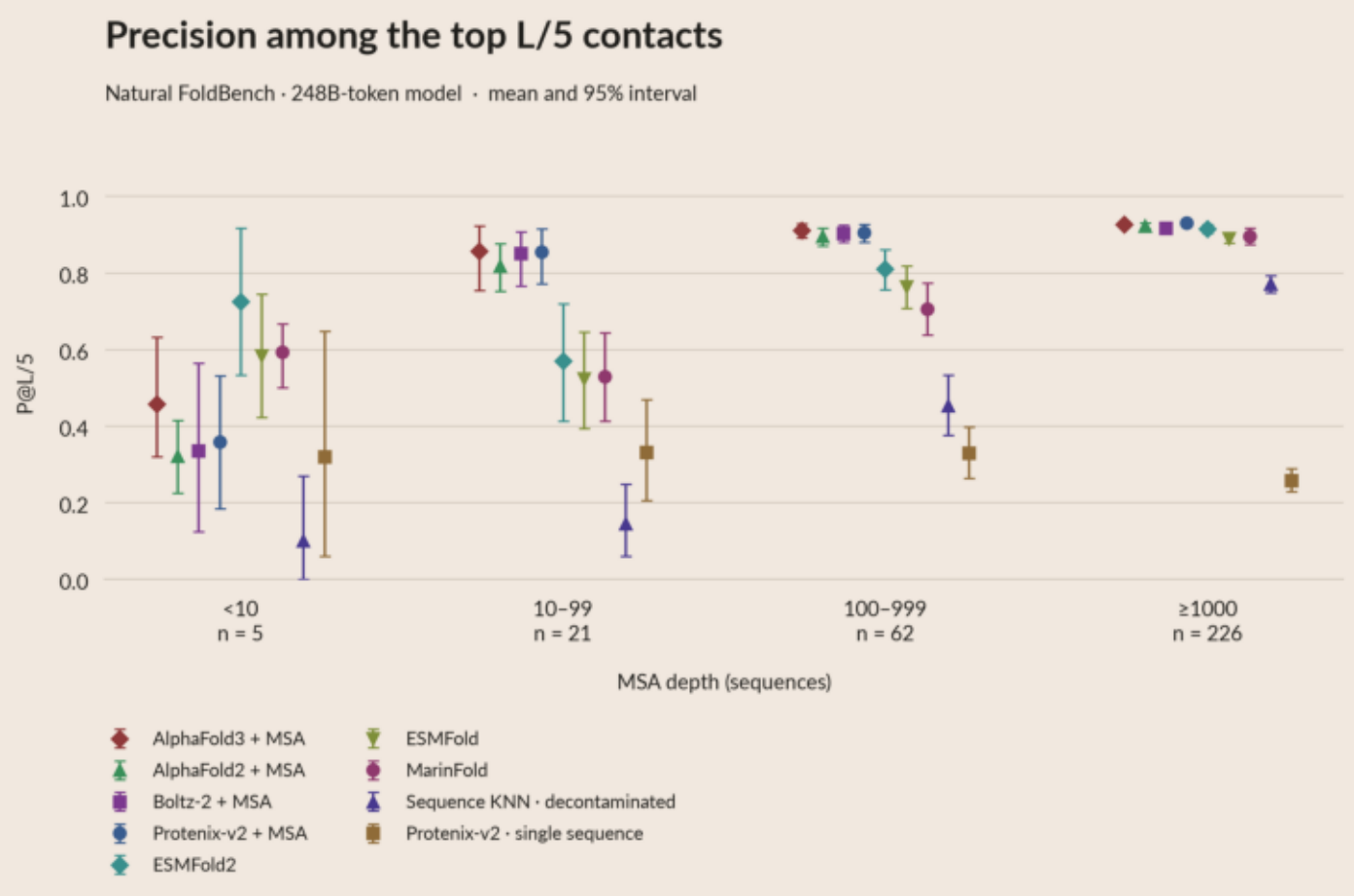
Same nine contact predictors and 314 natural proteins; Precision @ L/2 contacts, L=frozen input length. Every natural row uses the full floor(L/2) denominator. See PL2_ANALYSIS.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.




```
$ render.py
```

04_contacts_pl5.png

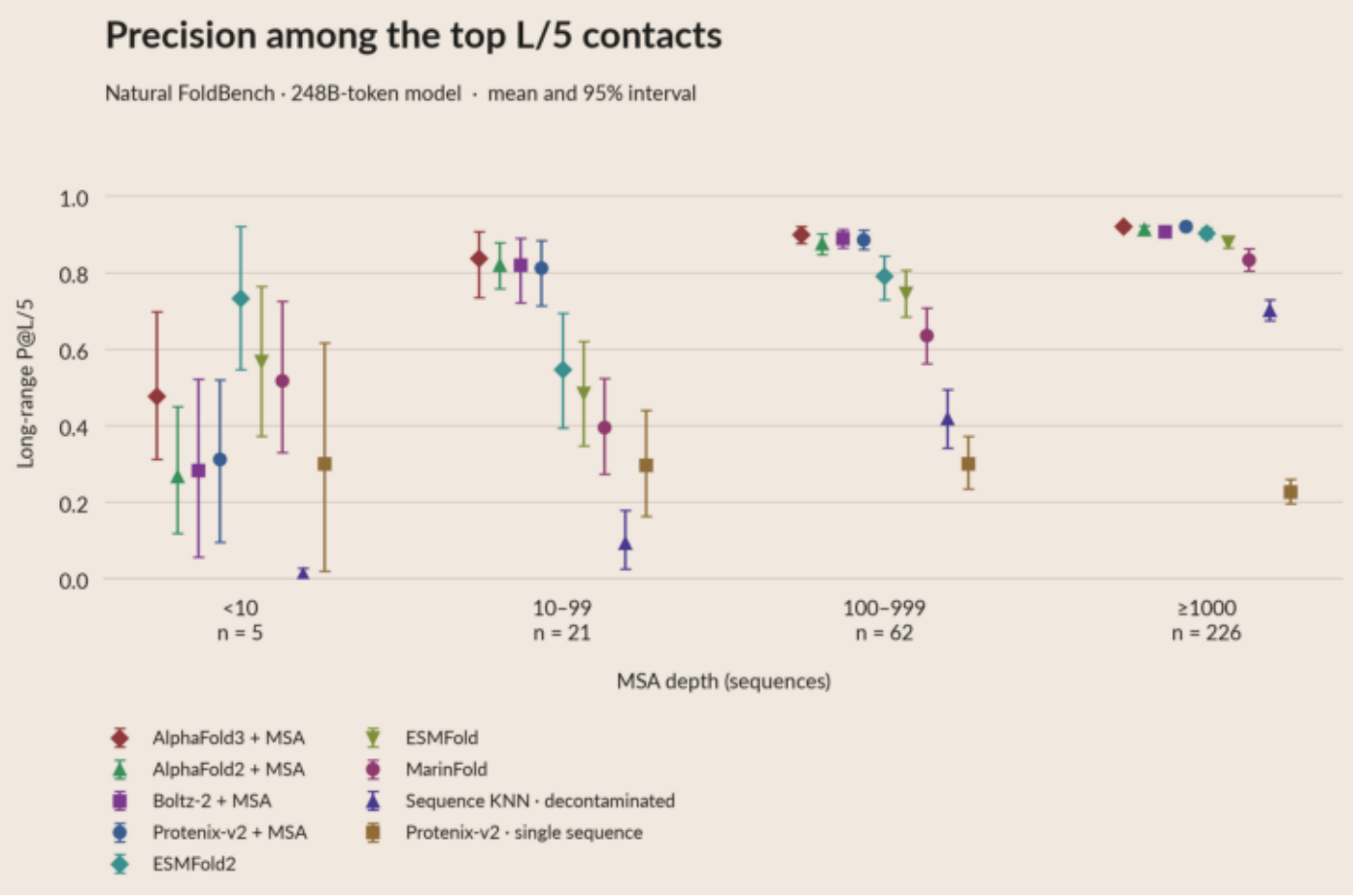
P@L/5 on 314 natural FoldBench proteins, using k=max(1,floor(L/5)) and L equal to the frozen input sequence length, not the number of resolved residues. Same nine predictors and same archived predictions as the R-precision panel, including AF2, AF3, Boltz-2 and decontaminated sequence-KNN. Ground truth and eligible pairs are unchanged: pyconfind degree >=0.001, resolved residues, separation >=6 for all-range or >=24 for long-range. Structure contacts rank by degree, MarinFold by votes; stable candidate-order ties. Means and 95% protein-bootstrap intervals. The menu exposes both ranges and split/design views. Preprocessing reproduces every archived R-precision score before using its P@L/5 counterpart.



$ render.py

04_contacts_pl5_long.png

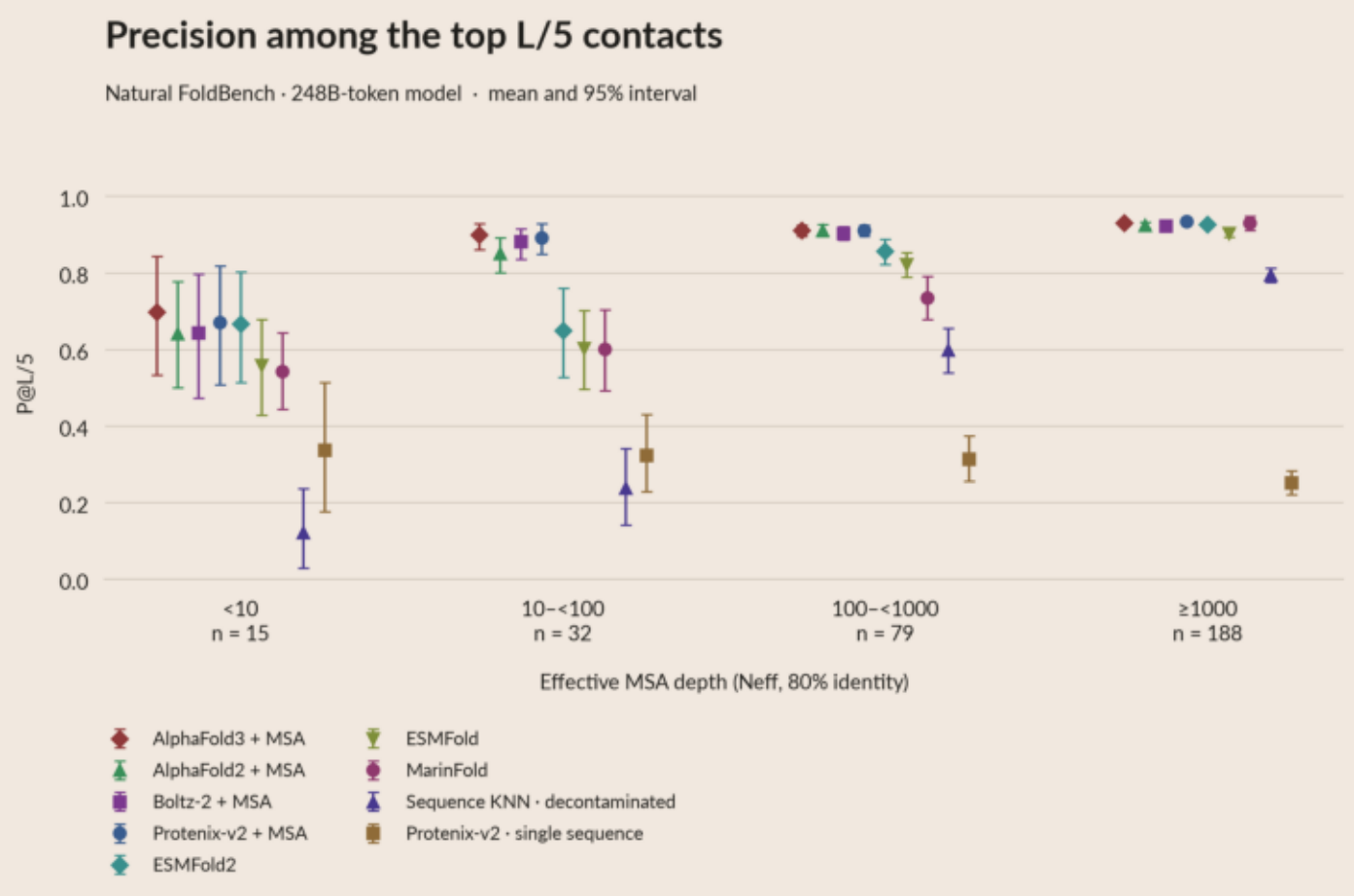
P@L/5 on 314 natural FoldBench proteins, using k=max(1,floor(L/5)) and L equal to the frozen input sequence length, not the number of resolved residues. Same nine predictors and same archived predictions as the R-precision panel, including AF2, AF3, Boltz-2 and decontaminated sequence-KNN. Ground truth and eligible pairs are unchanged: pyconfind degree >=0.001, resolved residues, separation >=6 for all-range or >=24 for long-range. Structure contacts rank by degree, MarinFold by votes; stable candidate-order ties. Means and 95% protein-bootstrap intervals. The menu exposes both ranges and split/design views. Preprocessing reproduces every archived R-precision score before using its P@L/5 counterpart.



$ render.py

04_contacts_pl5_neff.png

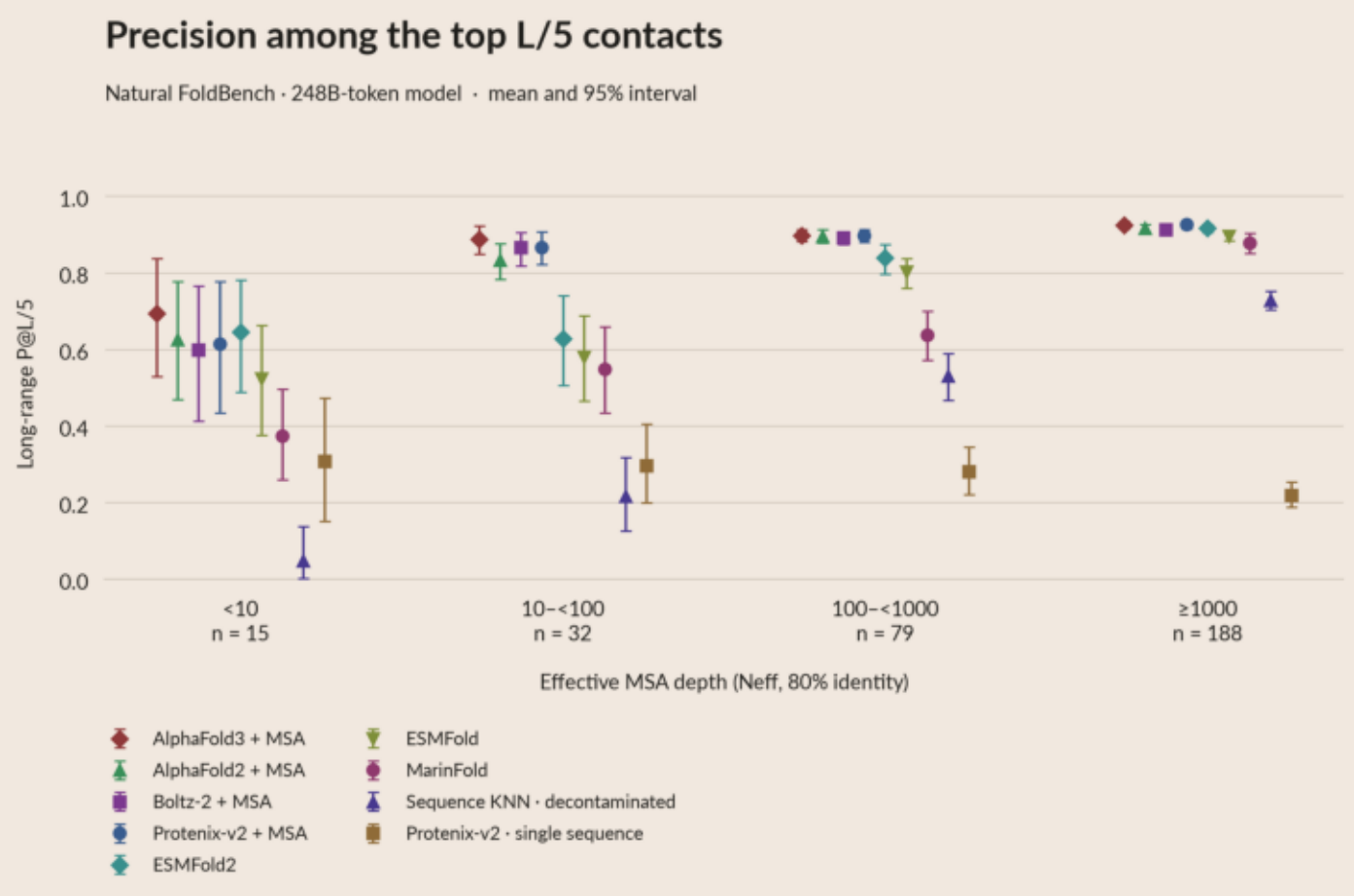
Same nine contact predictors and 314 natural proteins; precision among the top floor(L/5) contacts, L=frozen input length. See PL5_ANALYSIS.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.




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$ render.py
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04_contacts_pl5_neff_long.png

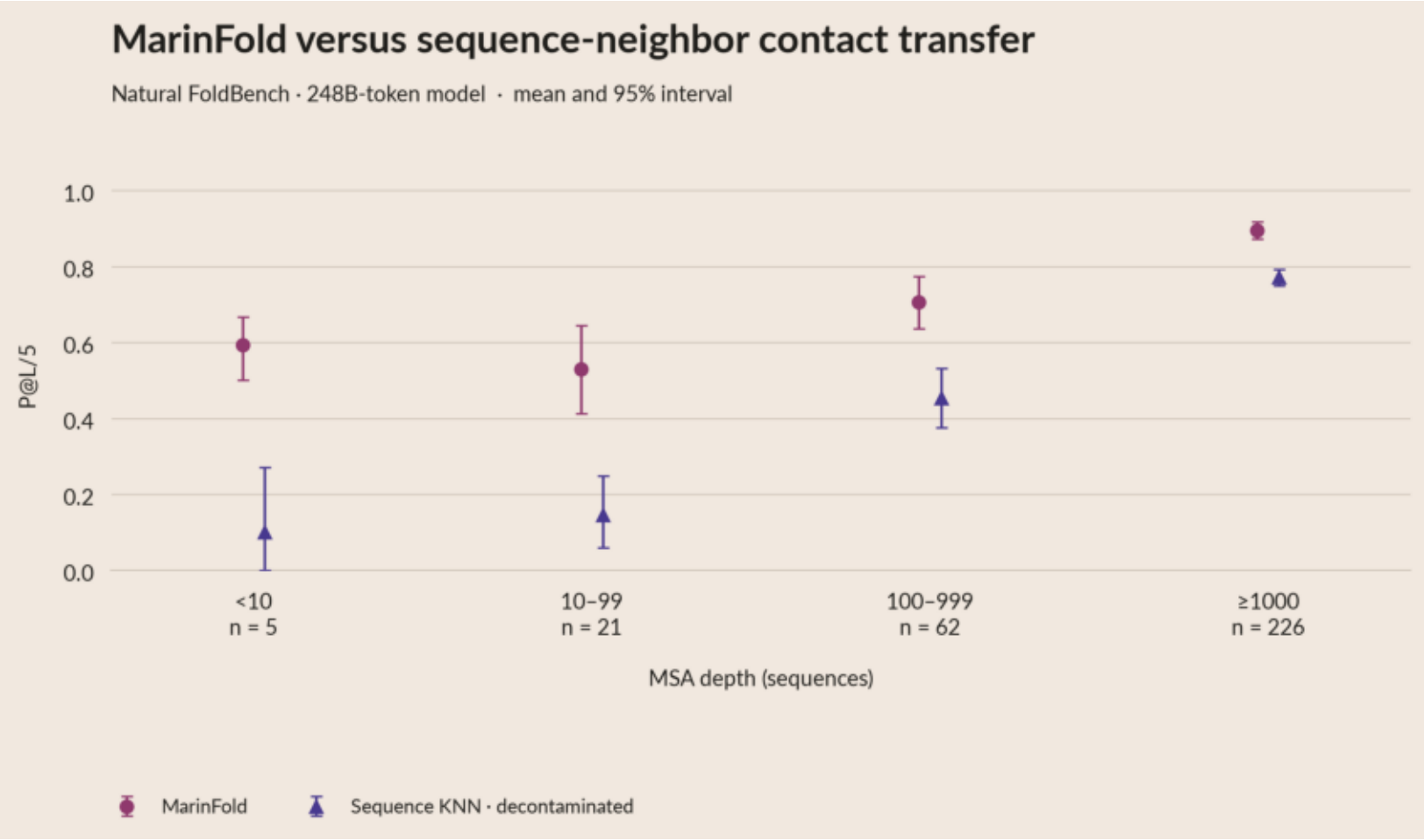
Same nine contact predictors and 314 natural proteins; precision among the top floor(L/5) contacts, L=frozen input length. See PL5_ANALYSIS.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.




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$ render.py
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04b_knn.png

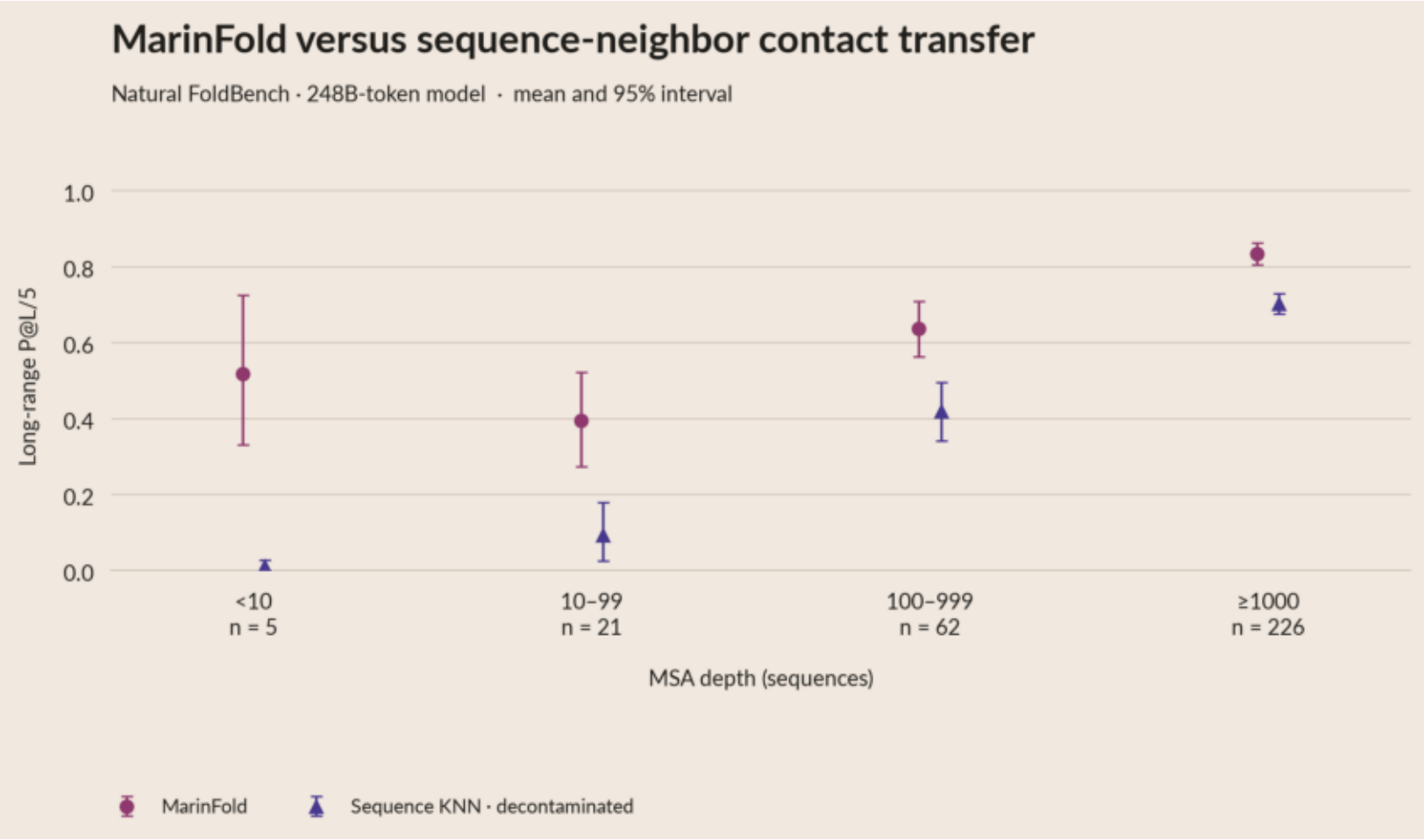
Same 314 natural proteins and fixed 248B-token MarinFold checkpoint. Sequence-KNN transfers contacts from the ten nearest sequences in the native decontaminated training corpus; it does not index the additional ProteinMPNN redesign sequences. Points are protein means with 95% protein-bootstrap intervals. The menu exposes all-range and long-range P@L/5 and R-precision. Paired MarinFold-minus-KNN differences and their intervals are in pl5_paired_deltas.csv. This is a contact-prediction baseline, not a structure predictor.




```
$ render.py
```

04b_knn_long.png

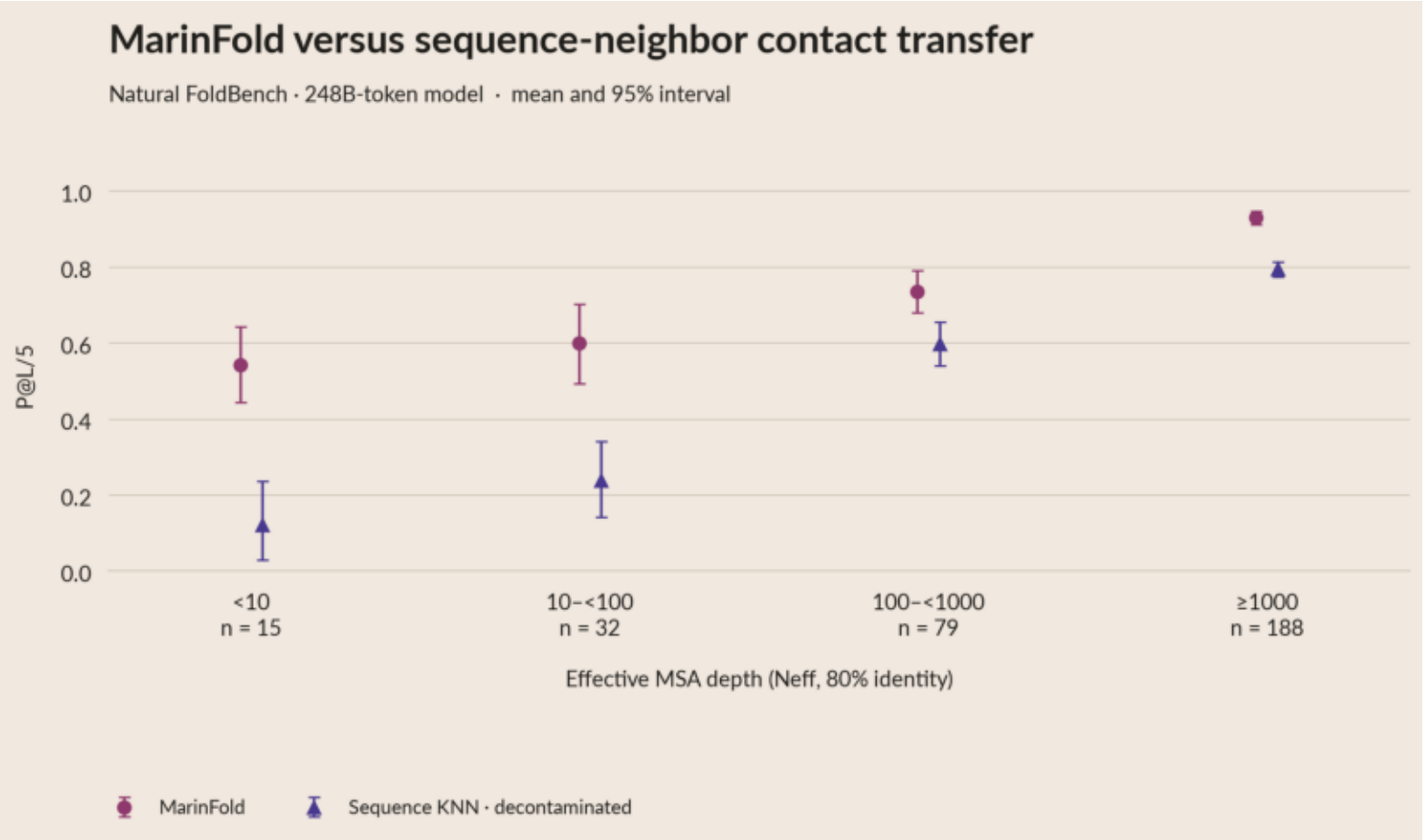
Same 314 natural proteins and fixed 248B-token MarinFold checkpoint. Sequence-KNN transfers contacts from the ten nearest sequences in the native decontaminated training corpus; it does not index the additional ProteinMPNN redesign sequences. Points are protein means with 95% protein-bootstrap intervals. The menu exposes all-range and long-range P@L/5 and R-precision. Paired MarinFold-minus-KNN differences and their intervals are in pl5_paired_deltas.csv. This is a contact-prediction baseline, not a structure predictor.



$ render.py

04b_knn_neff.png

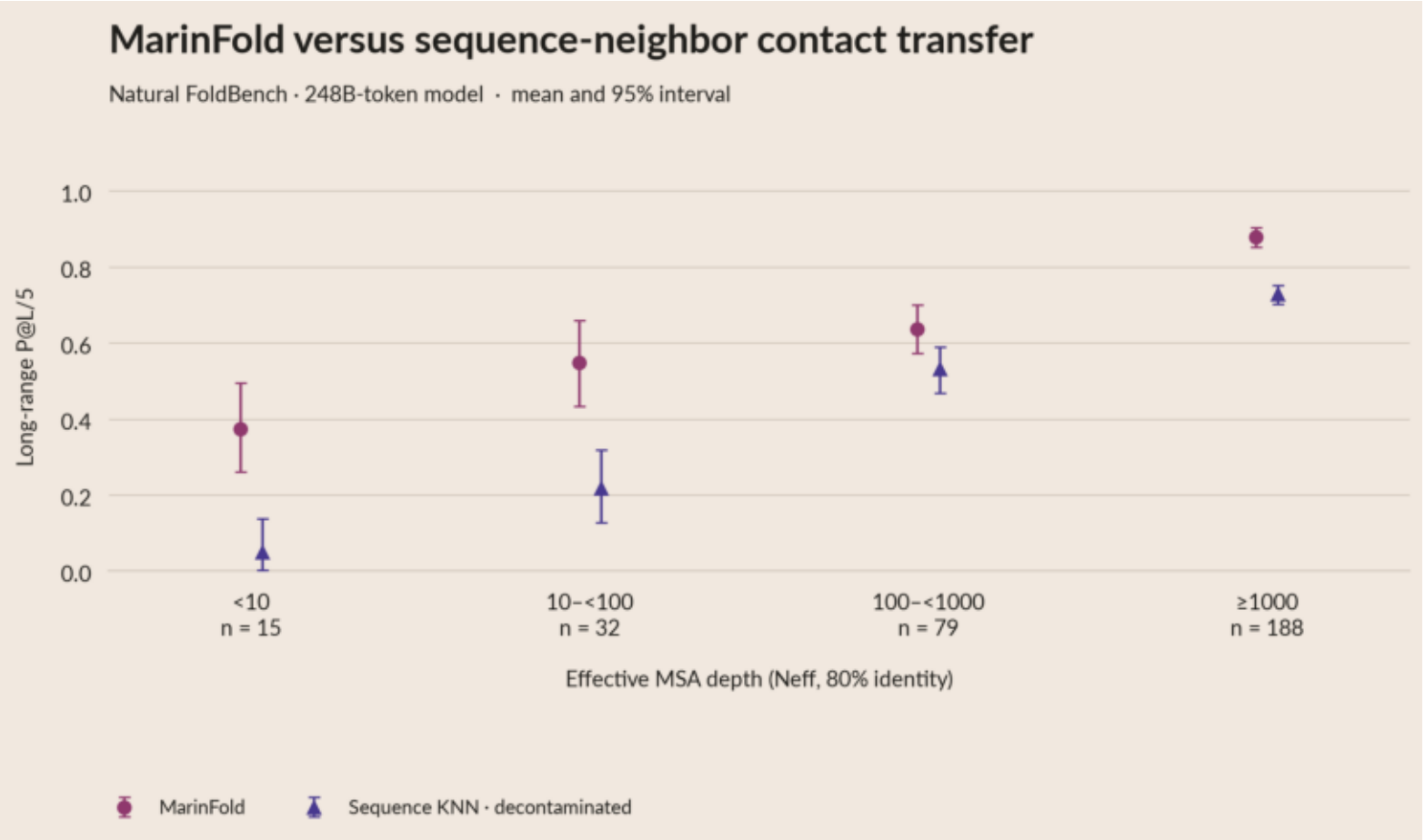
Same 314 natural proteins; 248B-token MarinFold versus k=10 sequence-KNN over the native decontaminated corpus. P@L/5 and R-precision, all and long-range; unchanged original metrics. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

04b_knn_neff_long.png

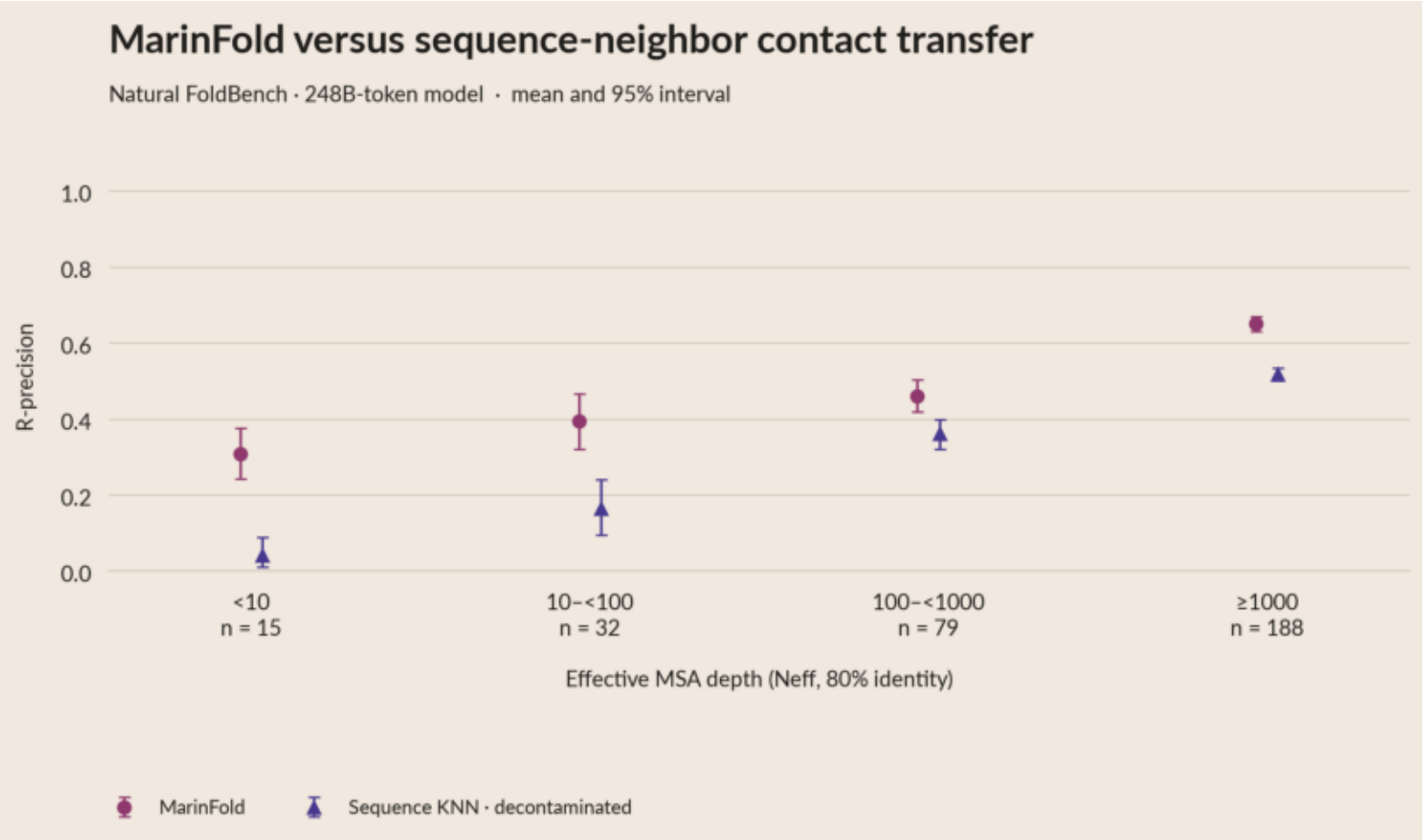
Same 314 natural proteins; 248B-token MarinFold versus k=10 sequence-KNN over the native decontaminated corpus. P@L/5 and R-precision, all and long-range; unchanged original metrics. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

04b_knn_neff_r_precision.png

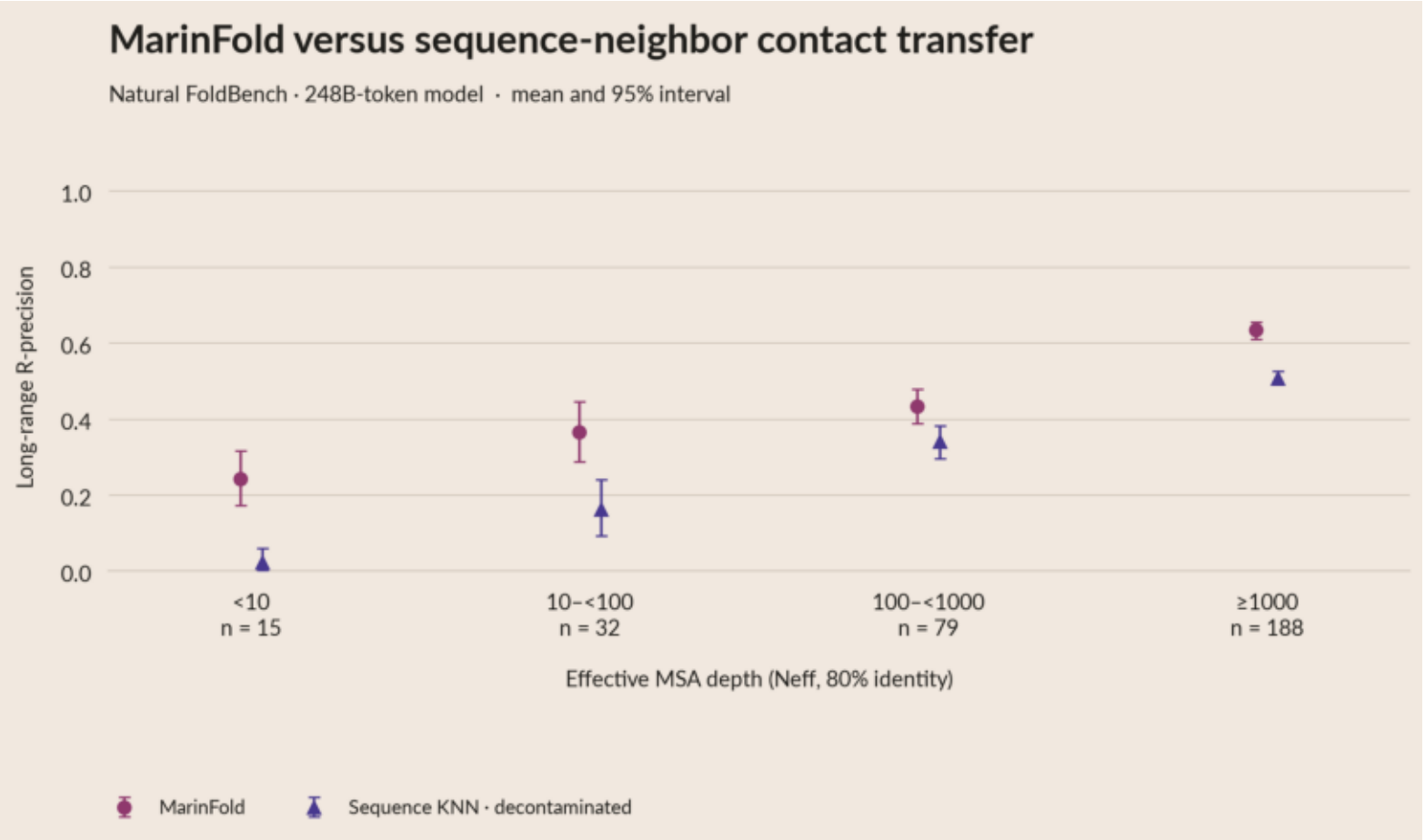
Same 314 natural proteins; 248B-token MarinFold versus k=10 sequence-KNN over the native decontaminated corpus. P@L/5 and R-precision, all and long-range; unchanged original metrics. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

04b_knn_neff_r_precision_long.png

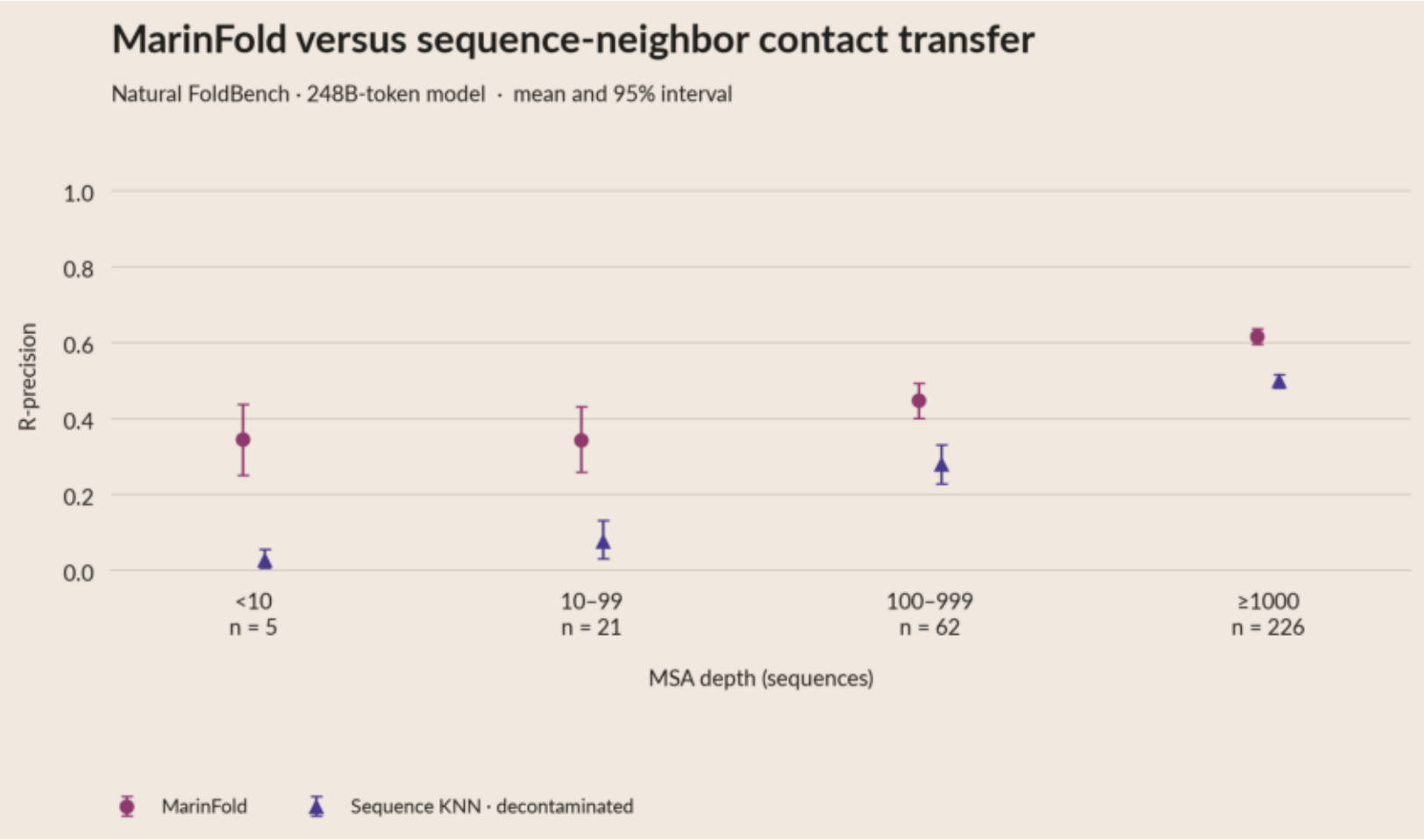
Same 314 natural proteins; 248B-token MarinFold versus k=10 sequence-KNN over the native decontaminated corpus. P@L/5 and R-precision, all and long-range; unchanged original metrics. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

04b_knn_r_precision.png

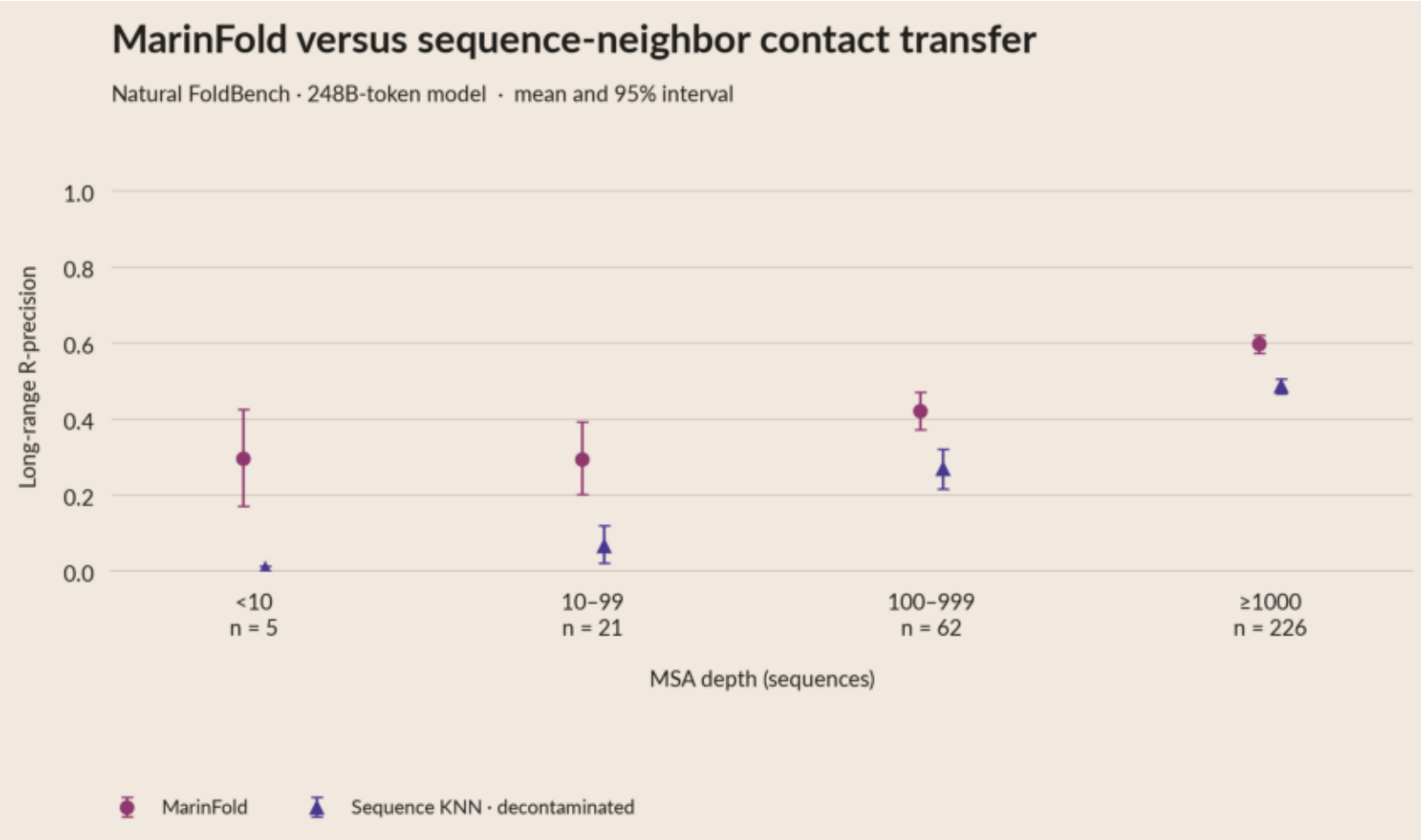
Same 314 natural proteins and fixed 248B-token MarinFold checkpoint. Sequence-KNN transfers contacts from the ten nearest sequences in the native decontaminated training corpus; it does not index the additional ProteinMPNN redesign sequences. Points are protein means with 95% protein-bootstrap intervals. The menu exposes all-range and long-range P@L/5 and R-precision. Paired MarinFold-minus-KNN differences and their intervals are in pl5_paired_deltas.csv. This is a contact-prediction baseline, not a structure predictor.



$ render.py

04b_knn_r_precision_long.png

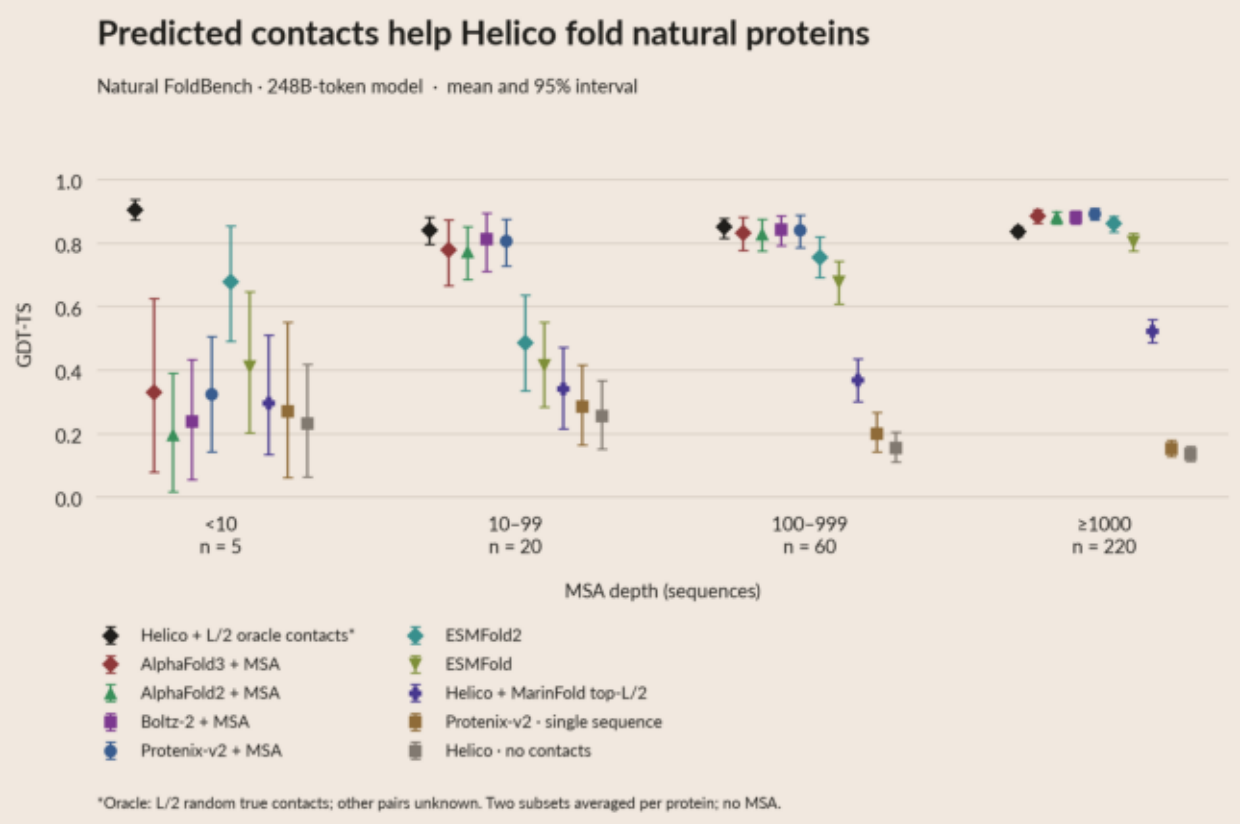
Same 314 natural proteins and fixed 248B-token MarinFold checkpoint. Sequence-KNN transfers contacts from the ten nearest sequences in the native decontaminated training corpus; it does not index the additional ProteinMPNN redesign sequences. Points are protein means with 95% protein-bootstrap intervals. The menu exposes all-range and long-range P@L/5 and R-precision. Paired MarinFold-minus-KNN differences and their intervals are in pl5_paired_deltas.csv. This is a contact-prediction baseline, not a structure predictor.



$ render.py

05_folding.png

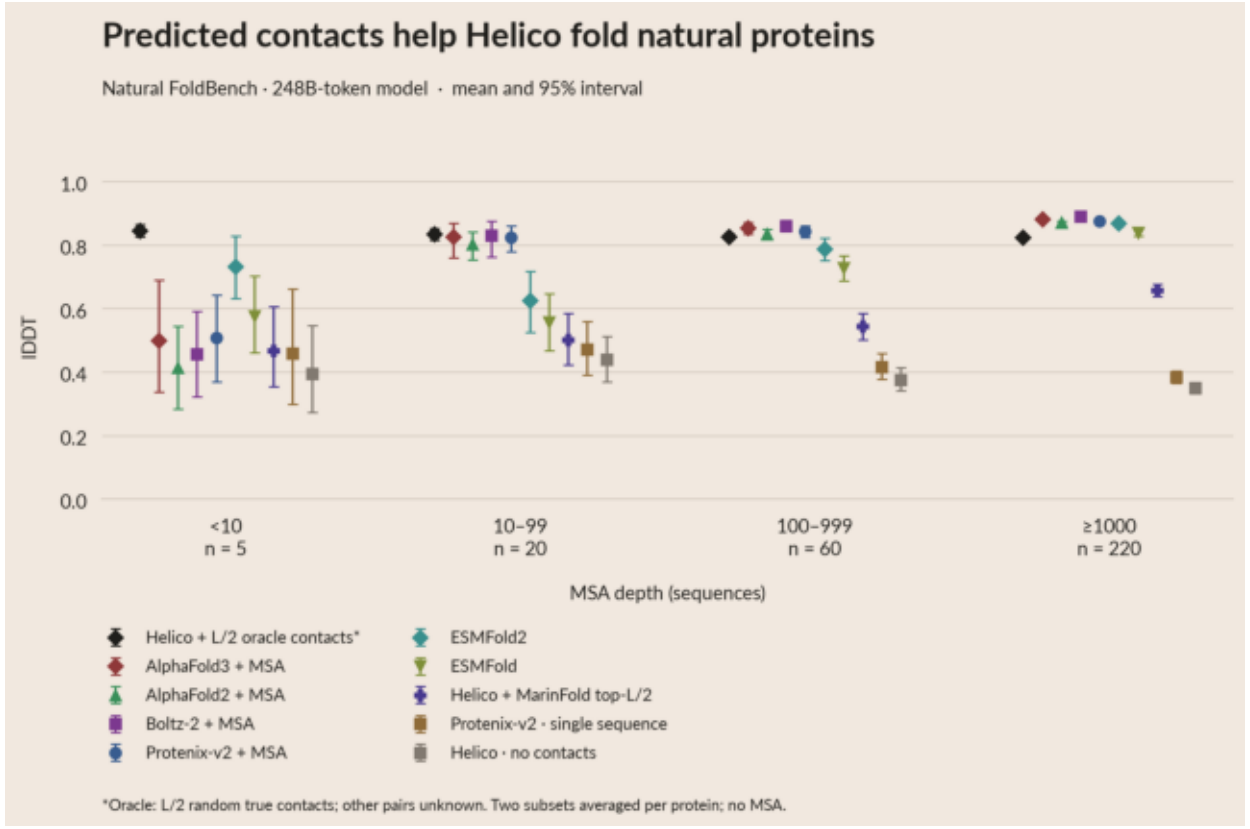
The 248B-token MarinFold model supplies exactly floor(L/2) predicted contacts to Helico, using archived 100-rollout consensus rankings. L is the frozen input sequence length. Filter mapped separation <6 before selecting contacts; every predicted map reaches its budget. Three diffusion samples/map, highest ranking_score selection; no MSA or oracle seed contacts. *Oracle supplies two random subsets of floor(L/2) ground-truth contacts, selects one of three samples per subset, then averages subset accuracies. All other pairs stay unknown in both contact arms. Same 305 natural proteins (95 validation + 210 test), with 95% protein-bootstrap intervals. Oracle caps at 109 versus 110 requested contacts for one protein, and uses two maps rather than one. See MARINFOLD_L2.md, ORACLE_L2.md and coverage.csv. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



$ render.py

05_folding_lddt.png

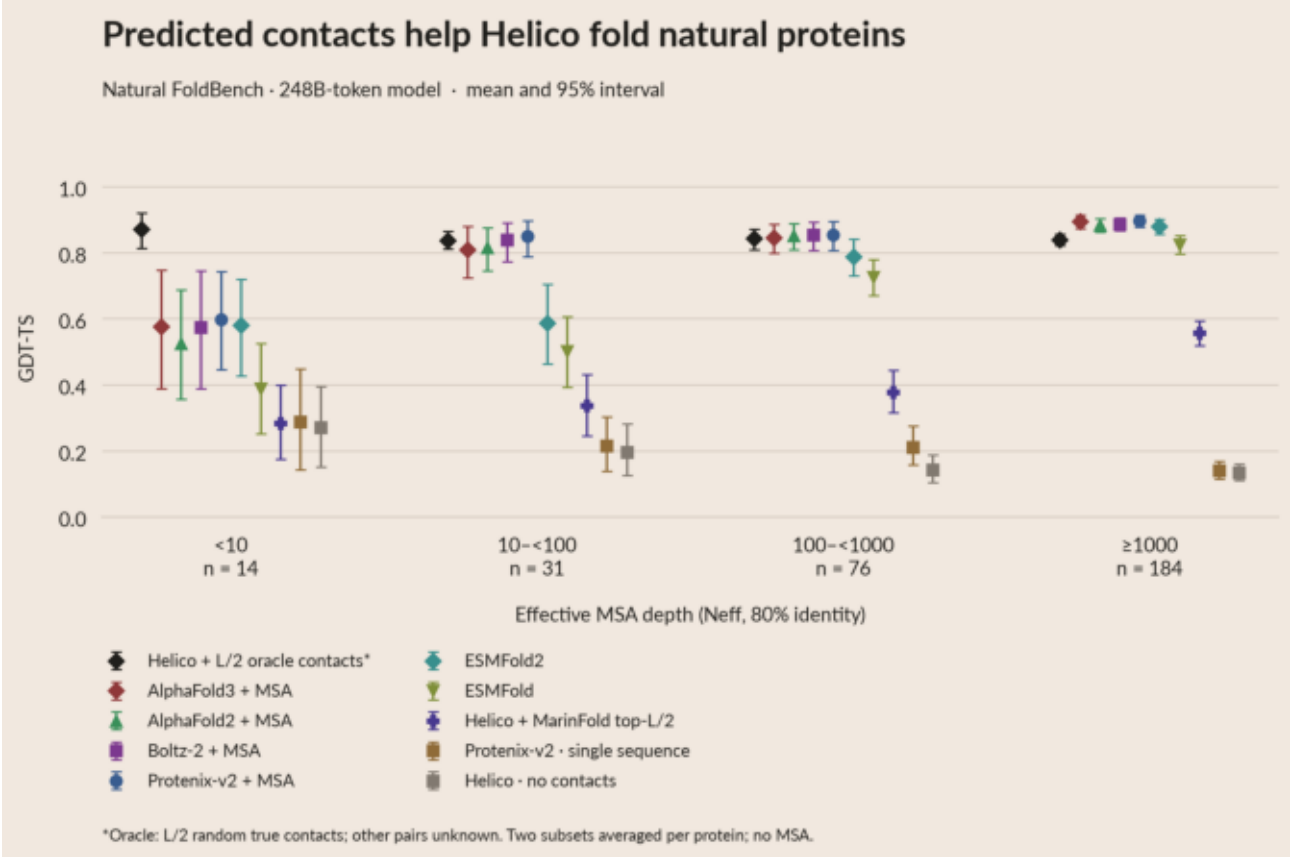
The 248B-token MarinFold model supplies exactly floor(L/2) predicted contacts to Helico, using archived 100-rollout consensus rankings. L is the frozen input sequence length. Filter mapped separation <6 before selecting contacts; every predicted map reaches its budget. Three diffusion samples/map, highest ranking_score selection; no MSA or oracle seed contacts. *Oracle supplies two random subsets of floor(L/2) ground-truth contacts, selects one of three samples per subset, then averages subset accuracies. All other pairs stay unknown in both contact arms. Same 305 natural proteins (95 validation + 210 test), with 95% protein-bootstrap intervals. Oracle caps at 109 versus 110 requested contacts for one protein, and uses two maps rather than one. See MARINFOLD_L2.md, ORACLE_L2.md and coverage.csv. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



$ render.py

05_folding_neff.png

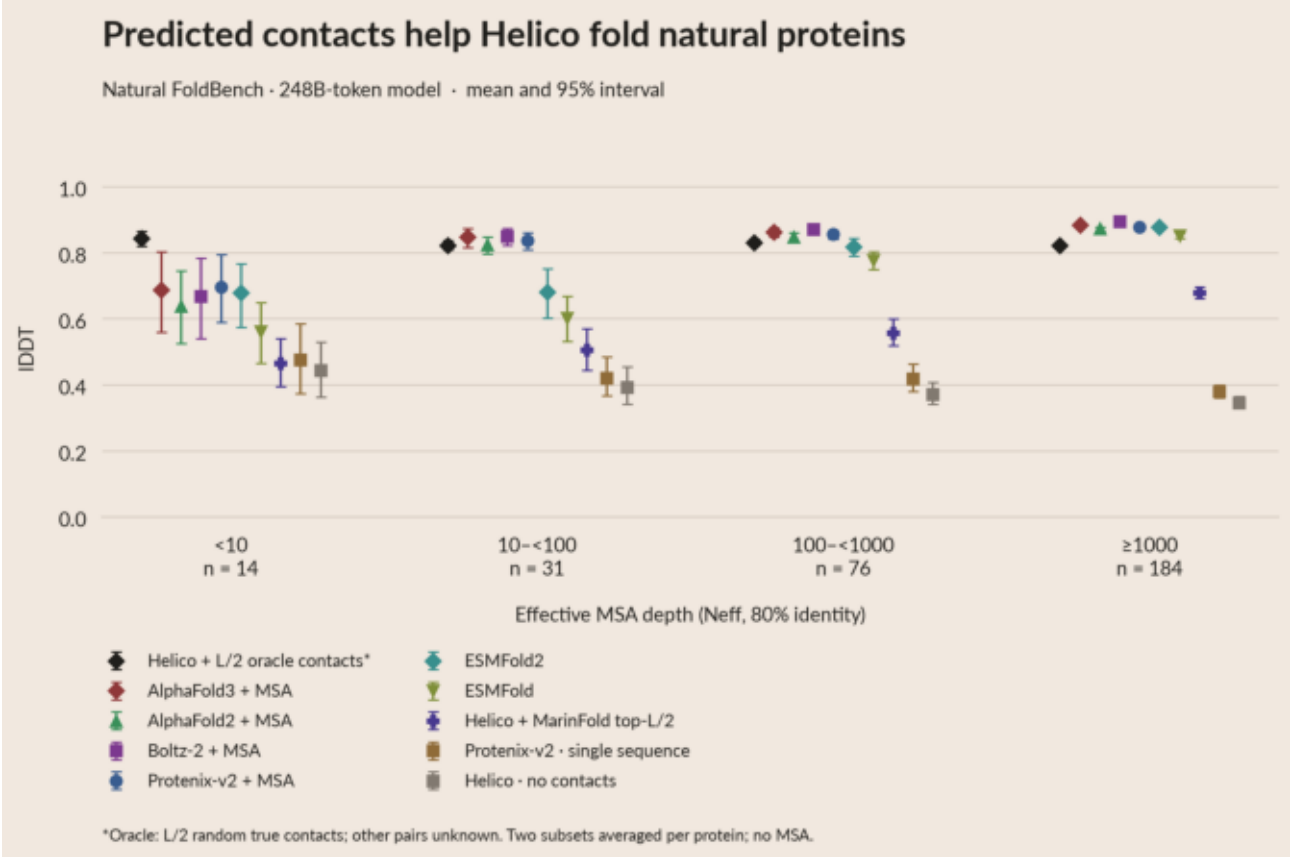
Same 305 natural proteins. MarinFold supplies exact top-L/2 predicted contacts, with no MSA or oracle seeds; three Helico samples, ranking_score selection. Oracle supplies two random L/2 true-contact subsets, selecting within each and averaging; one target has 109 rather than 110 oracle contacts. All seven predictors and no-contact Helico retained. See MARINFOLD_L2.md and ORACLE_L2.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

05_folding_neff_lddt.png

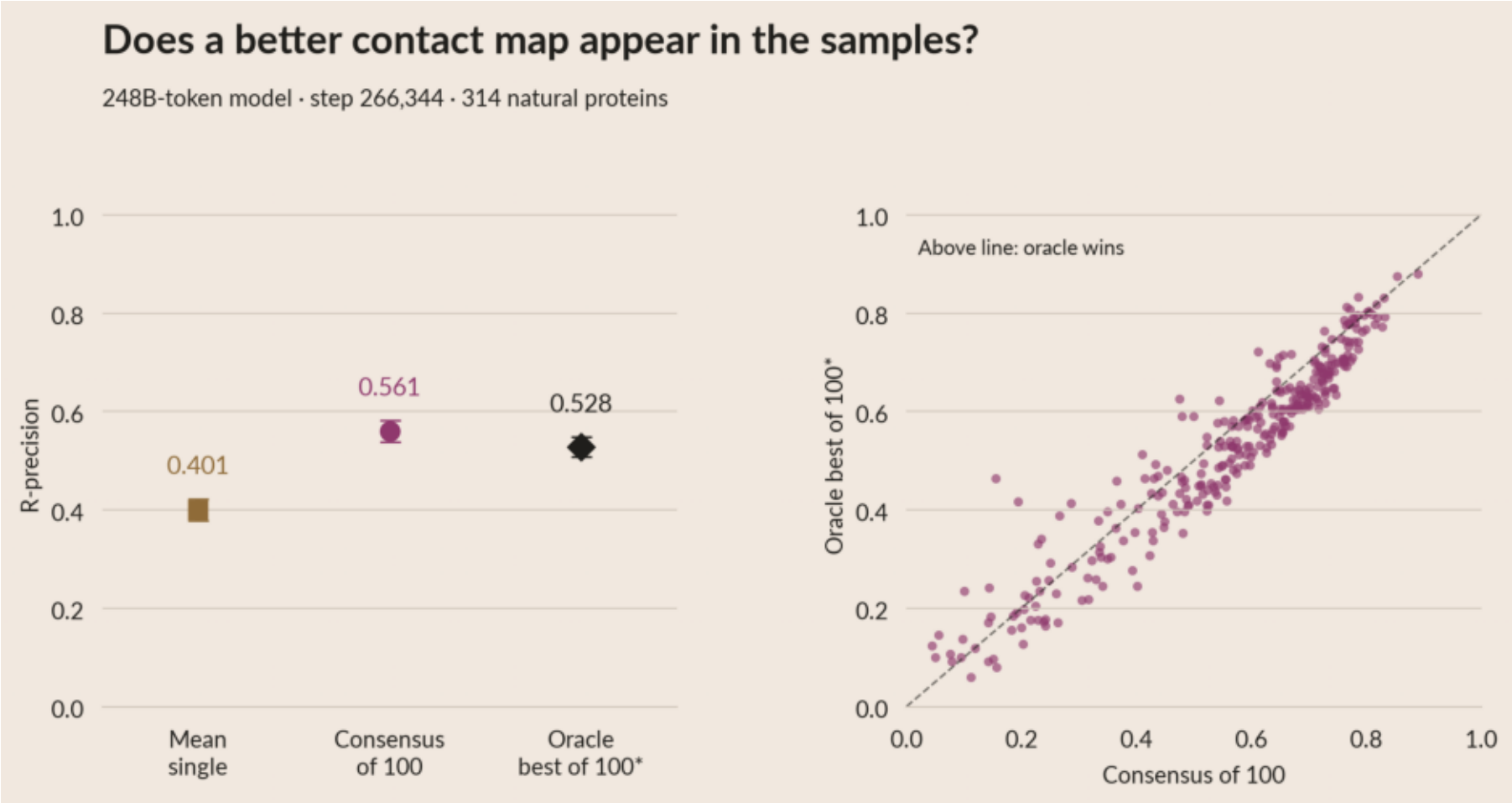
Same 305 natural proteins. MarinFold supplies exact top-L/2 predicted contacts, with no MSA or oracle seeds; three Helico samples, ranking_score selection. Oracle supplies two random L/2 true-contact subsets, selecting within each and averaging; one target has 109 rather than 110 oracle contacts. All seven predictors and no-contact Helico retained. See MARINFOLD_L2.md and ORACLE_L2.md. Neff is recomputed on the exact frozen predictor alignments: sum of inverse neighbor counts at >=80% identity over jointly non-gap columns. Repeated sequences are downweighted; no alignment or prediction is changed. Bins are <10, 10–<100, 100–<1000 and >=1000. Means and 95% intervals bootstrap proteins; cohort membership changes relative to raw depth. Split, viral and design views remain separate. Every score, Neff measurement and contributing protein traces through neff_figure_rows.csv, neff_summary.csv and NEFF_ANALYSIS.md.



$ render.py

06_sampling.png

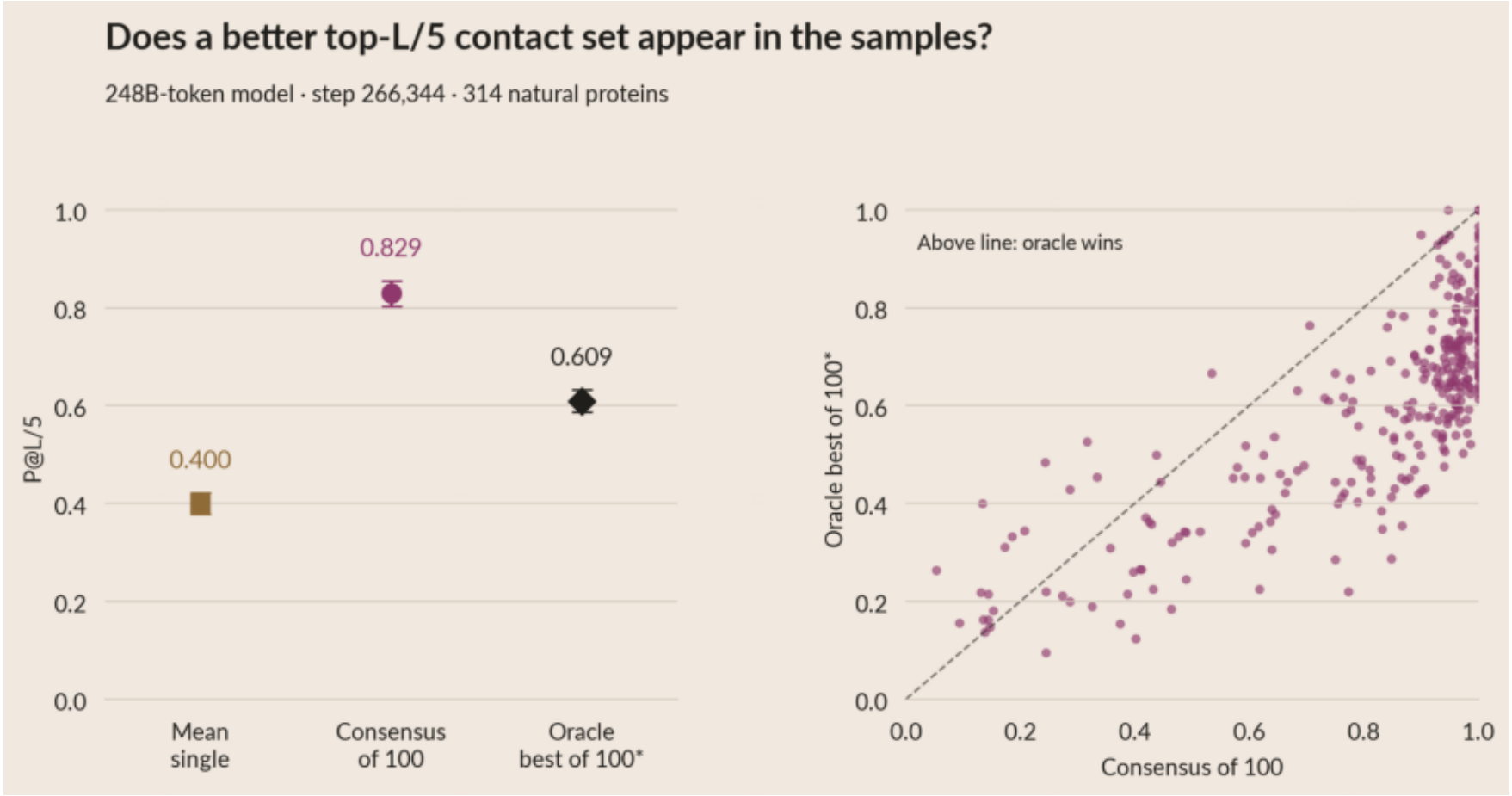
Exp277 step 266,344, 314 natural proteins (97 validation + 217 test), 100 iid rollouts per protein. Consensus and oracle use the same sample pool within each protein. Individual maps are ranked by emission order; short maps retain denominator R, and malformed or unfinished individual maps score zero. The diagnostic consensus uses all parsed maps, following exp321. Oracle selection uses ground truth. This measures contact accuracy, not distinct folds. Scatter colors identify MSA tiers in the interactive view.




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$ render.py
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06_sampling_pl5.png

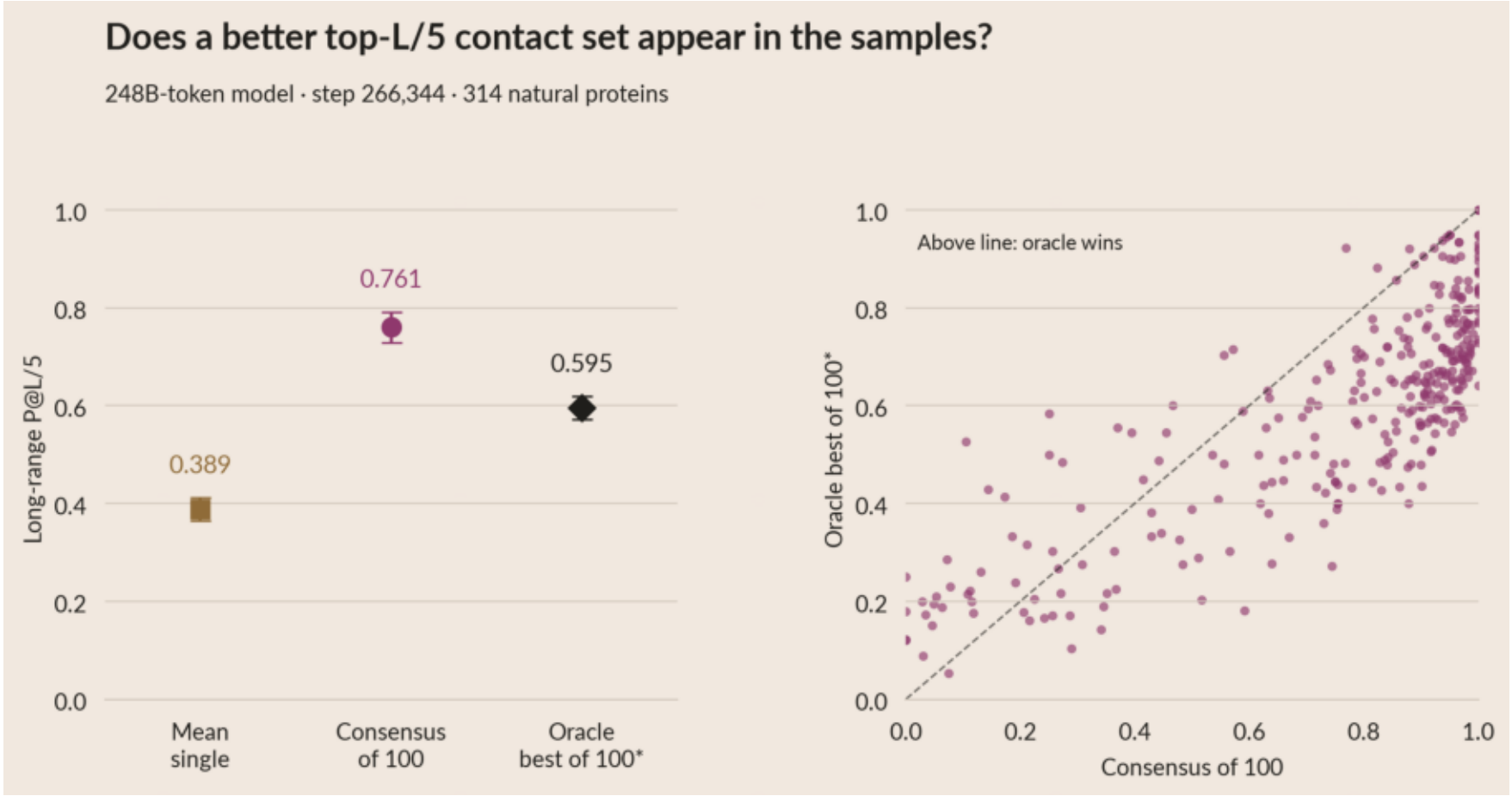
P@L/5 counterpart of the sampling diagnostic: the exact same 314 natural proteins and first 100 iid rollouts per protein. L is frozen input sequence length; k=max(1,floor(L/5)). Individual maps retain emission order after eligibility filtering and deduplication; unfilled ranks receive zero credit and invalid/unfinished maps score zero. Consensus ranks votes from all parsed maps, matching the original diagnostic. Oracle best-of-100 is reselected using P@L/5 and uses ground truth. The scatter pairs consensus with this newly selected oracle per protein. Main contact-panel votes instead omit unfinished maps. This measures contact accuracy, not distinct folds.



$ render.py

06_sampling_pl5_long.png

P@L/5 counterpart of the sampling diagnostic: the exact same 314 natural proteins and first 100 iid rollouts per protein. L is frozen input sequence length; k=max(1,floor(L/5)). Individual maps retain emission order after eligibility filtering and deduplication; unfilled ranks receive zero credit and invalid/unfinished maps score zero. Consensus ranks votes from all parsed maps, matching the original diagnostic. Oracle best-of-100 is reselected using P@L/5 and uses ground truth. The scatter pairs consensus with this newly selected oracle per protein. Main contact-panel votes instead omit unfinished maps. This measures contact accuracy, not distinct folds.




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$ render.py
```